

EM Toolkit (Tintinalli-mapped) – full toolkit

S1 Prehospital Care · S2 Disaster Management · S3 Resuscitation · S4 Resuscitative Procedures · S5 Analgesia, Anesthesia & Procedural Sedation · S6 Wound Management · S7 Cardiovascular Disease · S8 Pulmonary Disorders · S9 Gastrointestinal Disorders · S10 Renal & Genitourinary Disorders · S11 Obstetrics & Gynecology · S12 Pediatrics · S13 Infectious Diseases · S14 Neurology · S15 Toxicology · S16 Environmental Injuries · S17 Endocrine Disorders · S18 Hematologic & Oncologic Disorders · S19 Eye, Ear, Nose, Throat & Oral Disorders · S20 Dermatology · S21 Trauma · S22 Orthopedics · S23 Musculoskeletal Disorders · S24 Psychosocial Disorders · S25 Abuse & Assault · S26 Special Situations · printed 04/10/2026, 16:37:33 · generic per-kg doses (no patient weight entered). Reference aid only – verify clinically. Every **△ VERIFY** item is pending owner review (listed at the end of each section). Original summaries citing public guidelines (AHA, ERC, ESC, ACOG, WHO, ADA/JBDS, ATA, Endocrine Society, WMS, ILCOR, AACT/EAPCCT, DOH/RITM/NPMCC); Tintinalli 9e used only as a topic map.

S1 · Prehospital Care **Built**

EMS systems & handover

Philippine EMS is a mix of LGU/DRRMO rescue units, private and hospital ambulances, and volunteer groups (e.g. Red Cross). Call **911** (national) or **143** (Red Cross) (**△ VERIFY**). Scope of crews varies from first aider to EMT – ask what was done.

Medical oversight

- **Offline** (indirect): written protocols, standing orders, training, quality review
- **Online** (direct): real-time physician advice by radio/phone – document the name, time and orders given
- The receiving physician takes over responsibility on arrival; crews may not leave until handover is complete

Structured handover (ATMIST)

Age / sex	Time of onset/injury
Mechanism / medical complaint	Injuries / findings (head to toe)
Signs: vitals, GCS, trends	Treatment given (drugs, doses, times)

One person talks, the team listens, then questions; hands-off for ~30 s unless critical intervention needed.

Interfacility transfer & referral

- 1 **Stabilise first** within capability: airway, breathing, hemorrhage control, IV access, glucose, analgesia; never delay life-saving care for paperwork or deposit (**△ VERIFY**)
 - 2 Call the receiving physician: **SBAR** summary; confirm bed/specialist availability and record the name and time of acceptance. Use the DOH regional/provincial referral network or One Hospital Command centre where available
 - 3 Choose escort level by risk: BLS crew vs nurse vs physician; ventilated/vasopressor patients need a physician or critical-care nurse
 - 4 Send: referral letter, copies of labs/imaging (CD/phone images), medication sheet with times, consent, blood products if needed
 - 5 Equipment check: O₂ for 1.5× trip duration, suction, defibrillator, drugs, spare batteries; secure ETT, lines, drains before moving
- Patient/relative informed consent for transfer; document reasons (higher level of care, no specialist, no ICU bed)
 - Refusal of transfer: document capacity, risks explained, alternatives offered

Air medical transport

Air ambulance capacity in the Philippines is limited (private operators, AFP/PCG assets for disasters and islands). Consider when ground/sea transfer time is prohibitive.

- Boyle's law: trapped gas expands with altitude (**△ VERIFY**) → decompress pneumothorax/insert drain, vent NG tube, saline-filled ETT/Foley cuffs, check splints and casts

- Hypoxia worsens at altitude: increase FiO₂, consider pre-flight intubation in borderline airways (no room to intubate in a small cabin)
- Relative contraindications: undrained pneumothorax, recent diving (decompression illness – fly low/pressurised), combative patient, uncontrolled hemorrhage needing surgery the aircraft cannot reach faster
- Noise/vibration: monitor alarms invisible – rely on waveform capnography and SpO₂ displays

Mass gatherings

PH examples: Traslación of the Black Nazarene, Sinulog, Ati-Atihan, concerts, election rallies. Expect heat illness, syncope, crush/trample injuries, minor trauma, intoxication and chronic disease exacerbations

(**△ VERIFY**)

- Planning: medical command post, mobile teams, first-aid stations within ~5 min walk, ambulance access routes, referral hospitals pre-alerted, communications
- Crowd crush: compressive asphyxia – rapid extraction, high-flow O₂, treat as mass-casualty (see Disaster: triage)
- Heat: shaded cooling areas, water, early cold-water immersion for heat stroke (Environmental: heat)

Prehospital equipment & spinal motion restriction

Selective spinal motion restriction (**△ VERIFY**)

- Apply if blunt trauma with: altered consciousness/intoxication, midline spinal pain or tenderness, focal neuro deficit, spinal deformity, or distracting injury
- Penetrating trauma alone: **no** spinal immobilisation (delays care, no benefit)
- Use scoop stretcher or vacuum mattress for transfer; remove from backboard on arrival

Bleeding control

- Direct pressure → wound packing → commercial tourniquet 5–7 cm above the wound (not over a joint); note time
- Suspected pelvic fracture with shock: pelvic binder centred on the greater trochanters

Minimum kit checks

- BVM sizes, OPA/NPA, suction, SGA
- Glucometer, naloxone, epinephrine auto-injector/ampoules
- AED/defibrillator, O₂ cylinders with reserve, length-based pediatric tape

Prehospital sources

NAEMSP/ACS-COT/ACEP Spinal motion restriction position statement 2018 · NAEMSP medical oversight position statements · AAMS/CAMTS air medical accreditation standards · WHO Public health for mass gatherings 2015 · Philippine RA 8344 as amended by RA 10932 (2017) · RA 11223 Universal Health Care Act (Health Care Provider Networks) · EO 56 (2018) E911. Tintinalli 9e Section 1 used only as a topic map.

⚠ Items pending owner review in this section (5)

1. PH law: RA 8344 as amended by RA 10932 (2017) – hospitals/clinics may not demand deposits or refuse/delay emergency care; transfer only after stabilisation with documented acceptance

Summary of statute for orientation only; confirm penalties/details with hospital legal office.

2. PH emergency numbers: 911 (national), Philippine Red Cross 143, local LGU/DRRMO rescue lines

Numbers change locally; post the verified local list.

3. Air transport physiology: gas volume expands ~30% at 8,000 ft cabin altitude – drain pneumothorax before flight, fill ETT cuff with saline or monitor pressure, vent NG tube

AAMS/CAMTS teaching values; aircraft and cabin pressure vary.

4. Selective spinal motion restriction (2018 consensus): long backboards are not for transport immobilisation; collar + scoop/vacuum mattress when criteria present

NAEMSP/ACS-COT/ACEP 2018; local EMS protocols may still mandate boards.

5. Mass-gathering planning: typical patient presentation rate ~0.5–2 per 1,000 attendees (higher with heat, alcohol, crowd density)

Observational literature; PH religious processions (e.g. Traslación) may far exceed this.

 Mass-casualty triage

Answer top to bottom; START (adult) logic active – switch to Pediatric mode for JumpSTART.

△ VERIFY

- Life-saving interventions allowed during triage: open airway, control major bleeding (tourniquet), decompress tension pneumothorax, autoinjector antidotes
- **SALT:** Sort (walk / wave / still), Assess, Life-saving interventions, Treatment/Transport – adds an "expectant" category based on resources
- Re-triage at every step (scene, casualty collection point, ED door); under-triage kills, over-triage overwhelms
- Hospital: activate the code (e.g. Code Orange), incident command, surge beds, cancel elective cases, call-back lists; DOH HEMB / regional operations centre informed

 Typhoons, floods, earthquakes**Typhoon / flood**

- Injuries: lacerations, drowning, electrocution (downed lines), CO poisoning from generators (Environmental: CO)
- Infections 1–3 weeks later: **leptospirosis**, diarrhea/cholera, dengue, wound infections (Vibrio, Aeromonas), tetanus
- Interrupted dialysis, insulin and TB/HIV medicines – plan supplies; evacuation-centre outbreaks (measles, ARI)

Doxycycline – leptospirosis prophylaxis (flood exposure)△ VERIFY

Low risk

200 mg PO single dose within 24–72 h

Moderate risk

200 mg PO once daily × 3–5 days

High risk (continuous)

200 mg PO once weekly until exposure ends

Earthquake / collapse – crush syndrome

- 1 Start IV fluid **before extrication**: 0.9% NaCl ~1 L/hr adult
△ VERIFY
- 2 After release: cardiac monitor – sudden hyperkalemia/arrhythmia; treat (hyperkalemia)
- 3 Target urine output ~200–300 mL/hr in adults; if anuric, stop forcing fluids – dialysis △ VERIFY
- 4 Compartment syndrome: measure/suspect early; fasciotomy decisions by surgeon (infection risk in disaster settings)
- 5 Tetanus prophylaxis and wound care (tetanus table)

Volcanic eruption (e.g. Taal, Mayon)

- Ash: respiratory irritation, asthma/COPD flares, eye abrasions – N95 masks, eye irrigation
- Lahar/pyroclastic flow injuries: burns, inhalation injury, trauma

 Blast injuries

Mechanism	Typical injuries
Primary – pressure wave	Blast lung, TM rupture, bowel perforation, eye globe rupture, TBI

Mechanism	Typical injuries
Secondary – projectiles	Penetrating wounds anywhere (most common)
Tertiary – body thrown	Blunt trauma, fractures, amputations
Quaternary – everything else	Burns, crush, inhalation (smoke, CO, cyanide), psychological

- Blast lung: hypoxia, hemoptysis, "butterfly" CXR; may appear up to 48 h; lung-protective ventilation, judicious fluids; air embolism risk with positive pressure
- TM rupture alone does not rule in or out other primary blast injury; observe symptomatic patients ≥ 6–8 h (CXR, SpO₂)
- Expect a surge of walking wounded to the nearest hospital before the severely injured arrive – keep resus bays free
- Possible secondary device / contamination (dirty bomb) – security and radiation screening

 Chemical incidents & decontamination

- 1 Protect staff first: PPE, zones (hot – warm/decontamination – cold), decontaminate **outside** the ED
- 2 Remove all clothing (removes most contamination), bag it, then copious water rinse ± mild soap 3–5 min; dry decontamination (blotting) for limited water
- 3 Life-saving care in the warm zone: airway, autoinjector antidotes, hemorrhage control

Agent	Clues	Treatment
Nerve agents / organophosphates	SLUDGE + miosis, bronchorrhea, fasciculations, seizures	Atropine to dry secretions, pralidoxime, benzodiazepine (Tox: pesticides)
Chlorine, ammonia (industrial, pool)	Eye/throat burning, cough, wheeze, delayed pulmonary edema	O ₂ , bronchodilators, observe 6–24 h
Cyanide (fires, industrial)	Sudden collapse, seizures, lactate > 8	Hydroxocobalamin (antidotes)
Vesicants (mustard)	Delayed blistering, eye, airway injury	Decontaminate early; burn-type care

Atropine – nerve agent / mass organophosphate △ VERIFY

IV/IM

2 – 6 mg, repeat q5 min (double if no effect) until secretions dry

Pralidoxime (2-PAM) **VERIFY**

IV over 15–30 min

1 – 2 g, then infusion ~ 8 mg/kg/hr or repeat per protocol

Radiation & biological threats

Radiation

- Life-threatening injuries first – contaminated patients rarely endanger staff with standard precautions; remove clothing (removes ~90% external contamination), wash skin, cover wounds
- Acute radiation syndrome severity: time to vomiting < 1 h and falling absolute lymphocyte count at 24–48 h = high dose; repeat CBC q6–12h
- Internal contamination: potassium iodide (radioiodine only), Prussian blue (cesium/thallium), DTPA (plutonium/amerium) – via national authority
- PH: Philippine Nuclear Research Institute (PNRI) and DOH for advice

Potassium iodide (radioiodine) **VERIFY**

Single dose

Adult 130 mg · 3–18 y 65 mg · 1 mo–3 y 32 mg · < 1 mo 16 mg

Bioterrorism / unusual outbreaks

- Clues: clusters of severe illness in healthy people, unusual season/geography, simultaneous animal deaths
- Inhalational anthrax (widened mediastinum, sepsis), plague (pneumonic – droplet precautions), smallpox (synchronous centrifugal vesicles – airborne/contact)
- Isolate, notify DOH Epidemiology Bureau / RITM immediately; specimens via RITM

Disaster sources

Romig, JumpSTART pediatric MCI triage · Lerner et al., SALT mass casualty triage (Disaster Med Public Health Prep 2008) · CDC Explosions and blast injuries primer · Sever & Vanholder, ISN Renal Disaster Relief Task Force recommendations for crush casualties (NDT 2012) · DOH leptospirosis prevention/prophylaxis guidance · CHEMM (HHS) chemical emergencies · REMM (HHS) radiation emergency medical management · FDA potassium iodide guidance 2001 / WHO iodine thyroid blocking 2017 · WHO/DOH HEMB disaster health guidance. Tintinalli 9e Section 2 used only as a topic map.

Items pending owner review in this section (5)

1. START triage cut-offs: RR > 30 → immediate; radial pulse absent or capillary refill > 2 s → immediate; cannot follow simple commands → immediate; JumpSTART: RR < 15 or > 45, 5 rescue breaths if apneic with pulse

START/JumpSTART published algorithms; SALT is the US national guideline alternative.

2. Leptospirosis post-flood prophylaxis (DOH): low risk – doxycycline 200 mg single dose within 24–72 h of exposure; moderate risk (wounds, repeated wading, swallowed water) – 200 mg once daily for 3–5 days; high risk (continuous exposure) – 200 mg once weekly until exposure ends

DOH categories recalled; pregnant women/children: azithromycin or amoxicillin per DOH – confirm current DOH memorandum.

3. Crush syndrome: start 0.9% NaCl ~1 L/hr (children 15–20 mL/kg/hr) before/during extrication, then titrate to urine output ~200–300 mL/hr; avoid K⁺-containing fluids; treat hyperkalemia

ISN Renal Disaster Relief Task Force 2012; volumes must be reduced in elderly/anuric patients.

4. Potassium iodide (radioiodine release only): adult 130 mg; 3–18 y 65 mg (adolescents ≥ 70 kg 130 mg); 1 month–3 y 32 mg; < 1 month 16 mg – single daily dose, only on public-health instruction

FDA 2001 / WHO 2017; protects thyroid only; PH: on PNRI/DOH advice.

5. Nerve agent mass casualty: atropine 2–6 mg IV/IM (peds 0.05–0.1 mg/kg) repeated every 5 min to dry secretions; pralidoxime 1–2 g IV over 15–30 min (peds 25–50 mg/kg); diazepam/midazolam for seizures

CHEMM/US Army MRICD values; autoinjector dosing differs.

Anaphylaxis

Airway/breathing/circulation problem + likely allergen → **IM epinephrine first**, lie flat (legs up), high-flow O₂.
Antihistamines/steroids are not first-line.

Epinephrine IM – anaphylaxis **△ VERIFY**

IM anterolateral thigh

0.01 mg/kg, max 0.5 mg

1 mg/mL (1:1,000). Repeat every 5–15 min if no improvement.

Crystalloid bolus **△ VERIFY**

Bolus

500 – 1,000 mL NS / balanced crystalloid, repeat PRN

Reassess after each bolus.

Epinephrine infusion – refractory anaphylaxis **△ VERIFY**

IV infusion

0.05 – 0.5 mcg/kg/min

1 mg in 100 mL NS = 10 mcg/mL. After ≥ 2 IM doses + fluids. No undiluted IV boluses.

Salbutamol neb (bronchospasm) **△ VERIFY**

Nebulized

2.5 – 5 mg, repeat / continuous PRN

Diphenhydramine (adjunct – not first-line) **△ VERIFY**

IV/IM

25 – 50 mg

50 mg/mL. Treats skin symptoms only.

Glucagon (anaphylaxis on β-blocker) **△ VERIFY**

IV over 5 min

1 – 5 mg, then 5 – 15 mcg/min

Vomiting common – protect airway.

△ Items pending owner review in this section (9)

1. Pediatric IM epinephrine max 0.3 mg

Set per owner spec. Several guidelines (WAO, ASCIA, RCUK) allow 0.5 mg for adolescents / ≥ 30–50 kg.

2. Epinephrine infusion for refractory anaphylaxis (adult 0.05–0.5, peds 0.1–1 mcg/kg/min @ 10 mcg/mL)

Starting rates differ between guidelines (e.g. RCUK starts ~0.1 mcg/kg/min).

3. Glucagon (anaphylaxis on β-blocker / β-blocker overdose): adult 1–5 mg, peds 20–30 mcg/kg (max 1 mg); tox 50 mcg/kg

Doses differ between allergy and toxicology references.

4. Pediatric salbutamol neb 2.5 mg (< 20 kg) / 5 mg (≥ 20 kg)

Weight/age cut-offs differ by protocol.

5. Pediatric salbutamol neb weight cut-off is inconsistent in this app: 20 kg (Tox/Pulm asthma) vs 25 kg (hyperkalemia, More tab)

Existing values kept as-is; owner to pick one cut-off (or an age-based rule) and harmonise both cards.

6. IM epinephrine 0.01 mg/kg (1 mg/mL) – max 0.5 mg adult, 0.3 mg child; repeat q5–15 min

v1 values (WAO/EAACI/ASCIA); reused here for sting anaphylaxis and now flagged everywhere it appears.

7. Crystalloid bolus in anaphylaxis (adult 0.5–1 L; peds 10–20 mL/kg) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

8. Diphenhydramine adjunct (25–50 mg; peds 1 mg/kg) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

9. Salbutamol neb in anaphylaxis bronchospasm (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

 Undifferentiated shock

- 1 Recognise: hypotension or hypoperfusion (mottling, cap refill > 3 s, altered mentation, oliguria, lactate > 2–4) – normotensive shock exists
- 2 Two large-bore IVs/IO, monitor, lactate, VBG, glucose, ECG, cultures if sepsis possible
- 3 **RUSH** ultrasound – **Pump** (effusion/tamponade, LV function, RV dilation), **Tank** (IVC, FAST, pleural fluid, lung B-lines), **Pipes** (aortic aneurysm/dissection, DVT)
- 4 Treat by type: hypovolemic/hemorrhagic → blood (MTP); distributive → fluids + norepinephrine (pressors), anaphylaxis (anaphylaxis), adrenal (adrenal crisis); cardiogenic → cautious fluid, inotrope, revascularization; obstructive → decompress PTX, drain tamponade, lyse massive PE (PE)
- 5 Fluids in small boluses with reassessment ; start norepinephrine peripherally if needed (large proximal vein, check site)
- 6 Push-dose epinephrine only as a bridge during intubation or while preparing infusion

Push-dose epinephrine (10 mcg/mL) 

IV

5 – 20 mcg (0.5–2 mL) every 2–5 min **Acid-base interpreter****Enter pH, PaCO₂ and HCO₃⁻**Venous pH runs ~0.03 lower and venous HCO₃⁻ ~1–2 higher than arterial; VBG PCO₂ is unreliable for exact CO₂ in shock. **Hyperbaric oxygen**

- Emergency indications: CO poisoning with neurologic signs, LOC, pregnancy or very high COHb (CO card); decompression sickness and arterial gas embolism (diving – 100 % O₂, supine, fluids, diving/drowning); necrotizing soft-tissue infection (adjunct to surgery); crush injury; central retinal artery occlusion (early); severe anemia when transfusion impossible

- Contraindication: untreated pneumothorax (insert chest tube before chamber)
- Transfer: keep on 100 % O₂; for divers, fly low (cabin pressure ≤ 300 m) or ground transport
- PH: very few chambers (naval/military and some private hospitals) – identify the nearest facility in advance and call before transfer 

 Sudden cardiac death & inherited arrhythmias

- Survivor of cardiac arrest: post-arrest care (post-arrest card), 12-lead ECG, echo, coronary angiography if ischemic cause likely
- Look on the ECG for: **Brugada** type 1 pattern (coved ST elevation ≥ 2 mm V1–V2, unmasked by fever), long/short QT, pre-excitation (WPW), epsilon wave/T inversion V1–V3 (ARVC), LVH with deep Q waves (HCM)
- PH: **bangungot** (sudden unexplained nocturnal death in young Southeast Asian men) is linked to Brugada syndrome – treat fever aggressively, avoid sodium-channel blockers and other Brugada-provoking drugs (see brugadadrugs.org), refer for ICD evaluation; screen first-degree relatives
- Syncope with exertion, while supine, preceded by palpitations, or family history of sudden death → admit/cardiology
- Electrical storm: arrhythmia card

 Resuscitation (extended) sources

Evans et al., Surviving Sepsis Campaign 2021 · Perera et al., RUSH exam (Emerg Med Clin North Am 2010) · Berend, Seifter et al., Physiological approach to assessment of acid-base disturbances (NEJM 2014) · Albert, Dell & Winters (1967) · UHMS Hyperbaric Oxygen Therapy Indications, 15th ed. · Zeppenfeld et al., ESC ventricular arrhythmias and SCD 2022 · Antzelevitch et al., J-wave syndromes expert consensus 2016 · Gervacio-Domingo et al., Brugada/bangungot in Filipinos (various). Tintinalli 9e Section 3 used only as a topic map.

 Items pending owner review in this section (27)**1. Epinephrine via ETT (adult 2–2.5 mg; peds 0.1 mg/kg, capped here at 2.5 mg)**

Evidence weak; IV/IO preferred. The 2.5 mg pediatric cap is borrowed from the adult ETT dose.

2. Atropine ampule strength assumed 1 mg/mL

Philippine stock may be 0.5 or 0.6 mg/mL – check the ampule before drawing up.

3. Pediatric atropine 0.02 mg/kg with NO minimum dose (max 0.5 mg)

PALS 2020 removed the 0.1 mg minimum; some local protocols still use it.

4. Pediatric lidocaine 1 mg/kg, capped at 100 mg

100 mg cap is from common references, not stated explicitly in PALS.

5. Pediatric magnesium sulfate 25–50 mg/kg (max 2 g) and stock concentrationPhilippine MgSO₄ comes as 50%, 25% and 20% – confirm which is on hand.**6. Procainamide (adult 20–50 mg/min to 17 mg/kg; peds 15 mg/kg over 30–60 min)**

Availability in the Philippines is limited; confirm product/concentration (100 mg/mL assumed).

7. Adult synchronized cardioversion energies (AHA 2025: AF/flutter 200 J; narrow-complex & monomorphic VT 100 J)

Device-specific; local protocols may still use 2020 values (50–100 J narrow, 120–200 J AF).

8. Double sequential / vector-change defibrillation for refractory VF

AHA 2025 “may be considered”; requires local policy and two defibrillators.

9. Pediatric vasoactive infusion ranges (epinephrine/norepinephrine 0.05–1 mcg/kg/min; epi for bradycardia 0.1–1)

Titration ranges vary by PICU protocol; standard 16 mcg/mL bag assumed.

10. Pediatric calcium (CaCl₂ 10% 20 mg/kg max 2 g; gluconate 10% 60 mg/kg max 3 g)Maximum single doses differ (some cap CaCl₂ at 1 g).**11. Pediatric doses are capped at an adult dose (engine-wide). Caps added where v1 had none: procainamide 1 g, bicarbonate 50 mEq, ketamine/propofol 200 mg, etomidate 30 mg, rocuronium 120 mg, succinylcholine 150 mg, fentanyl 100 mcg, phenobarbital 1.5 g, mannitol 100 g, fomepizole 1.5 g, methylene blue 200 mg, atropine (OP) 2 mg**

For drugs dosed per kg in adults too, the cap is the adult dose at a ~100 kg reference weight – a policy choice for owner sign-off, not a guideline value.

12. ACLS arrhythmia drug doses reused from v1 in the Cardiovascular section (adenosine, amiodarone, lidocaine, magnesium, bradycardia infusions)

AHA 2020/2025 values; unchanged from v1 but now flagged wherever shown.

13. Dextrose doses reused from v1 in the Hypoglycemia card (adult 25 g D50; peds 0.5–1 g/kg D10/D25; neonate 0.2 g/kg)

Unflagged in v1; peds range differs from ISPAD (0.2–0.5 g/kg) – see en-hypo.

14. Adult defibrillation energy: biphasic 120–200 J per manufacturer (maximum if unknown), monophasic 360 J

AHA 2020/2025; device-specific.

15. Pediatric energies: defibrillation 2 J/kg → 4 J/kg → ≥ 4 J/kg (max 10 J/kg or adult dose); synchronized cardioversion 0.5–1 J/kg then 2 J/kg

PALS 2020/2025; v1 values reused, now flagged.

16. Epinephrine – cardiac arrest (1 mg / 0.01 mg/kg q3–5 min) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

17. Calcium chloride/gluconate in arrest (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

18. Sodium bicarbonate 1 mEq/kg in arrest (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

19. ACLS/PALS algorithm step doses and shock energies (epinephrine, amiodarone, lidocaine, atropine; 120–200 J / 2 → 4 J/kg)

Inline values from v1 / merged card, previously unflagged.

20. Defibrillation / cardioversion energies (adult and pediatric)

Inline values from v1 / merged card, previously unflagged.

21. Push-dose epinephrine (bridge only): 10 mcg/mL, give 0.5–2 mL (5–20 mcg) IV q2–5 min; peds 1 mcg/kg (max 10–20 mcg); start norepinephrine infusion early

Common ED practice; dilution errors reported – use pre-made syringes if available.

22. Undifferentiated shock: crystalloid boluses 250–500 mL (peds 10–20 mL/kg) with reassessment after each; norepinephrine 0.05–0.5 mcg/kg/min early if MAP < 65 despite fluids

SSC 2021; avoid fixed large volumes in cardiogenic shock.

23. HBO indications per UHMS (CO poisoning with LOC/neurologic signs/pregnancy/COHb > 25 %, decompression sickness, arterial gas embolism, necrotizing infections, crush, CRAO, etc.); PH chambers are few – contact facility early

UHMS 15th ed; PH chamber locations must be confirmed (Navy/AFP and a few private hospitals).

24. IO in conscious patient: 2 % lidocaine (preservative-free) 40 mg (2 mL) slowly over 2 min via IO, wait 60 s, flush, then 20 mg; peds 0.5 mg/kg (max 40 mg)

Manufacturer (EZ-IO) guidance.

25. Post-arrest targets: SpO₂ 92–98 %, PaCO₂ 35–45 mmHg, MAP ≥ 65 (consider higher), glucose 140–180 mg/dL, prevent fever (≤ 37.5 °C) for ≥ 72 h (TTM 32–36 °C optional per AHA 2023), prognosticate ≥ 72 h after normothermia

AHA 2023 / ILCOR 2023 / ERC-ESICM 2025.

26. Monitoring targets: MAP ≥ 65 mmHg, capillary refill ≤ 3 s, urine output ≥ 0.5 mL/kg/h, PLR stroke-volume rise ≥ 10–15 %, ScvO₂ ≥ 70 %

SSC 2021 / ANDROMEDA-SHOCK.

27. Transcutaneous pacing: rate 60–80/min, increase output from ~50 mA until electrical capture then add ~10 % (often 50–100 mA); confirm mechanical capture by pulse/ultrasound; analgesia/sedation (e.g. fentanyl 0.5–1 mcg/kg, midazolam 1–2 mg)

ACC/AHA 2018 bradycardia guideline.

🕒 CPR code timer & event log

2-min CPR cycle
Press Start when CPR begins

Epinephrine (q3–5 min)

No events logged.

Alerts at end of each 2-min cycle (switch compressor, rhythm/pulse check ≤ 10 s) and when epinephrine enters the 3–5 min window. Timer and log are saved on this device.

ACLS algorithms

- 1 Start CPR (30:2 until advanced airway), give O₂, attach monitor/defib
- 2 **Shock** biphasic 120–200 J (device) / monophasic 360 J △ VERIFY
- 3 CPR 2 min · IV/IO access
- 4 Rhythm check → **Shock** biphasic 120–200 J (device) / monophasic 360 J △ VERIFY
- 5 CPR 2 min · **Epinephrine 1 mg** IV/IO q3–5 min (after initial shocks fail) · consider advanced airway + waveform capnography △ VERIFY
- 6 Rhythm check → **Shock** biphasic 120–200 J (device) / monophasic 360 J △ VERIFY
- 7 CPR 2 min · **Amiodarone 300 mg** or **lidocaine 1–1.5 mg/kg** · treat reversible causes (Hs & Ts) △ VERIFY
- 8 Repeat: shock → CPR 2 min · epinephrine q3–5 min · **Amiodarone 150 mg** or **lidocaine 0.5–0.75 mg/kg** △ VERIFY
- 9 No organized rhythm → asystole/PEA arm. Organized rhythm with pulse/ETCO₂ jump → post-ROSC care

💉 Arrest & arrhythmia drugs

Epinephrine – cardiac arrest △ VERIFY

IV/IO q3–5 min

1 mg

= 10 mL of 0.1 mg/mL (1:10,000), or 1 mL of 1 mg/mL flushed. Non-shockable: ASAP. Shockable: after initial defibrillation attempts fail (AHA 2025).

Endotracheal (only if no IV/IO) △ VERIFY

2 – 2.5 mg in 5–10 mL NS

Amiodarone △ VERIFY

VF/pVT – 1st dose

300 mg IV/IO push

= 6 mL of 50 mg/mL

VF/pVT – 2nd dose

150 mg IV/IO

= 3 mL of 50 mg/mL

Stable wide-complex tachycardia

150 mg IV over 10 min (repeat PRN), then 1 mg/min × 6 h

Max ≈ 2.2 g / 24 h.

Lidocaine (alternative to amiodarone) △ VERIFY

VF/pVT – 1st dose

1 – 1.5 mg/kg

Then 0.5–0.75 mg/kg q5–10 min; max total 3 mg/kg.

Adenosine – SVT △ VERIFY

1st dose

6 mg rapid IV push + 20 mL NS flush

= 2 mL of 3 mg/mL. Antecubital or proximal vein.

2nd dose

12 mg rapid IV push + flush

= 4 mL of 3 mg/mL

Atropine – bradycardia

Symptomatic bradycardia △ VERIFY

1 mg IV q3–5 min, max 3 mg

= 1 mL of 1 mg/mL. Unlikely to work in infranodal block / transplanted heart.

Magnesium sulfate – torsades △ VERIFY

Torsades / hypomagnesemia

1 – 2 g IV

= 2–4 mL of 50% (500 mg/mL). Pulseless: dilute in 10 mL, over 1–2 min. With pulse: over 5–20 min.

Bradycardia – second line △ VERIFY

Dopamine

5 – 20 mcg/kg/min

400 mg in 250 mL = 1,600 mcg/mL.

Epinephrine infusion

2 – 10 mcg/min

1 mg in 250 mL = 4 mcg/mL → 30 – 150 mL/hr. Titrate.

Transcutaneous pacing

If atropine ineffective / unstable

Rate 60–80/min; increase mA to capture; confirm pulse; sedate/analgesia.

Procainamide △ VERIFY

Stable wide-complex tachycardia

20 – 50 mg/min IV

Until arrhythmia suppressed, hypotension, QRS widens > 50% or max 17 mg/kg. Maintenance 1–4 mg/min. Avoid in long QT / CHF.

Calcium △ VERIFY

Calcium chloride 10%

1 g = 10 mL slow IV/IO

Central/large vein. HyperK, hypoCa, Ca-blocker or Mg toxicity.

Calcium gluconate 10%

3 g = 30 mL IV over 5–10 min

≈ same elemental Ca as 1 g CaCl₂.

Sodium bicarbonate 8.4% △ VERIFY

HyperK / TCA / severe acidosis

1 mEq/kg

Not routine in arrest. 50 mL ampule = 50 mEq.

Dextrose – hypoglycemia ⚠️ VERIFY

Glucose < 70 mg/dL (3.9 mmol/L)

25 g IV = 50 mL D50

Or 250 mL D10. Thiamine 100 mg IV first if malnourished/alcohol use.

⚡ Defibrillation & cardioversion

Defibrillation – VF / pVT (unsynchronized)

Biphasic 120 – 200 J (per manufacturer; max if unknown)

Monophasic: 360 J

Subsequent shocks: same or higher energy

⚠️ VERIFY

Synchronized cardioversion – unstable tachyarrhythmia ⚠️ VERIFY

AF / flutter 200 J · narrow regular 100 J · monomorphic VT 100 J

Polymorphic VT / unable to sync → unsynchronized defibrillation

Escalate if no response; re-sync after each shock; sedate if feasible

Refractory VF after ≥ 3 shocks: consider vector-change or double sequential defibrillation ⚠️ VERIFY

CPR quality & post-ROSC

- Rate **100–120/min**; depth **5–6 cm**; full recoil; minimize pauses (< 10 s); switch compressor every 2 min
- Ratio **30:2**; with advanced airway: continuous compressions + **1 breath every 6 s (10/min)**
- Waveform capnography: confirm tube; ETCO₂ < 10 mmHg → improve CPR; abrupt rise → suspect ROSC
- Avoid hyperventilation

Post-ROSC

- SpO₂ 92–98%; PaCO₂ 35–45 mmHg
- SBP ≥ 90 mmHg, MAP ≥ 65 mmHg
- 12-lead ECG – STEMI → emergent cath; glucose; temperature control (avoid fever, target ≤ 37.5 °C)

Reversible causes – Hs & Ts

Point-of-care ultrasound during pulse checks only (do not prolong pauses beyond 10 s).

⚠️ Items pending owner review in this section (28)

1. Epinephrine via ETT (adult 2–2.5 mg; peds 0.1 mg/kg, capped here at 2.5 mg)

Evidence weak; IV/IO preferred. The 2.5 mg pediatric cap is borrowed from the adult ETT dose.

2. Atropine ampule strength assumed 1 mg/mL

Philippine stock may be 0.5 or 0.6 mg/mL – check the ampule before drawing up.

3. Pediatric atropine 0.02 mg/kg with NO minimum dose (max 0.5 mg)

PALS 2020 removed the 0.1 mg minimum; some local protocols still use it.

4. Pediatric lidocaine 1 mg/kg, capped at 100 mg

100 mg cap is from common references, not stated explicitly in PALS.

5. Pediatric magnesium sulfate 25–50 mg/kg (max 2 g) and stock concentration

Philippine MgSO₄ comes as 50%, 25% and 20% – confirm which is on hand.

6. Procainamide (adult 20–50 mg/min to 17 mg/kg; peds 15 mg/kg over 30–60 min)

Availability in the Philippines is limited; confirm product/concentration (100 mg/mL assumed).

7. Adult synchronized cardioversion energies (AHA 2025: AF/flutter 200 J; narrow-complex & monomorphic VT 100 J)

Device-specific; local protocols may still use 2020 values (50–100 J narrow, 120–200 J AF).

8. Double sequential / vector-change defibrillation for refractory VF

AHA 2025 “may be considered”; requires local policy and two defibrillators.

9. Pediatric vasoactive infusion ranges (epinephrine/norepinephrine 0.05–1 mcg/kg/min; epi for bradycardia 0.1–1)

Titration ranges vary by PICU protocol; standard 16 mcg/mL bag assumed.

10. Pediatric IM epinephrine max 0.3 mg

Set per owner spec. Several guidelines (WAO, ASCIA, RCUK) allow 0.5 mg for adolescents / ≥ 30–50 kg.

11. Pediatric calcium (CaCl₂ 10% 20 mg/kg max 2 g; gluconate 10% 60 mg/kg max 3 g)

Maximum single doses differ (some cap CaCl₂ at 1 g).

12. Pediatric doses are capped at an adult dose (engine-wide). Caps added where v1 had none: procainamide 1 g, bicarbonate 50 mEq, ketamine/propofol 200 mg, etomidate 30 mg, rocuronium 120 mg, succinylcholine 150 mg, fentanyl 100 mcg, phenobarbital 1.5 g, mannitol 100 g, fomepizole 1.5 g, methylene blue 200 mg, atropine (OP) 2 mg

For drugs dosed per kg in adults too, the cap is the adult dose at a ~100 kg reference weight – a policy choice for owner sign-off, not a guideline value.

13. ACLS arrhythmia drug doses reused from v1 in the Cardiovascular section (adenosine, amiodarone, lidocaine, magnesium, bradycardia infusions)

AHA 2020/2025 values; unchanged from v1 but now flagged wherever shown.

14. Dextrose doses reused from v1 in the Hypoglycemia card (adult 25 g D50; peds 0.5–1 g/kg D10/D25; neonate 0.2 g/kg)

Unflagged in v1; peds range differs from ISPAD (0.2–0.5 g/kg) – see en-hypo.

15. Adult defibrillation energy: biphasic 120–200 J per manufacturer (maximum if unknown), monophasic 360 J

AHA 2020/2025; device-specific.

16. Pediatric energies: defibrillation 2 J/kg → 4 J/kg → ≥ 4 J/kg (max 10 J/kg or adult dose); synchronized cardioversion 0.5–1 J/kg then 2 J/kg

PALS 2020/2025; v1 values reused, now flagged.

17. Epinephrine – cardiac arrest (1 mg / 0.01 mg/kg q3–5 min) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

18. Calcium chloride/gluconate in arrest (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

19. Sodium bicarbonate 1 mEq/kg in arrest (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

20. ACLS/PALS algorithm step doses and shock energies (epinephrine, amiodarone, lidocaine, atropine; 120–200 J / 2 → 4 J/kg)

Inline values from v1 / merged card, previously unflagged.

21. Defibrillation / cardioversion energies (adult and pediatric)

Inline values from v1 / merged card, previously unflagged.

22. Push-dose epinephrine (bridge only): 10 mcg/mL, give 0.5–2 mL (5–20 mcg) IV q2–5 min; peds 1 mcg/kg (max 10–20 mcg); start norepinephrine infusion early

Common ED practice; dilution errors reported – use pre-made syringes if available.

23. Undifferentiated shock: crystalloid boluses 250–500 mL (peds 10–20 mL/kg) with reassessment after each; norepinephrine 0.05–0.5 mcg/kg/min early if MAP < 65 despite fluids

SSC 2021; avoid fixed large volumes in cardiogenic shock.

24. HBO indications per UHMS (CO poisoning with LOC/neurologic signs/pregnancy/COHb > 25 %, decompression sickness, arterial gas embolism, necrotizing infections, crush, CRAO, etc.); PH chambers are few – contact facility early

UHMS 15th ed; PH chamber locations must be confirmed (Navy/AFP and a few private hospitals).

25. IO in conscious patient: 2 % lidocaine (preservative-free) 40 mg (2 mL) slowly over 2 min via IO, wait 60 s, flush, then 20 mg; peds 0.5 mg/kg (max 40 mg) Manufacturer (EZ-IO) guidance.

26. Post-arrest targets: SpO₂ 92–98 %, PaCO₂ 35–45 mmHg, MAP ≥ 65 (consider higher), glucose 140–180 mg/dL, prevent fever (≤ 37.5 °C) for ≥ 72 h (TTM 32–36 °C optional per AHA 2023), prognosticate ≥ 72 h after normothermia

AHA 2023 / ILCOR 2023 / ERC-ESICM 2025.

27. Monitoring targets: MAP ≥ 65 mmHg, capillary refill ≤ 3 s, urine output ≥ 0.5 mL/kg/h, PLR stroke-volume rise ≥ 10–15 %, ScvO₂ ≥ 70 %

SSC 2021 / ANDROMEDA-SHOCK.

28. Transcutaneous pacing: rate 60–80/min, increase output from ~50 mA until electrical capture then add ~10 % (often 50–100 mA); confirm mechanical capture by pulse/ultrasound; analgesia/sedation (e.g. fentanyl 0.5–1 mcg/kg, midazolam 1–2 mg)

ACC/AHA 2018 bradycardia guideline.

✓ RSI checklist (SOAP-ME)

- ☐ **S Suction** – Yankauer on, tested (2nd suction if soiled airway)
- ☐ **O Oxygen** – preoxygenate ≥ 3 min (NRB 15 L/min + nasal cannula 15 L/min, or BVM with PEEP / NIV); apneic oxygenation via NC
- ☐ **A Airway equipment** – laryngoscope/VL + spare blade, ETT (chosen size and ½ size smaller), stylet, bougie, 10 mL syringe, SGA, BVM, OPA/NPA, cric kit
- ☐ **P Positioning** – ear-to-sternal notch / ramp, head-up 20–30° (infants: shoulder roll)
- ☐ **M Monitoring & Meds** – SpO₂, ECG, BP cycling q2–3 min; drugs drawn up, labeled, doses double-checked
- ☐ **E EtCO₂** – waveform capnography connected and ready
- ☐ + IV/IO × 2 working; fluids running
- ☐ + Hemodynamics optimized (shock index, push-dose pressor / infusion ready)
- ☐ + Plan A/B/C/D verbalized; roles assigned; difficulty predictors reviewed
- ☐ + Post-intubation sedation/analgesia ready; OG tube; CXR ordered

RSI drug doses

Induction

Ketamine △ VERIFY

Induction

1 – 2 mg/kg

Shock: reduce to 0.5–1 mg/kg.

Etomidate △ VERIFY

Induction

0.3 mg/kg

Hemodynamically stable-ish. Single dose adrenal suppression – usually acceptable.

Propofol △ VERIFY

Induction

1 – 2 mg/kg

Hypotension – 0.5–1 mg/kg in shock/elderly, or choose ketamine/etomidate.

Paralysis

Rocuronium △ VERIFY

RSI

1.2 mg/kg

Onset ~45–60 s. Reversal: sugammadex 16 mg/kg.

Succinylcholine △ VERIFY

RSI

1.5 mg/kg

Avoid: hyperK, MH history, burns/crush/denervation > 48–72 h, myopathy.

Adjuncts

Fentanyl (pre-treatment / analgesia) △ VERIFY

IV over 30–60 s

1 – 3 mcg/kg

Omit or reduce in shock.

Push-dose pressors △ VERIFY

Phenylephrine

50 – 200 mcg IV q2–5 min

100 mcg/mL (10 mg in 100 mL NS): 0.5 – 2 mL.

Epinephrine

10 – 20 mcg IV q2–5 min

10 mcg/mL (1 mL of 0.1 mg/mL + 9 mL NS): 1 – 2 mL.

ETT size & depth

Adult ETT (ID)

Female 7.0 – 7.5 • Male 7.5 – 8.0 mm

Depth at lips ≈ 21 cm (F) / 23 cm (M). Suction catheter 12–14 Fr.

SGA / blade

LMA 4 (50–70 kg) • 5 (70–100 kg)

Mac 3–4 / Miller 2–3. Confirm: waveform EtCO₂ + bilateral breath sounds + CXR.

Post-intubation sedation

Infusions △ VERIFY

Propofol

5 – 50 mcg/kg/min

Hypotension; avoid high doses > 48 h (PRIS).

Fentanyl

0.5 – 2 mcg/kg/hr

Midazolam

20 – 100 mcg/kg/hr

= 0.02–0.1 mg/kg/hr.

Ketamine bolus (if hypotensive)

0.5 – 1 mg/kg

Failed airway plan (A → D)

- 1 Plan A – Laryngoscopy/VL:** max 3 attempts (+1 by most experienced). Optimize position, bougie, external laryngeal manipulation. Oxygenate between attempts.
- 2 Plan B – 2nd-generation SGA:** max 3 attempts; if oxygenating, stop and think (wake, intubate via SGA, or proceed).
- 3 Plan C – Facemask:** 2-person BVM + OPA/NPA, full paralysis. If oxygenating → wake / plan.
- 4 Plan D – CICO:** declare “Can’t intubate, can’t oxygenate”. Adult: scalpel–bougie–tube cricothyroidotomy (size 6.0 cuffed). Young children: needle cricothyroidotomy / surgical airway per local protocol △ VERIFY.

△ Items pending owner review in this section (12)

1. Pediatric atropine 0.02 mg/kg with NO minimum dose (max 0.5 mg)

PALS 2020 removed the 0.1 mg minimum; some local protocols still use it.

2. Push-dose pressors (phenylephrine 100 mcg/mL; epinephrine 10 mcg/mL)

Dilution recipes vary; dilution errors are common – use local standard preparation.

3. Pediatric RSI propofol 1–2 mg/kg

Elective anesthesia uses 2.5–3.5 mg/kg; emergency dosing depends on hemodynamics.

4. Post-intubation sedation/analgesia infusion ranges (propofol, fentanyl, midazolam)

Titrate to local ICU sedation protocol and targets (RASS).

5. Pediatric CICO technique (needle vs surgical cricothyroidotomy) and age cut-off

Age threshold (~8–12 y) and kit vary by institution / ENT availability.

6. Pediatric doses are capped at an adult dose (engine-wide). Caps added where v1 had none: procainamide 1 g, bicarbonate 50 mEq, ketamine/propofol 200 mg, etomidate 30 mg, rocuronium 120 mg, succinylcholine 150 mg, fentanyl 100 mcg, phenobarbital 1.5 g, mannitol 100 g, fomepizole 1.5 g, methylene blue 200 mg, atropine (OP) 2 mg

For drugs dosed per kg in adults too, the cap is the adult dose at a ~100 kg reference weight – a policy choice for owner sign-off, not a guideline value.

7. Etomidate 0.3 mg/kg induction (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

8. Ketamine 1–2 mg/kg induction (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

9. Propofol 1–2 mg/kg induction (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

10. Rocuronium 1.2 mg/kg RSI (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

11. Succinylcholine 1.5 mg/kg (peds 2 mg/kg) RSI (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

12. Fentanyl 1–3 mcg/kg pre-treatment/analgesia (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

S4 • Resuscitative Procedures – Access, monitoring & post-arrest Built

Vascular access

Route	Key points
Peripheral IV	Two large-bore (16–18 gauge) for resuscitation; ultrasound-guided long catheters for difficult access
Intraosseous	Fastest in arrest/shock: proximal tibia (2 cm below tibial tuberosity, medial), proximal humerus (adults – faster flow, arm adducted, hand on abdomen), distal femur in infants; all drugs/blood can be given; pressure bag; avoid fractured bone, prosthesis, infected site; remove within 24 h △ VERIFY
Central venous	Ultrasound-guided IJ or femoral (femoral fastest in arrest – compressions not interrupted); confirm venous placement (ultrasound/manometry) before dilating; CXR for IJ/subclavian
Arterial line	Radial (ultrasound); for vasopressors, frequent ABGs, labile BP
Children	IO after 2 failed IV attempts or 90 s in arrest; umbilical venous catheter in neonates

Lidocaine 2 % (preservative-free) – IO infusion pain △ VERIFY

IO

40 mg (2 mL) over 2 min, wait 60 s, flush; then 20 mg

Hemodynamic monitoring

- MAP ≥ 65 mmHg for most shock (higher if chronic hypertension; permissive lower targets in uncontrolled hemorrhage until control, except head injury)
- Fluid responsiveness: passive leg raise with real-time stroke volume/VTI (≥ 10 –15 % rise), pulse pressure variation in ventilated patients in sinus rhythm; IVC size/collapse is a weak guide

- Perfusion targets: capillary refill ≤ 3 s (ANDROMEDA-SHOCK), lactate clearance, urine output ≥ 0.5 mL/kg/h, mentation △ VERIFY
- ScvO₂ < 70 % suggests inadequate delivery (anemia, low output); very high ScvO₂ with high lactate – mitochondrial/shunt (sepsis, cyanide)
- Stop fluids when no longer responsive or signs of congestion (B-lines, rising JVP, hypoxemia)

Pericardiocentesis

- Indication: tamponade with hemodynamic compromise (non-traumatic, or traumatic as a bridge to surgery – chest trauma)
- Ultrasound to find the largest, closest pocket: parasternal (5th ICS, lateral to sternum – avoid internal mammary 1 cm lateral) or apical or subxiphoid
- Sterile prep, local anesthesia; 18-gauge needle/catheter; aspirate as you advance under ultrasound; agitated saline confirms position
- Seldinger pigtail catheter left for drainage; removing even a small volume may restore BP
- Complications: ventricular puncture, coronary laceration, pneumothorax, arrhythmia; ECG monitoring throughout

Post-arrest care

- Airway secured, ETCO₂ waveform; ventilate to normocapnia; titrate FiO₂ to SpO₂ 92–98 % (avoid hyperoxia and hypoxia) △ VERIFY
- 12-lead ECG: STEMI → emergent coronary angiography; no STEMI but likely cardiac cause → individualised
- Hemodynamics: MAP ≥ 65 , avoid hypotension; norepinephrine first-line; echo
- Temperature: actively prevent fever (≤ 37.5 °C) for at least 72 h in comatose patients; avoid prophylactic prehospital cold fluids
- Seizures: EEG if comatose; treat clinical seizures; glucose control; sedation/analgesia
- Find and treat cause (H's and T's – ACLS card)
- Prognostication: multimodal, not before 72 h after normothermia/sedation clearance; avoid premature withdrawal

Pacing & implanted devices

- Unstable bradycardia unresponsive to atropine: transcutaneous pacing or chronotropic infusion (epinephrine/dopamine – arrest drugs)
- Pads anterior–posterior; set rate, increase output until capture; confirm mechanical capture △ VERIFY
- Transvenous pacing (right IJ/left subclavian) as soon as possible if ongoing pacing needed
- Magnet** over a pacemaker → asynchronous pacing (useful for pacemaker-mediated tachycardia); over an ICD → suspends shocks (inappropriate shocks), pacing unaffected
- Pacemaker malfunction: failure to pace, failure to capture, under/over-sensing – CXR (lead position), electrolytes, device interrogation
- Defibrillating patients with devices: pads ≥ 8 cm from the generator; interrogate afterwards

Ethics of resuscitation

- Respect valid patient wishes (advance directives, DNR orders); when unknown, resuscitate and clarify
- BLS termination-of-resuscitation rule (out-of-hospital): arrest not witnessed by EMS, no shock delivered, no ROSC before transport → consider stopping (validated in adults)
- Factors favouring stopping: prolonged downtime without CPR, persistent asystole, ETCO₂ < 10 mmHg after 20 min of high-quality ALS (one factor, never alone)

- Factors favouring continuing: hypothermia, drowning in cold water, toxicologic arrest, pregnancy, ECPR candidate
- Offer family presence; designate a staff member to support them
- PH: no specific advance-directive law – follow hospital DNR policy, document discussion with patient/family; death notification

Resuscitative procedures (extended) sources

Perman et al., AHA 2023 focused update on adult advanced cardiovascular life support · Nolan et al., ERC-ESICM post-resuscitation

care guidelines 2021/2025 · ILCOR temperature control CoSTR 2023 · Morrison et al., BLS termination-of-resuscitation rule (NEJM 2006) · Gorski et al., INS Infusion Therapy Standards of Practice 2021 · Adler et al., ESC pericardial diseases 2015 · Kusumoto et al., ACC/AHA/HRS bradycardia guideline 2018 · Hernández et al., ANDROMEDA-SHOCK (JAMA 2019) · Monnet et al., passive leg raising (Crit Care 2015). Tintinalli 9e Section 4 used only as a topic map.

⚠ Items pending owner review in this section (4)

- 1. IO in conscious patient: 2 % lidocaine (preservative-free) 40 mg (2 mL) slowly over 2 min via IO, wait 60 s, flush, then 20 mg; peds 0.5 mg/kg (max 40 mg)**
Manufacturer (EZ-IO) guidance.
- 2. Post-arrest targets: SpO₂ 92–98 %, PaCO₂ 35–45 mmHg, MAP ≥ 65 (consider higher), glucose 140–180 mg/dL, prevent fever (≤ 37.5 °C) for ≥ 72 h (TTM 32–36 °C optional per AHA 2023), prognosticate ≥ 72 h after normothermia**
AHA 2023 / ILCOR 2023 / ERC-ESICM 2025.
- 3. Monitoring targets: MAP ≥ 65 mmHg, capillary refill ≤ 3 s, urine output ≥ 0.5 mL/kg/h, PLR stroke-volume rise ≥ 10–15 %, ScvO₂ ≥ 70 %**
SSC 2021 / ANDROMEDA-SHOCK.
- 4. Transcutaneous pacing: rate 60–80/min, increase output from ~50 mA until electrical capture then add ~10 % (often 50–100 mA); confirm mechanical capture by pulse/ultrasound; analgesia/sedation (e.g. fentanyl 0.5–1 mcg/kg, midazolam 1–2 mg)**
ACC/AHA 2018 bradycardia guideline.

S5 • Analgesia, Anesthesia & Procedural Sedation Built

Acute pain management

Assess pain with a score (0–10, Wong–Baker faces, FLACC in preverbal children); treat early and re-assess within 30–60 min. Combine agents with different mechanisms; use non-pharmacological measures (splint, ice, elevation, distraction).

Acetaminophen (paracetamol) △ VERIFY

PO / IV q6h

1 g

Max 4 g/day; ≤ 3 g/day if liver disease, chronic alcohol, malnutrition, < 50 kg

Ibuprofen △ VERIFY

PO

400 mg q6–8h (max 1.2 g/day OTC, 2.4 g/day Rx)

Ketorolac △ VERIFY

IV/IM

10 – 15 mg single dose (no extra analgesia above 15 mg)

≤ 5 days total; avoid > 65 y high-risk, AKI, bleeding, dengue.

Morphine IV △ VERIFY

IV titration

2.5 – 5 mg q5–10 min to effect (elderly 1–2 mg)

Fentanyl △ VERIFY

IV

0.5 – 1 mcg/kg

Repeat q5–10 min; 50 mcg/mL.

Tramadol (commonly used in PH) △ VERIFY

PO / IV

50 – 100 mg q6h (max 400 mg/day; ≥ 75 y max 300 mg)

Avoid: seizures, SSRIs/MAOI, CYP2D6 ultra-rapid metabolisers.

Ketamine – sub-dissociative analgesia △ VERIFY

IV over 10–15 min

0.1 – 0.3 mg/kg

- Renal colic: NSAID first-line (better than opioids), add IV acetaminophen; avoid if AKI
- Dengue (common in PH): **avoid NSAIDs and IM injections** – acetaminophen only
- Elderly: start low (morphine 1–2 mg), avoid tramadol/codeine combinations with confusion risk; constipation prophylaxis △ VERIFY
- Codeine: avoid in children and breastfeeding mothers (ultrarapid metabolisers)

Procedural sedation

Before

- ASA class I–II ideal for ED; III–IV and difficult airways → senior/anesthesia
- Recent food intake is not a reason to delay ED sedation (ACEP 2014) – weigh urgency vs aspiration risk
- Consent, two clinicians (one dedicated to sedation/airway), monitoring incl. **capnography**, suction, BVM, reversal agents, resuscitation drugs

Ketamine – dissociative sedation △ VERIFY

IV over 30–60 s

1 – 2 mg/kg

Repeat 0.5 mg/kg q5–10 min. IM 4–5 mg/kg.

Propofol – sedation △ VERIFY

IV load

0.5 – 1 mg/kg

Then 0.25–0.5 mg/kg q1–3 min. Hypotension, apnea.

Midazolam (+ fentanyl) – moderate sedation △ VERIFY

IV

0.02 – 0.05 mg/kg

Max ~ 2.5 mg per dose in adults; elderly 0.5–1 mg.

Etomidate – brief sedation (e.g. cardioversion) △ VERIFY

IV

0.1 – 0.15 mg/kg

Myoclonus common.

- Reversal: naloxone (dosing); flumazenil only for iatrogenic benzodiazepine sedation in benzodiazepine-naïve patients
- Ketamine cautions: age < 3 months, active psychosis; emergence reactions – calm environment, small midazolam dose if severe
- Discharge: back to baseline mental status, tolerating fluids, responsible adult, written instructions

Local & regional anesthesia

Enter weight to calculate maximum local anesthetic volumes.

- Lidocaine with epinephrine is safe in fingers, toes, nose and ears in healthy patients
- Less painful injection: warm, buffer (lidocaine : 8.4% bicarbonate = 10 : 1), small needle, inject slowly through wound edges
- **Digital block**: ring or single-injection volar (transthecal/subcutaneous) 2–3 mL per side △ VERIFY
- **Hematoma block** (distal radius): aspirate hematoma, inject 10 mL 1% lidocaine △ VERIFY
- **Fascia iliaca block** for hip/femur fracture: large-volume dilute LA under ultrasound △ VERIFY
- **LAST**: perioral numbness, tinnitus, seizures, arrhythmia/arrest → stop injection, airway, benzodiazepine, 20% lipid emulsion (antidotes – lipid), prolonged CPR

Chronic pain & opioid use disorder

- Look for a new acute cause behind a chronic pain flare (red flags: fever, weight loss, neuro deficit, cancer, IV drug use)
- Prefer non-opioid multimodal therapy; if opioids needed, shortest course (≤ 3 days), lowest dose, no long-acting formulations from the ED; check for other prescribers
- Opioid withdrawal: COWS score; buprenorphine initiation from the ED where available (limited in PH – refer to DOH-accredited treatment and rehabilitation centres)
- Offer naloxone and overdose education to people who use opioids
- Sick cell crisis, cancer pain: treat promptly and adequately – do not label as drug-seeking

Analgesia & sedation sources

ACEP Clinical policy: procedural sedation and analgesia in the ED (2014) • Green et al., Clinical practice guideline for emergency department ketamine dissociative sedation (Ann Emerg Med 2011) • ASA practice guidelines for moderate procedural sedation 2018 • ACEP policy on opioid prescribing • ASRA checklist for local anesthetic systemic toxicity 2020 • AAGBI/NICE hip fracture analgesia • WHO cancer pain guidelines 2018 • Philippine FDA product information. Tintinalli 9e Section 5 used only as a topic map.

⚠ Items pending owner review in this section (7)

1. Acetaminophen: adult 1 g PO/IV q6h, max 4 g/day (≤ 3 g/day if liver disease, alcohol use, malnutrition, < 50 kg use 15 mg/kg); peds 15 mg/kg q6h, max 75 mg/kg/day (max 1 g/dose)

Common label/guideline values; IV dosing < 50 kg weight-based.

2. NSAIDs: ibuprofen 400 mg PO q6–8h (peds 10 mg/kg, max 400 mg); ketorolac 10–15 mg IV/IM single dose (analgesic ceiling; peds 0.5 mg/kg, max 15 mg) – avoid in AKI, GI bleed, late pregnancy, dengue

ACEP/ED studies show ketorolac 10 mg ≈ 30 mg; dengue warning is PH-specific practice.

3. Opioids: morphine 0.05–0.1 mg/kg IV (adult 2.5–5 mg increments); fentanyl 1 mcg/kg IV, intranasal 1.5 mcg/kg (max 100 mcg); tramadol 50–100 mg PO q6h (max 400 mg/day; not < 12 y)

Titrate to effect; elderly/renal: lower doses. Tramadol: seizures, serotonin syndrome, CYP2D6 variability.

4. Sub-dissociative ketamine analgesia 0.1–0.3 mg/kg IV over 10–15 min (or 0.15 mg/kg/hr infusion); intranasal 0.5–1 mg/kg

ACEP/ED trials; dysphoria with fast push.

5. Procedural sedation: ketamine 1–2 mg/kg IV (peds 1.5–2) or 4–5 mg/kg IM; propofol 0.5–1 mg/kg then 0.25–0.5 mg/kg q1–3 min; ketofol 1:1 0.5 mg/kg each; etomidate 0.1–0.15 mg/kg; midazolam 0.02–0.05 mg/kg + fentanyl 0.5–1 mcg/kg

ACEP 2014 policy and ketamine guideline (Green 2011); fasting need not delay ED sedation.

6. Local anesthetic maximum doses: lidocaine 4.5 mg/kg plain (max 300 mg), 7 mg/kg with epinephrine (max 500 mg); bupivacaine 2.5 mg/kg plain (max 175 mg), 3 mg/kg with epinephrine (max 225 mg)

Classic textbook maxima; ASRA emphasises site and patient factors. 1% = 10 mg/mL.

7. Fascia iliaca block for hip fracture: ~ 30–40 mL of dilute long-acting LA (e.g. bupivacaine 0.25% or levobupivacaine) within weight maxima, ultrasound-guided

AAGBI/NICE hip fracture; volume must respect mg/kg limits.

Wound evaluation & preparation

- History: mechanism (crush, bite, glass, organic matter), time since injury, tetanus status, diabetes/immunosuppression
 - Examine with good light and bloodless field: distal neurovascular function **before** anesthesia; tendon function against resistance; look into the base through full range of motion
 - Imaging: X-ray for glass (most ≥ 2 mm visible) and metal; ultrasound for wood/plastic
 - Closure timing: most wounds can be closed up to ~12–18 h (face/scalp up to 24 h or more); contaminated/late wounds → delayed primary closure at 3–5 days
- 1 Anesthetise first (max dose calculator); clip – do not shave – hair; never shave eyebrows
 - 2 Irrigate with clean tap water or saline (△ VERIFY); remove debris and devitalised tissue
 - 3 Explore for foreign bodies and deep-structure injury; extend the wound if needed

Wound closure

Site	Material	Removal △ VERIFY
Face	6-0 non-absorbable (or 5-0/6-0 fast-absorbing); adhesive for low tension	5 days
Lip	Align vermilion border first; 6-0 skin, 5-0 absorbable mucosa/muscle	5 days
Scalp	Staples or 4-0; hair apposition with adhesive	7–10 days
Trunk, arms	4-0 (deep 4-0 absorbable if gaping)	7–10 days
Legs	4-0	10–14 days
Hands, feet, over joints	4-0 / 5-0; splint joints	10–14 days

- Tissue adhesive: clean, linear, low-tension wounds; not on mucosa, joints, hands with movement
- Children: absorbable sutures avoid painful removal
- Do not close: infected wounds, high-risk bites on hands, puncture wounds, very contaminated wounds > 24 h

Special wounds – hands, face, legs

- Clenched-fist ("fight bite")** over MCP joint: assume joint penetration – explore, X-ray, no primary closure, amoxicillin-clavulanate, hand surgery if joint/tendon involved
- High-pressure injection injury** (paint, grease): tiny wound, catastrophic damage – emergency surgical referral, no digital block
- Partial tendon lacerations > 50%, flexor tendon injuries, digital nerve injuries → hand surgeon (repair within days)

- Nail bed: repair lacerations; drain subungual hematoma by trephination
- Ear/nose cartilage: cover cartilage, pressure dressing for auricular hematoma
- Lower leg pretibial flaps in elderly: Steri-Strips, avoid tension sutures

Bites & puncture wounds

- Irrigate thoroughly; debride devitalised tissue; facial dog bites may be closed primarily; hand bites and cat punctures left open
- Prophylactic antibiotics for: cat bites, hand/foot/genital/face bites, deep punctures, crush injury, immunocompromise/asplenia, wounds needing closure, human bites
- Plantar puncture through a shoe: Pseudomonas risk if infected – treat infection (adults ciprofloxacin), no routine prophylaxis
- Report animal bites; assess rabies risk on every mammal bite (rabies PEP)

Amoxicillin-clavulanate – bite prophylaxis △ VERIFY

PO

875/125 mg BID × 3–5 days

Rabies post-exposure prophylaxis

Select the exposure category.

- 1 **Wash every bite/scratch immediately for 15 min with soap and running water**, then povidone-iodine or alcohol – the single most effective step
- 2 Vaccine (PCECV/PVRV): **intradermal 0.1 mL at 2 sites on days 0, 3 and 7** (PH Animal Bite Treatment Centre regimen), or IM 1 vial on days 0, 3, 7 and 14–28 (△ VERIFY)
- 3 Category III (or II if immunocompromised): RIG into and around all wounds on day 0 (up to day 7) (△ VERIFY)
- 4 Do not delay PEP for animal observation or lab confirmation; PEP is never contraindicated (pregnancy, infants included)
- 5 Delay suturing large bites if possible; if closure needed, infiltrate RIG first

Refer to the nearest DOH-accredited Animal Bite Treatment Centre (ABTC) for the series; rabies is nearly 100% fatal once symptomatic (hydrophobia, aerophobia, agitation) – palliative care and isolation.

Tetanus prophylaxis

Doses of tetanus toxoid	Clean minor wound	All other wounds
Unknown or < 3	Td/Tdap (DTaP < 7 y)	Td/Tdap + TIG
≥ 3 , last < 5 y	Nothing	Nothing
≥ 3 , last 5–10 y	Nothing	Td/Tdap
≥ 3 , last > 10 y	Td/Tdap	Td/Tdap

△ VERIFY

- "Other" wounds: contaminated (dirt, feces, saliva), punctures, avulsions, crush, burns, frostbite, missiles
- Pregnancy: Tdap is safe and recommended (also protects the newborn against pertussis)
- Clinical tetanus: see Infectious Diseases (metronidazole, TIG, magnesium, airway)

Tetanus immunoglobulin (human) **VERIFY**

IM (different site from toxoid)

250 IU (500 IU if > 24 h old, heavy contamination or burns – some protocols)

Soft-tissue foreign bodies

- Image first: X-ray for glass/metal/stone; ultrasound (hyperechoic with shadow) for wood/thorns
- Set a time limit (~ 20–30 min); remove only if accessible – deep or near nerves/vessels/joints → refer
- Fish hook: advance-and-cut, string-yank or needle-cover techniques

- Organic material (wood, thorns, sea urchin spines) causes infection/granulomas – remove completely; consider antibiotics
- Tell patients a retained foreign body is possible; return if pain, swelling, discharge

Wound sources

WHO Rabies vaccines position paper (WER 2018) and WHO Expert Consultation on Rabies, 3rd report (TRS 1012, 2018) · DOH National Rabies Prevention and Control Program Manual of Procedures · ACIP recommendations for tetanus/diphtheria/pertussis vaccines (2018–2020) · WHO tetanus vaccines position paper 2017 · Stevens et al., IDSA SSTI guideline 2014 · Fernandez et al., Water for wound cleansing (Cochrane) · ACEP/AAFP laceration repair reviews. Tintinalli 9e Section 6 used only as a topic map.

Items pending owner review in this section (7)

1. Local anesthetic maximum doses: lidocaine 4.5 mg/kg plain (max 300 mg), 7 mg/kg with epinephrine (max 500 mg); bupivacaine 2.5 mg/kg plain (max 175 mg), 3 mg/kg with epinephrine (max 225 mg)

Classic textbook maxima; ASRA emphasises site and patient factors. 1% = 10 mg/mL.

2. Rabies PEP (WHO 2018, DOH): Category II → vaccine; Category III → vaccine + rabies immunoglobulin. Vaccine: intradermal 2-site on days 0, 3, 7 (PH ABTC regimen) or IM 1 dose on days 0, 3, 7 and 14–28; previously vaccinated → 2 doses (days 0, 3), no RIG

Schedules per WHO 2018 / DOH NRPCP; confirm current DOH memorandum and product (PCECV/PVRV) ID volume (0.1 mL per site).

3. Rabies immunoglobulin: HRIG 20 IU/kg or ERIG 40 IU/kg, infiltrated into and around all wounds as far as anatomically feasible (no need to inject the remainder IM); given up to day 7 after first vaccine dose

WHO 2018; ERIG skin testing not required by WHO but some PH protocols still perform it.

4. Tetanus prophylaxis: < 3 or unknown doses → Td/Tdap (DTaP if < 7 y) for all wounds + TIG 250 IU IM for dirty/major wounds; ≥ 3 doses → booster if > 10 y (clean minor) or > 5 y (other wounds)

ACIP; PH: TIG supply limited – equine antitoxin (ATS 1,500 IU after sensitivity test) still used in some facilities.

5. Irrigation: potable tap water as effective as sterile saline; ~ 50–100 mL per cm of wound, 5–8 psi (e.g. 30–60 mL syringe + 18–19 G cannula)

Cochrane/ED trials; volumes are expert consensus.

6. Bite prophylaxis (dog/cat/human, high-risk wounds): amoxicillin–clavulanate 875/125 mg PO BID × 3–5 days (peds 25 mg/kg/dose amoxicillin component BID, max 875 mg); penicillin allergy: doxycycline or TMP-SMX + metronidazole/clindamycin

IDSA 2014; peds dose formulations vary (7:1 vs 4:1).

7. Suture removal times: face 5 d, scalp 7–10 d, trunk/arms 7–10 d, legs 10–14 d, over joints/hands/feet 10–14 d; children slightly earlier

Consensus ranges; absorbable sutures avoid removal visits in children.

 ACS / STEMI timeline

Enter first medical contact (and onset) to get live deadlines.

- 1 ECG within 10 min;** repeat q15–30 min if non-diagnostic and still in pain. STEMI: new ST elevation at J point in ≥ 2 contiguous leads – ≥ 1 mm (V2–V3; ≥ 2 mm men ≥ 40 y, ≥ 2.5 mm men < 40 y, ≥ 1.5 mm women) **△ VERIFY**
- 2 STEMI equivalents / occlusion MI:** posterior MI (ST depression V1–V3 → record V7–V9, ≥ 0.5 mm), new LBBB or paced rhythm with modified Sgarbossa (concordant STE ≥ 1 mm, concordant STD V1–V3, or discordant STE/S ≥ 25%), de Winter T waves, STE aVR + diffuse STD with ongoing ischemia (left main/multivessel), hyperacute T waves – discuss with the cath lab
- 3 Reperfusion:** primary PCI if FMC-to-device ≤ 120 min achievable; otherwise **fibrinolysis within 30 min** (onset < 12 h, no contraindication) then transfer – angiography 2–24 h (pharmaco-invasive) or **rescue PCI** if ST resolution < 50% at 60–90 min, ongoing pain, or instability **△ VERIFY**
- 4 Cardiogenic shock or arrest with ROSC + STEMI** → immediate angiography regardless of delay; culprit-only PCI in shock
- 5 NSTEMI-ACS:** high-sensitivity troponin 0/1-h or 0/2-h algorithm; immediate invasive (< 2 h) if unstable, refractory pain, arrhythmia, mechanical complication, acute HF; early (< 24 h) if NSTEMI confirmed or GRACE > 140

Antithrombotics & anti-ischemic**Aspirin** **△ VERIFY**

Chewed (non-enteric) loading

162 – 325 mg

Then 75–100 mg daily. Rectal 300–600 mg if unable to swallow.

P2Y12 inhibitor (load) **△ VERIFY**

Ticagrelor

180 mg PO

or Prasugrel (PCI)

60 mg PO – avoid prior stroke/TIA

or Clopidogrel

600 mg (PCI) · 300 mg with fibrinolysis (75 mg, no load, if > 75 y)

Unfractionated heparin **△ VERIFY**

Primary PCI bolus

70 – 100 units/kg

Cath-lab ACT-guided.

With fibrinolysis

60 U/kg (max 4,000 U) then 12 U/kg/hr (max 1,000 U/hr)

Enoxaparin **△ VERIFY**

With fibrinolysis (age-adjusted)

< 75 y: 30 mg IV + 1 mg/kg SC q12h

≥ 75 y: no IV bolus, 0.75 mg/kg SC q12h

CrCl < 30: 1 mg/kg SC q24h

NSTEMI-ACS

1 mg/kg SC q12h

Nitroglycerin (ischemic pain) **△ VERIFY**

SL

0.4 mg q5 min × 3

IV infusion

5 – 10 mcg/min, titrate by 10 mcg/min q3–5 min

Avoid: SBP < 90, RV infarct (inferior STEMI – check V4R), PDE-5 inhibitor (sildenafil 24 h, tadalafil 48 h).

- O₂ only if SpO₂ < 90% · morphine may delay oral P2Y12 absorption – use sparingly · no NSAIDs · β-blocker orally in first 24 h only if no HF/shock signs
- Inferior STEMI: right-sided leads (V4R); RV infarct → avoid nitrates, fluids for preload

 STEMI fibrinolysis

Tenecteplase (single IV bolus over 5–10 s) **△ VERIFY**

Weight band

< 60 kg 30 mg · 60–69 35 · 70–79 40 · 80–89 45 · ≥ 90 kg 50 mg

Age ≥ 75 y: half dose

Alteplase – accelerated (90 min) **△ VERIFY**

Regimen

15 mg bolus; 0.75 mg/kg over 30 min (max 50); 0.5 mg/kg over 60 min (max 35)

Streptokinase **△ VERIFY**

IV infusion

1.5 million units over 30–60 min

Hypotension → slow/pause infusion, fluids. Never re-use (antibodies) – prior exposure → fibrin-specific agent.

Adjuncts: aspirin + clopidogrel (300 mg; 75 mg if > 75 y) + UFH or enoxaparin (Antithrombotics above). Transfer to a PCI centre for all lysed patients.

Absolute contraindications

- Any prior intracranial hemorrhage
- Ischemic stroke < 3 months (except within 4.5 h)
- Known intracranial neoplasm / AVM
- Suspected aortic dissection
- Active bleeding (excl. menses) or bleeding diathesis
- Significant head/face trauma or intracranial/spinal surgery < 2–3 months
- Severe uncontrolled HTN unresponsive to treatment (streptokinase: prior use < 6 months)

Relative

- SBP > 180 or DBP > 110
- Ischemic stroke > 3 months, dementia
- Traumatic/prolonged CPR (> 10 min), major surgery < 3 weeks
- Internal bleeding 2–4 weeks; non-compressible punctures
- Pregnancy; active peptic ulcer; oral anticoagulant

△ VERIFY

 HEART score

– Enter age in the patient bar (age points: < 45 = 0, 45–64 = 1, ≥ 65 = 2)

 Hypertensive emergencies

Emergency = severe BP (often > 180/120) **with** acute target-organ damage (encephalopathy, stroke/ICH, ACS, pulmonary edema,

dissection, AKI/TMA, eclampsia, retinal hemorrhage/papilledema). Without damage = severe hypertension/urgency: oral agents, recheck, outpatient follow-up – no IV boluses.

Condition	Target ⚠️ VERIFY	Preferred
Most emergencies	MAP ↓ ≤ 25% in 1st h → 160/100–110 over 2–6 h → normal over 24–48 h	Labetalol, nicardipine
Aortic dissection	SBP ≤ 120, HR ≤ 60 within 20 min	Esmolol/labetalol first, then nicardipine/nitroprusside
Pulmonary edema / ACS	Symptom relief; ~25% ↓	Nitroglycerin (+ NIV); avoid hydralazine
Ischemic stroke	Lysis/thrombectomy: < 185/110 before, < 180/105 × 24 h · no reperfusion: treat if > 220/120 (↓ 15%)	Labetalol, nicardipine
Intracerebral hemorrhage	SBP 140 (avoid < 130, avoid big swings)	Nicardipine, labetalol
Preeclampsia / eclampsia	≥ 160/110 treated within 30–60 min; < 140/90 (see OB)	OB: labetalol, hydralazine, nifedipine + Mg
Sympathomimetic (cocaine, amphetamine) / pheochromocytoma	SBP < 140 (pheo)	Benzodiazepines first; phentolamine; α before β (no unopposed β-blocker)

Labetalol ⚠️ VERIFY

IV bolus

20 mg over 2 min, then 20 – 80 mg q10 min (max 300 mg)

or infusion

0.5 – 2 mg/min

Nicardipine ⚠️ VERIFY

Infusion

5 mg/hr, increase 2.5 mg/hr q5–15 min (max 15 mg/hr)

25 mg in 250 mL = 0.1 mg/mL. Reflex tachycardia; avoid in severe aortic stenosis.

Esmolol ⚠️ VERIFY

Load (over 1 min)

500 mcg/kg

Infusion

50 – 300 mcg/kg/min

10 mg/mL. Titrate q4 min to HR/BP.

Nitroglycerin infusion ⚠️ VERIFY

IV

5 – 200 mcg/min (titrate q3–5 min)

50 mg in 250 mL = 200 mcg/mL. Preferred for ACS and pulmonary edema.

Sodium nitroprusside ⚠️ VERIFY

Infusion

0.3 – 10 mcg/kg/min

50 mg in 250 mL D5W = 200 mcg/mL, light-protected. Arterial line. > 2 mcg/kg/min or > 72 h: cyanide/thiocyanate risk; max rate ≤ 10 min.

Hydralazine ⚠️ VERIFY

IV

10 – 20 mg q4–6h

Unpredictable, prolonged effect – not first-line except pregnancy.

Phentolamine (catecholamine excess: pheochromocytoma, cocaine/MAOI crisis) ⚠️ VERIFY

IV

1 – 5 mg, repeat q5–15 min

Pediatric

0.05 – 0.1 mg/kg (max 5 mg)

Acute heart failure & cardiogenic shock

Profile at the bedside: **warm/wet** (most) → vasodilator + diuretic · **cold/wet** → inotrope ± pressor, cath lab if ACS · **cold/dry** → cautious fluid. Find the trigger (CHAMPIT: ACS, Hypertensive emergency, Arrhythmia, Mechanical cause, Pulmonary embolism, Infection, Tamponade).

SCAPE – hypertensive flash pulmonary edema

① Sit upright; **NIV immediately** (CPAP 5–10 cmH₂O or BiPAP) – reduces intubation

② **High-dose nitroglycerin** is the main treatment; diuretic is secondary (patients are often euvolemic, fluid redistributed) ⚠️ VERIFY

High-dose nitroglycerin – SCAPE ⚠️ VERIFY

SL (while setting up)

0.4 – 0.8 mg, repeat q3–5 min

IV bolus

400 – 1,000 mcg over 2 min (may repeat)

IV infusion

Start 100 – 200 mcg/min, titrate rapidly (up to ~400 mcg/min) then wean as BP falls

Hold if SBP < 100–110 or MAP drop > 30%. Avoid in severe AS/RV infarct/PDE-5.

Congestion (warm/wet)

Furosemide – acute HF ⚠️ VERIFY

Diuretic-naïve

20 – 40 mg IV

On oral loop diuretic

1 – 2.5 × total daily oral dose, given IV

Check spot urine Na at 2 h (> 50–70 mmol/L) / urine output at 6 h (> 100–150 mL/hr); else double the dose.

- Add a thiazide (metolazone/HCTZ) or acetazolamide for diuretic resistance; daily weight, K, Mg, creatinine
- Avoid NSAIDs, non-DHP calcium blockers; continue GDMT unless shock or bradycardia

Cardiogenic shock (SCAI stages)

A At risk	Large MI, HF – normal perfusion
B Beginning	Hypotension/tachycardia, perfusion preserved (lactate < 2)
C Classic	Hypoperfusion requiring intervention (inotrope/pressor/MCS); lactate ≥ 2
D Deteriorating	Failing initial therapy
E Extremis	Arrest/CPR, ECMO, refractory collapse

- 1 Treat the cause: **STEMI** → **emergent PCI**; mechanical complications (VSR, papillary rupture – echo) → surgery; tamponade → drain; arrhythmia → cardiovert
- 2 MAP ≥ 65: **norepinephrine** first-line; add **dobutamine** for low output once MAP adequate ⚠️ VERIFY
- 3 Arterial line, lactate, urine output, echo; early shock team / mechanical support referral (IABP, Impella, VA-ECMO where available)

Norepinephrine (MAP ≥ 65)

Infusion ⚠️ VERIFY

0.05 – 0.5 mcg/kg/min

4 mg in 250 mL = 16 mcg/mL. Peripheral OK short-term via good proximal vein.

Dobutamine (low output) ⚠️ VERIFY

Infusion

2 – 20 mcg/kg/min

250 mg in 250 mL = 1,000 mcg/mL. Start 2–5; arrhythmias, hypotension.

Arrhythmias

Buttons open the shared algorithm and code timer in Resuscitation – same doses, one source.

Unstable? (hypotension, altered mentation, shock, ischemic pain, acute HF)

Synchronized cardioversion ⚠️ VERIFY

AF / flutter 200 J · narrow regular 100 J · monomorphic VT 100 J

Polymorphic VT / unable to sync → unsynchronized defibrillation
Escalate if no response; re-sync after each shock; sedate if feasible

Pulseless VT/VF – defibrillation ⚠️ VERIFY

Biphasic 120 – 200 J (per manufacturer; max if unknown)

Bradycardia – pacing

Transcutaneous pacing if atropine fails

Rate 60–80/min; raise mA until capture, confirm mechanical pulse (femoral); analgesia/sedation. Transvenous via cardiology.

Stable tachycardia – by QRS and regularity

Narrow, regular (SVT, flutter)	Modified Valsalva (strain 15 s, then supine + legs raised) → adenosine 6 → 12 (→ 12) mg rapid push with flush → diltiazem/β-blocker; flutter waves revealed → rate control
Narrow, irregular (AF, MAT)	Rate control (diltiazem or metoprolol; amiodarone/digoxin if HFrEF/hypotension); onset < 24–48 h or anticoagulated → rhythm control option; CHA ₂ DS ₂ -VA for anticoagulation
Wide, regular	Assume VT. Procainamide or amiodarone; adenosine only if regular + monomorphic and SVT with aberrancy likely; expert consult; cardioversion if any doubt
Wide, irregular	Pre-excited AF (WPW): no AV-nodal blockers (adenosine, diltiazem, verapamil, β-blocker, digoxin, amiodarone) → procainamide (or ibutilide) or cardioversion. Polymorphic VT/torsades: Mg, defibrillate if unstable

Adenosine – SVT ⚠️ VERIFY

1st dose

6 mg rapid IV push + 20 mL NS flush

= 2 mL of 3 mg/mL. Antecubital or proximal vein.

2nd dose

12 mg rapid IV push + flush

= 4 mL of 3 mg/mL

Diltiazem – AF/flutter rate control ⚠️ VERIFY

IV over 2 min

0.25 mg/kg, max 25 mg

Repeat 0.35 mg/kg after 15 min if needed; infusion 5–15 mg/hr. Not in HFrEF, WPW, hypotension.

Metoprolol – rate control ⚠️ VERIFY

IV over 2 min

2.5 – 5 mg, repeat q5 min to 15 mg

Avoid in decompensated HF, bronchospasm, WPW.

Amiodarone ⚠️ VERIFY

VF/pVT – 1st dose

300 mg IV/IO push

= 6 mL of 50 mg/mL

VF/pVT – 2nd dose

150 mg IV/IO

= 3 mL of 50 mg/mL

Stable wide-complex tachycardia

150 mg IV over 10 min (repeat PRN), then 1 mg/min × 6 h

Max ≈ 2.2 g / 24 h.

Procainamide ⚠️ VERIFY

Stable wide-complex tachycardia

20 – 50 mg/min IV

Until arrhythmia suppressed, hypotension, QRS widens > 50% or max 17 mg/kg. Maintenance 1–4 mg/min. Avoid in long QT / CHF.

Lidocaine (alternative to amiodarone) ⚠️ VERIFY

VF/pVT – 1st dose

1 – 1.5 mg/kg

Then 0.5–0.75 mg/kg q5–10 min; max total 3 mg/kg.

Magnesium sulfate – torsades ⚠️ VERIFY

Torsades / hypomagnesemia

1 – 2 g IV

= 2–4 mL of 50% (500 mg/mL). Pulseless: dilute in 10 mL, over 1–2 min. With pulse: over 5–20 min.

Bradycardia

Atropine – bradycardia

Symptomatic bradycardia ⚠️ VERIFY

1 mg IV q3–5 min, max 3 mg

= 1 mL of 1 mg/mL. Unlikely to work in infranodal block / transplanted heart.

Bradycardia – second line ⚠️ VERIFY

Dopamine

5 – 20 mcg/kg/min

400 mg in 250 mL = 1,600 mcg/mL.

Epinephrine infusion

2 – 10 mcg/min

1 mg in 250 mL = 4 mcg/mL → 30 – 150 mL/hr. Titrate.

Transcutaneous pacing

If atropine ineffective / unstable

Rate 60–80/min; increase mA to capture; confirm pulse; sedate/analgesia.

- High-grade AV block (Mobitz II, 3rd degree) → pacing pads on; atropine often ineffective (infranodal)
- Look for causes: hyperkalemia, β-blocker / CCB / digoxin toxicity, inferior MI, hypothermia, hypothyroidism, raised ICP, Lyme/Chagas
- QT prolongation → stop QT drugs, correct K > 4 and Mg > 2; torsades → magnesium, overdrive pacing/isoproterenol

Aortic dissection

- ☐ **High-risk condition:** Marfan / connective tissue disease, family history of aortic disease, known aortic valve disease or aneurysm, recent aortic manipulation
- ☐ **High-risk pain:** abrupt onset, severe, ripping/tearing (chest, back or abdomen)
- ☐ **High-risk exam:** pulse deficit, SBP differential > 20 mmHg, focal neuro deficit + pain, new aortic regurgitation murmur + pain, hypotension/shock

0 ADD-RS (0–3)

ADD-RS 0: low risk – D-dimer can help rule out (ADD-RS 0–1) ⚠️ VERIFY

- 1** Analgesia (IV opioid) – pain drives catecholamines
- 2** **Impulse control first:** IV β-blocker to **HR ≤ 60**, then vasodilator for **SBP 100–120** within ~20 min (never vasodilator first – reflex tachycardia) ⚠️ VERIFY
- 3** **Type A** (ascending) → emergency cardiothoracic surgery; **Type B** → medical therapy ± endovascular if complicated (malperfusion, rupture, refractory pain/HTN)
- 4** Hypotension = tamponade, rupture, severe AR, or coronary involvement → surgery, not β-blockers; no fibrinolysis/anticoagulation (STEMI with dissection: inferior STEMI is the classic trap)

Esmolol ⚠️ VERIFY

Load (over 1 min)

500 mcg/kg

Infusion

50 – 300 mcg/kg/min

10 mg/mL. Titrate q4 min to HR/BP.

Labetalol ⚠️ VERIFY

IV bolus

20 mg over 2 min, then 20 – 80 mg q10 min (max 300 mg)

or infusion

0.5 – 2 mg/min

Nicardipine ⚠️ VERIFY

Infusion

5 mg/hr, increase 2.5 mg/hr q5–15 min (max 15 mg/hr)

25 mg in 250 mL = 0.1 mg/mL. Reflex tachycardia; avoid in severe aortic stenosis.

Abdominal aortic aneurysm – suspected rupture

- Pain (abdomen/back/flank) + hypotension or syncope in > 50–60 y, smoker: bedside POCUS (aorta > 3 cm), vascular surgery immediately – CT only if stable
- Permissive hypotension (target SBP ~70–90 / conscious and mentating) until control; activate massive transfusion (MTP)
- Mimics renal colic – image the aorta in older first-time "colic"

♥ Pericarditis, myocarditis, tamponade

Acute pericarditis – ≥ 2 of 4 (ESC)

- Pleuritic chest pain, better sitting forward
- Pericardial friction rub
- New widespread ST elevation / PR depression (PR elevation in aVR)
- New or worsening pericardial effusion

High-risk features → **admit:** fever > 38 °C, subacute onset, large effusion (> 20 mm) or tamponade, no response to NSAID after 7 days, myopericarditis (troponin ↑), immunosuppression, trauma, oral anticoagulant. Look for TB (common in PH: large effusion, constitutional symptoms), uremia, malignancy, autoimmune.

Acute pericarditis – anti-inflammatory △ VERIFY

Ibuprofen

600 mg PO q8h × 1–2 wk, then taper (+ PPI)

or Aspirin

750 – 1,000 mg PO q8h × 1–2 wk, then taper (preferred post-MI)

+ Colchicine (3 months)

0.5 mg BID (once daily if < 70 kg)

Reduce in CKD; interactions (clarithromycin, azoles, statins)

Myocarditis red flags

- Chest pain + troponin rise with non-obstructed coronaries / viral prodrome; new HF, syncope, ventricular arrhythmia, AV block
- **Fulminant:** cardiogenic shock within days → ICU, echo, early mechanical support referral; avoid NSAIDs in myocarditis-predominant disease; dengue and COVID/vaccine-related cases reported locally
- Exercise restriction ≥ 3–6 months; cardiac MRI

Tamponade – recognise early

- Hypotension, tachycardia, JVD (absent if hypovolemic), muffled sounds, pulsus paradoxus > 10 mmHg, electrical alternans/low voltage
- POCUS: effusion + RA systolic collapse / RV diastolic collapse + plethoric IVC
- Fluid bolus only as a bridge; avoid positive-pressure ventilation and diuretics/vasodilators; intubation can precipitate arrest
- **Pericardiocentesis** (ultrasound-guided, subxiphoid or parasternal) for tamponade; traumatic or dissection-related → surgery (needle only as a bridge in arrest)

📖 Cardiovascular sources

Rao et al., 2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for acute coronary syndromes · Byrne et al., 2023 ESC ACS guidelines · Thygesen et al., Fourth Universal Definition of MI 2018 · STREAM-2 (2023) · Six/Backus HEART score (2008/2013) · Whelton et al., 2017 ACC/AHA hypertension guideline · Flynn et al., AAP 2017 pediatric hypertension guideline · McDonagh et al., 2021 ESC heart failure guidelines (+2023 update) · Heidenreich et al., 2022 AHA/ACC/HFSA HF guideline · SCAI SHOCK stage classification 2019/2022 · AHA 2020 and 2025 ACLS/PALS guidelines · Joglar et al., 2023 ACC/AHA AF guideline · Isselbacher et al., 2022 ACC/AHA aortic disease guideline · Rogers et al., ADD-RS (Circulation 2011) & ADVISED (2018) · Adler et al., 2015 ESC pericardial diseases guidelines and the 2025 ESC myocarditis & pericarditis guidelines · Greenberg et al., AHA 2022 ICH guideline · Powers et al., AHA/ASA 2019 acute ischemic stroke. Tintinalli 9e Section 7 used only as a topic map.

△ Items pending owner review in this section (32)

1. Atropine ampule strength assumed 1 mg/mL

Philippine stock may be 0.5 or 0.6 mg/mL – check the ampule before drawing up.

2. Pediatric atropine 0.02 mg/kg with NO minimum dose (max 0.5 mg)

PALS 2020 removed the 0.1 mg minimum; some local protocols still use it.

3. Pediatric lidocaine 1 mg/kg, capped at 100 mg

100 mg cap is from common references, not stated explicitly in PALS.

4. Pediatric magnesium sulfate 25–50 mg/kg (max 2 g) and stock concentration

Philippine MgSO₄ comes as 50%, 25% and 20% – confirm which is on hand.

5. Procainamide (adult 20–50 mg/min to 17 mg/kg; peds 15 mg/kg over 30–60 min)

Availability in the Philippines is limited; confirm product/concentration (100 mg/mL assumed).

6. Adult synchronized cardioversion energies (AHA 2025: AF/flutter 200 J; narrow-complex & monomorphic VT 100 J)

Device-specific; local protocols may still use 2020 values (50–100 J narrow, 120–200 J AF).

7. Pediatric vasoactive infusion ranges (epinephrine/norepinephrine 0.05–1 mcg/kg/min; epi for bradycardia 0.1–1)

Titration ranges vary by PICU protocol; standard 16 mcg/mL bag assumed.

8. Adult norepinephrine range 0.05–0.5 mcg/kg/min

Many units titrate in mcg/min; ceiling and peripheral-line policies vary.

9. UFH: VTE 80 U/kg bolus + 18 U/kg/hr; ACS 60 U/kg (max 4,000 U) + 12 U/kg/hr (max 1,000 U/hr)

Titrate to aPTT/anti-Xa by local nomogram.

10. ACLS arrhythmia drug doses reused from v1 in the Cardiovascular section (adenosine, amiodarone, lidocaine, magnesium, bradycardia infusions)

AHA 2020/2025 values; unchanged from v1 but now flagged wherever shown.

11. Adult defibrillation energy: biphasic 120–200 J per manufacturer (maximum if unknown), monophasic 360 J

AHA 2020/2025; device-specific.

12. Pediatric energies: defibrillation 2 J/kg → 4 J/kg → ≥ 4 J/kg (max 10 J/kg or adult dose); synchronized cardioversion 0.5–1 J/kg then 2 J/kg

PALS 2020/2025; v1 values reused, now flagged.

13. STEMI time targets: ECG ≤ 10 min of arrival; FMC-to-device ≤ 90 min (≤ 120 min including transfer, else lyse); door-to-needle ≤ 30 min; lysis → transfer, angiography 2–24 h, rescue PCI if < 50% ST resolution at 60–90 min; primary PCI still reasonable 12–48 h with ongoing symptoms/instability
2025 ACC/AHA ACS and 2023 ESC (ESC uses ≤ 10 min diagnosis-to-lysis bolus). Local STEMI network protocol governs.

14. STEMI ECG criteria (4th UDMI): new ST elevation at J point in ≥ 2 contiguous leads – ≥ 1 mm all leads except V2–V3: ≥ 2 mm men ≥ 40 y, ≥ 2.5 mm men < 40 y, ≥ 1.5 mm women; posterior V7–V9 ≥ 0.5 mm

Thresholds from the 4th Universal Definition (2018); OMI/STEMI-equivalent patterns are guideline-endorsed to varying degrees.

15. Aspirin 162–325 mg chewed (non-enteric) loading, then 75–100 mg daily

AHA 162–325 mg; ESC 150–300 mg PO (75–250 mg IV).

16. P2Y₁₂ loading: ticagrelor 180 mg; prasugrel 60 mg (PCI only; avoid prior stroke/TIA; ≥ 75 y or < 60 kg 5 mg maintenance); clopidogrel 600 mg for PCI, 300 mg with fibrinolysis (75 mg, no load, if > 75 y)

Timing of pre-treatment differs (ESC: not routine in NSTEMI-ACS when early angiography planned).

17. Anticoagulation: primary PCI UFH 70–100 U/kg IV; with fibrinolysis UFH 60 U/kg (max 4,000 U) then 12 U/kg/hr (max 1,000 U/hr, aPTT 1.5–2×) or enoxaparin < 75 y 30 mg IV + 1 mg/kg SC q12h (max 100 mg first 2 doses), ≥ 75 y no IV bolus, 0.75 mg/kg SC q12h (max 75 mg first 2 doses), CrCl < 30: 1 mg/kg q24h; NSTEMI-ACS enoxaparin 1 mg/kg SC q12h

2013/2025 ACC/AHA; ESC 2023 similar. Renal dosing and bleeding risk decide.

18. Nitroglycerin 0.4 mg SL q5 min × 3 for ischemic pain (avoid SBP < 90, RV infarct, PDE-5 inhibitor within 24–48 h); IV 5–10 mcg/min titrated

ACC/AHA. Not shown to reduce mortality; do not use to mask pain before ECG.

19. Tenecteplase single bolus by weight: < 60 kg 30 mg · 60–69 35 mg · 70–79 40 mg · 80–89 45 mg · ≥ 90 kg 50 mg; half dose if ≥ 75 y

Half dose ≥ 75 y comes from STREAM/STREAM-2 and ESC 2023; not in all labels. TNK may not be stocked in PH hospitals.

20. Alteplase accelerated STEMI regimen: 15 mg IV bolus, 0.75 mg/kg over 30 min (max 50 mg), 0.5 mg/kg over 60 min (max 35 mg); total ≤ 100 mg
GUSTO accelerated regimen.

21. Streptokinase 1.5 million units IV over 30–60 min; do not repeat (antibodies) – use a fibrin-specific agent if previously exposed

Still used where fibrin-specific agents are unavailable (relevant to PH public hospitals). Hypotension common – slow infusion.

22. HEART score bands: 0–3 low (MACE ≈ 1–2%), 4–6 moderate, 7–10 high

Use with serial or high-sensitivity troponin pathway; HEART alone is not a 0/1-h hs-troponin algorithm.

23. BP targets: general hypertensive emergency – lower MAP ≤ 25% in 1st hour, then to 160/100–110 over 2–6 h, then normal over 24–48 h; aortic dissection SBP ≤ 120 (100–120) and HR ≤ 60 within 20 min; preeclampsia/eclampsia, pheochromocytoma SBP < 140 in 1st hour; ischemic stroke: < 185/110 before and < 180/105 for 24 h after lysis (no lysis: treat only > 220/120); ICH: target SBP 140 (avoid < 130); ACS/AHF: vasodilator to symptom relief

2017 ACC/AHA HTN, 2019 AIS, 2022 ICH. Do not treat asymptomatic severe hypertension with rapid IV drops.

24. IV antihypertensives: labetalol 20 mg IV then 20–80 mg q10 min (max 300 mg) or 0.5–2 mg/min; nicardipine 5 mg/hr, + 2.5 mg/hr q5–15 min (max 15); esmolol 500 mcg/kg load then 50–300 mcg/kg/min; NTG 5–200 mcg/min; nitroprusside 0.3–10 mcg/kg/min (max 10 for ≤ 10 min); hydralazine 10–20 mg IV; phentolamine 1–5 mg IV (repeat; for catecholamine excess)

2017 ACC/AHA HTN table; nicardipine/clevidipine availability varies in PH.

25. Pediatric hypertensive emergency: lower BP ≤ 25% of planned reduction in first 8 h, rest over 24–48 h; labetalol 0.2–1 mg/kg (max 40 mg) or 0.25–3 mg/kg/hr; nicardipine 0.5–3 mcg/kg/min; esmolol 100–500 mcg/kg/min; hydralazine 0.1–0.2 mg/kg (max 20 mg); nitroprusside 0.3–8 mcg/kg/min

AAP 2017 Flynn guideline table (recalled) – pediatric nephrology/PICU input recommended. Nitroprusside peds max uncertain.

26. Acute HF: furosemide IV – diuretic-naïve 20–40 mg; on oral loop diuretic 1–2.5× the daily oral dose IV; reassess natriuresis (spot urine Na > 50–70 mmol/L at 2 h) / urine output > 100–150 mL/hr at 6 h; double dose if inadequate

ESC 2021 HF / 2019 HFA diuretic position paper.

27. SCAPE (sympathetic crashing acute pulmonary edema): high-dose nitroglycerin – SL 0.4–0.8 mg repeated, or IV bolus 400–1,000 mcg over 2 min and/or infusion starting 100–200 mcg/min titrated rapidly (up to ~400 mcg/min) + immediate NIV

Based on observational ED data (e.g. Mathew 2021, Wilson 2017); not in ESC/AHA guidelines in this form. Owner review essential.

28. Dobutamine 2–20 mcg/kg/min (start 2–5) – pump concentration 250 mg in 250 mL = 1,000 mcg/mL (premix 250 mg/250 mL or 500 mg/250 mL = 2,000 mcg/mL)

Concentration assumption must match local premix.

29. Rate control: diltiazem 0.25 mg/kg IV over 2 min (then 0.35 mg/kg after 15 min), infusion 5–15 mg/hr; metoprolol 2.5–5 mg IV over 2 min q5 min to 15 mg; avoid both in decompensated HFrEF, WPW, hypotension – use amiodarone/digoxin

2023 ACC/AHA AF guideline.

30. Aortic dissection: IV β-blocker first (esmolol 500 mcg/kg then 50–300 mcg/kg/min, or labetalol) to HR ≤ 60, then vasodilator (nicardipine/nitroprusside) for SBP 100–120; opioid analgesia

2022 ACC/AHA aortic guideline. If AR severe/shock, avoid β-blocker – surgery.

31. ADD-RS + D-dimer (ADvISED): ADD-RS 0–1 with D-dimer < 500 ng/mL FEU → dissection very unlikely; ADD-RS ≥ 2 → direct CT angiography

ADvISED (2018) failure rate ≈ 0.3% in ADD-RS ≤ 1 + D-dimer negative; not validated in all populations.

32. Acute pericarditis: ibuprofen 600 mg q8h (or aspirin 750–1,000 mg q8h) for 1–2 weeks then taper + colchicine 0.5 mg BID (0.5 mg daily if < 70 kg) for 3 months; avoid routine steroids; pediatric ibuprofen 30–50 mg/kg/day + q8h, colchicine < 5 y 0.5 mg/day, > 5 y 1–1.5 mg/day + 2–3

ESC 2015 doses; the 2025 ESC myocarditis-pericarditis guideline may refine – check current version. Diagnosis needs ≥ 2 of 4 criteria.

S8 • Pulmonary Disorders Merged

Uses weight, age, sex and height from the top of the page.

Asthma exacerbation

Severity	Features (adult, BTS/SIGN 2019 · GINA 2024)
Moderate	↑ symptoms, PEF > 50–75% best/predicted, no features of acute severe (GINA mild-moderate: talks in phrases, HR 100–120, SpO ₂ 90–95%)
Acute severe	Any one: PEF 33–50%, RR ≥ 25, HR ≥ 110, cannot complete sentences (GINA severe: talks in words, agitated, RR > 30, accessory muscles, HR > 120, SpO ₂ < 90%, PEF ≤ 50%)
Life-threatening	Any one: PEF < 33%, SpO ₂ < 92%, PaO ₂ < 8 kPa (60 mmHg), “normal” PaCO ₂ 4.6–6.0 kPa, silent chest, cyanosis, poor effort, exhaustion, altered consciousness, arrhythmia, hypotension
Near-fatal	Raised PaCO ₂ and/or need for mechanical ventilation with raised inflation pressures

O₂ to SpO₂ 93–98% (GINA 93–95%; children 94–98%). ABG if SpO₂ < 92% or life-threatening features.

- 1 **Salbutamol** 5 mg nebulised (O₂-driven), repeat every 15–30 min or back-to-back/continuous if severe; mild-moderate: pMDI + spacer 4–10 puffs q20 min △ VERIFY
- 2 **Ipratropium** 0.5 mg neb q4–6h for acute severe / life-threatening or poor response △ VERIFY
- 3 **Steroid within 1 h**: prednisolone 40–50 mg PO daily for ≥ 5 days, or hydrocortisone 100 mg IV q6h if unable to swallow △ VERIFY
- 4 **Magnesium sulfate** 1.2–2 g IV over 20 min (single dose) for acute severe not responding or life-threatening △ VERIFY
- 5 **Adrenaline** only if anaphylaxis / angioedema (IM 0.5 mg – see Tox: Anaphylaxis) or peri-arrest. IV aminophylline/salbutamol infusion: senior/ICU only △ VERIFY
- 6 **Escalate early** (ICU/anaesthesia) for life-threatening/near-fatal features, rising PaCO₂, exhaustion or drowsiness. **NIV is not routine** in asthma (only in ICU, never delaying intubation)
- 7 **Intubation cautions**: most experienced operator; ketamine induction; large ETT; low RR (8–10), low Vt 6–8 mL/kg PBW, long expiratory time (I:E ≥ 1:4), permissive hypercapnia, Pplat < 30. Hypotension/arrest after intubation → **disconnect the circuit** to release auto-PEEP, consider tension pneumothorax △ VERIFY

Discharge: PEF > 75% 1 h after treatment, inhaler technique, ICS-containing controller, written plan, follow-up within 2 days.

COPD exacerbation – GOLD 2025

- 1 **Controlled O₂** to SpO₂ **88–92%** (Venturi 24–28% or nasal 1–2 L/min); ABG within 30–60 min and after any FiO₂ change. Avoid high-flow O₂ (worsens hypercapnia)
- 2 **Bronchodilators**: salbutamol 2.5–5 mg ± ipratropium 0.5 mg neb q4–6h (more often initially) – **air-driven** nebulisers with nasal O₂ if hypercapnic △ VERIFY
- 3 **Prednisone/prednisolone 40 mg PO daily × 5 days** (GOLD); blood eosinophils may guide △ VERIFY
- 4 **Antibiotics** if ↑ sputum purulence + ↑ dyspnea and/or ↑ sputum volume, or if NIV/ventilation needed – 5 days; choice by local resistance (e.g. amoxicillin-clavulanate, doxycycline, macrolide; Pseudomonas risk → antipseudomonal) △ VERIFY
- 5 **NIV indications** (GOLD): acute respiratory acidosis **pH ≤ 7.35 and PaCO₂ ≥ 45 mmHg (6 kPa)**; severe dyspnea with fatigue/accessory muscle use; persistent hypoxemia despite O₂. BTS: start if pH < 7.35 and PaCO₂ > 6.5 kPa persist after ≤ 1 h optimal therapy. Start IPAP 15 → 20–30, EPAP 4–5 cmH₂O; ABG at 1–2 h △ VERIFY

- 6 **Intubate** if NIV fails/contraindicated (see Respiratory failure card), pH < 7.25 worsening on NIV, ↓ GCS, arrest. Discuss ceiling of care early
- 7 Look for triggers/mimics: pneumonia, PE, pneumothorax, ACS/HF, arrhythmia

Pneumonia (CAP) – CURB-65

- ☐ Confusion (new disorientation)
- ☐ Urea > 7 mmol/L (BUN > 19 mg/dL)
- ☐ Respiratory rate ≥ 30/min
- ☐ SBP < 90 or DBP ≤ 60 mmHg
- ☐ Age ≥ 65 y (auto from age if entered)

Low – consider outpatient (Lim 2003). Use with judgment, SpO₂, comorbidity and social factors.

- **PSI/PORT** (20 variables) is the IDSA/ATS-preferred tool for outpatient vs inpatient; CURB-65 is quicker
- **Severe CAP (IDSA/ATS 2019)** – 1 major: septic shock on vasopressors, or respiratory failure needing mechanical ventilation; *or* ≥ 3 minor: RR ≥ 30, PaO₂/FiO₂ ≤ 250, multilobar infiltrates, confusion, BUN ≥ 20 mg/dL, WBC < 4,000, platelets < 100,000, temp < 36 °C, hypotension needing aggressive fluids → ICU

Empiric antibiotics (PCAP 2016 / IDSA-ATS 2019) △ VERIFY

- **Low-risk (outpatient), no comorbidity**: amoxicillin 1 g PO TID, or azithromycin 500 mg daily × 3 d / clarithromycin 500 mg BID △ VERIFY
- **Low-risk with comorbidity**: amoxicillin-clavulanate (or sulfamonomethoxazole / cefuroxime) ± macrolide or doxycycline
- **Moderate-risk (ward)**: IV non-antipseudomonal β-lactam (e.g. ceftriaxone 1–2 g daily, ampicillin-sulbactam, amoxicillin-clavulanate) + macrolide/doxycycline, or respiratory FQ alone (levofloxacin 750 mg daily, moxifloxacin 400 mg daily) △ VERIFY
- **High-risk (ICU), no Pseudomonas risk**: IV β-lactam + IV macrolide (or + respiratory FQ)
- **Pseudomonas / MRSA risk** (prior isolation, recent hospitalization + IV antibiotics, structural lung disease): antipseudomonal β-lactam (pip-tazo, cefepime, meropenem) + macrolide/FQ; add vancomycin or linezolid if MRSA risk – adjust in Renal dosing
- Cultures (blood, sputum) before antibiotics in moderate/high risk, but do not delay; 5–7 days total if stable

Pulmonary embolism – ESC 2019

Wells score (PE)

- ☐ Clinical signs/symptoms of DVT **(+3)**
- ☐ PE is the most likely diagnosis **(+3)**
- ☐ Heart rate > 100 **(+1.5)**
- ☐ Immobilisation ≥ 3 d or surgery in previous 4 wk **(+1.5)**
- ☐ Previous DVT/PE **(+1.5)**
- ☐ Hemoptysis **(+1)**
- ☐ Malignancy (treatment within 6 mo / palliative) **(+1)**

0 PE unlikely (≤ 4) → D-dimer · three-tier: low (< 2)

YEARS items present (DVT signs, hemoptysis, PE most likely): **0** → PE excluded if D-dimer < 1,000 ng/mL △ VERIFY.

PERC rule (only if gestalt low risk < 15%)

- ☐ Age ≥ 50 (auto)

- ☐ HR ≥ 100
- ☐ SpO₂ < 95% on room air
- ☐ Unilateral leg swelling
- ☐ Hemoptysis
- ☐ Surgery or trauma ≤ 4 wk (requiring anesthesia)
- ☐ Prior PE or DVT
- ☐ Hormone use (OCP, HRT, estrogen)

0 positive – PERC negative: if gestalt < 15%, no further testing

sPESI

- ☐ Age > 80 (auto)
- ☐ Cancer
- ☐ Chronic heart failure or chronic lung disease
- ☐ HR ≥ 110
- ☐ SBP < 100 mmHg
- ☐ SpO₂ < 90%

sPESI 0 – low risk (30-day mortality $\approx$ 1%) – consider early discharge/outpatient if no RV dysfunction and social support (Hestia)

Testing

- **Age-adjusted D-dimer cut-off:** age $\times$ 10 μ g/L (ng/mL FEU) if > 50 y; 500 if $\leq$ 50 (A VERIFY)
- **CTPA** if PE likely or D-dimer positive. Unstable $\rightarrow$ bedside echo (RV dilatation) and treat as high-risk PE if CTPA not feasible. V/Q or CTPA in pregnancy per local pathway (YEARS adapted / leg ultrasound first)

ESC 2019 risk

- **High (massive):** arrest, obstructive shock, or SBP < 90 for > 15 min / vasopressors $\rightarrow$ reperfusion
- **Intermediate-high:** PESI III–V / sPESI $\geq$ 1 + RV dysfunction and raised troponin $\rightarrow$ anticoagulate, monitor, rescue lysis if decompensates
- **Intermediate-low:** one or none of RV dysfunction/troponin · **Low:** sPESI 0, no RV dysfunction

Anticoagulation (A VERIFY)

- Start while awaiting imaging if probability high/intermediate
- LMWH: enoxaparin 1 mg/kg SC q12h; **UFH** if high-risk, reperfusion likely, or CrCl < 30 (see Renal dosing) (A VERIFY)
- DOAC preferred for most: apixaban 10 mg BID $\times$ 7 d $\rightarrow$ 5 mg BID; rivaroxaban 15 mg BID $\times$ 21 d $\rightarrow$ 20 mg daily (avoid in APS, severe renal impairment, pregnancy) (A VERIFY)

Thrombolysis – high-risk PE (A VERIFY)

- **Alteplase 100 mg IV over 2 h** (ESC), or accelerated 0.6 mg/kg over 15 min (max 50 mg) (A VERIFY)
- **Cardiac arrest from suspected PE:** alteplase 50 mg IV bolus (repeat after 15 min if needed); continue CPR 60–90 min (A VERIFY)
- Contraindications: prior ICH, ischemic stroke < 6 mo, CNS neoplasm, major trauma/surgery/head injury < 3 wk, bleeding diathesis, active bleeding. If contraindicated/failed: catheter-directed therapy or surgical embolectomy; ECMO
- Avoid aggressive fluids (> 500 mL) in RV failure; norepinephrine for hypotension (A VERIFY)

Pneumothorax

Tension pneumothorax – clinical diagnosis, do not wait for imaging

- 1 Respiratory distress, hypoxia, hypotension, $\downarrow$ breath sounds, (late) tracheal deviation, distended neck veins; ventilated: $\uparrow$ airway pressures, falling SpO₂/BP, PEA
- 2 **Needle decompression** (A VERIFY): adult **4th/5th ICS, anterior-to-mid axillary line** (ATLS 10th) or **2nd ICS mid-clavicular line** (children; alternative in adults). Long cannula ($\geq$ 8 cm) in adults – kinks/failure common
- 3 Then **finger thoracostomy** (ventilated patients, arrest – bilateral in traumatic arrest) or **chest tube** 4th/5th ICS in the safe triangle

Spontaneous pneumothorax (BTS 2023)

- **Primary** (no lung disease) with minimal symptoms and no high-risk features $\rightarrow$ **conservative** (observation, follow-up, often outpatient) regardless of size
- **Significant symptoms:** needle **aspiration** (16–18 G, stop at 2.5 L or resistance/cough) (A VERIFY), ambulatory valve device, or small-bore chest drain – shared decision on patient priorities
- **High-risk features** $\rightarrow$ drain and admit: hemodynamic instability, significant hypoxia, bilateral, hemopneumothorax, underlying lung disease (secondary PTX, e.g. COPD, age > 50 with smoking history), mechanical ventilation
- **Secondary** pneumothorax: admit, usually small-bore drain; early respiratory referral. No air travel until resolved; no diving lifelong unless definitive surgery

ARDS & lung-protective ventilation

Enter PaO₂ and FiO₂ for P/F ratio.

Enter sex and height (cm) at the top for PBW / tidal volume

(A VERIFY)

- **Berlin 2012:** onset $\leq$ 1 wk, bilateral opacities not explained by effusion/collapse/nodules, not fully explained by cardiac failure/overload; P/F with PEEP $\geq$ 5: mild 200–300, moderate 100–200, severe $\leq$ 100
- **2023 global definition:** adds non-intubated ARDS (HFNC $\geq$ 30 L/min or NIV/CPAP $\geq$ 5: P/F $\leq$ 300 or SpO₂/FiO₂ $\leq$ 315 with SpO₂ $\leq$ 97%), allows lung ultrasound for bilateral opacities and a resource-limited variant (no PEEP/flow requirement)
- **Ventilate:** Vt 6 mL/kg PBW, **plateau $\leq$ 30 cmH₂O**, driving pressure $\leq$ 15, RR up to 35 (permissive hypercapnia, pH $\geq$ 7.25), SpO₂ 88–95% (A VERIFY)
- **PEEP** by ARDSNet FiO₂/PEEP table; higher PEEP for moderate–severe
- **Prone $\geq$ 16 h/day** if P/F < 150 on FiO₂ $\geq$ 0.6 (PROSEVA); conservative fluids once shock resolved; consider NMB in early severe; ECMO referral if refractory

Hemoptysis

Massive / life-threatening: definitions vary ($\geq$ 100–600 mL/24 h); any volume causing airway compromise, hypoxemia or instability is life-threatening (anatomic dead space only $\sim$ 150 mL).

- 1 **Position bleeding side down** (lateral decubitus) to protect the good lung; high-flow O₂; suction
- 2 **Airway:** intubate with a large ETT ($\geq$ 8.0) to allow bronchoscopy; consider selective mainstem intubation of the non-bleeding lung (left side bleeding $\rightarrow$ advance into right main bronchus); double-lumen tube only by experts
- 3 IV access $\times$ 2, crossmatch, Hb/coags; **reverse anticoagulation**, correct platelets; stop antiplatelets
- 4 **TXA:** nebulised 500 mg TID or IV 1 g (off-label, limited evidence) (A VERIFY)
- 5 **CT angiography** chest if stable enough $\rightarrow$ **bronchial artery embolization (IR)** first-line definitive therapy; **bronchoscopy** (localise, cold saline, adrenaline, balloon tamponade); thoracic surgery if refractory
- 6 Causes: TB (common locally), bronchiectasis, malignancy, aspergilloma, PE, anticoagulation, vasculitis, mitral stenosis

Pleural effusion – Light's criteria

Exudate if any: pleural/serum protein > 0.5 · pleural/serum LDH > 0.6 · pleural LDH > 2/3 of serum ULN.

Urgent tap / drain

- Large effusion with respiratory distress or hemodynamic compromise (therapeutic tap; stop at ~1.5 L or if chest tightness/cough – re-expansion edema) (**△ VERIFY**)
- Effusion + sepsis/pneumonia → diagnostic tap: **pus, pH < 7.2** (or glucose < 72 mg/dL / 4 mmol/L), or organisms on Gram stain → chest drain (empyema / complicated parapneumonic) (**△ VERIFY**)
- Suspected hemothorax (trauma, anticoagulation, post-procedure) → chest drain; massive (> 1,500 mL) → thoracic surgery (**△ VERIFY**)
- Use ultrasound guidance for all pleural procedures; send pH (blood-gas syringe), protein, LDH, glucose, cell count, Gram stain/culture (+ AFB/ADA for TB), cytology

Respiratory failure, NIV & HFNC

ROX index (HFNC)

$ROX = (SpO_2 / FiO_2) \div RR$. Enter FiO_2 as % (e.g. 50).

- **Type 1 (hypoxemic):** $PaO_2 < 60$ mmHg (8 kPa) with normal/low $PaCO_2$ – V/Q mismatch, shunt (pneumonia, edema, ARDS, PE)
- **Type 2 (hypercapnic):** $PaCO_2 > 45$ mmHg (6 kPa); acute if pH < 7.35 – pump failure (COPD, neuromuscular, sedation, obesity hypoventilation, chest wall)

NIV / CPAP – indications

- COPD with respiratory acidosis (see COPD card); cardiogenic pulmonary edema (CPAP or BiPAP); obesity hypoventilation; neuromuscular disease / chest wall; immunocompromised hypoxemia; post-extubation in high-risk; pre-oxygenation

NIV – contraindications

- Respiratory/cardiac arrest, need for immediate intubation; cannot protect airway, ↓ GCS (except hypercapnic drowsiness under close supervision), copious secretions, vomiting/upper GI bleed
- Hemodynamic instability, life-threatening arrhythmia; facial trauma/burns/surgery, fixed upper airway obstruction, recent upper GI/airway surgery; **undrained pneumothorax**; uncooperative/agitated

HFNC

- Flow 30–60 L/min (start ~50), FiO_2 titrated to SpO_2 target; reassess ROX at 2, 6 and 12 h – do not delay needed intubation (delay increases mortality)

Intubation thresholds (clinical, not a single number)

- Cannot maintain/protect airway, ↓ GCS, arrest/peri-arrest
- Failure of oxygenation despite HFNC/NIV (e.g. $SpO_2 < 88$ –90% on $FiO_2 \geq 0.6$ –1.0), $ROX < 2.85$ –3.85 by time point
- Failure of ventilation: pH < 7.25 and worsening on NIV after 1–2 h, rising $PaCO_2$, exhaustion, RR > 35 with accessory muscles/paradox
- Anticipated course (e.g. airway burn, shock needing procedures). Agree ceiling of care. See Airway tab: RSI

Pulmonary sources

BTS/SIGN British guideline on the management of asthma 2019 · GINA 2024 report · GOLD 2025 report · BTS/ICS acute hypercapnic respiratory failure (NIV) 2016 · Metlay et al., ATS/IDSA adult CAP guideline 2019 · Lim et al., CURB-65 (Thorax 2003) · Philippine Clinical Practice Guidelines on CAP in immunocompetent adults (PCAP) 2016 · Konstantinides et al., ESC/ERS acute PE guideline 2019 · Kline PERC 2004/2008 · van der Hulle, YEARS 2017 · Righini, ADJUST-PE 2014 · Roberts et al., BTS pleural disease guideline 2023 · ATLS 10th ed. · ARDS Definition Task Force (Berlin) 2012 · Matthay et al., global definition of ARDS 2023/2024 · ARDSNet ARMA 2000 · PROSEVA 2013 · Light 1972 · Roca et al., ROX index 2019 · Wand et al., nebulized TXA for hemoptysis 2018. Antibiotic choices depend on local resistance – **check the local antibiogram, formulary and pharmacist.**

△ Items pending owner review in this section (19)

1. Enoxaparin: treatment 1 mg/kg q12h → q24h if CrCl < 30; prophylaxis 40 mg → 30 mg daily; avoid in dialysis

Anti-Xa monitoring in severe CKD, obesity, low weight; local VTE protocol.

2. UFH: VTE 80 U/kg bolus + 18 U/kg/hr; ACS 60 U/kg (max 4,000 U) + 12 U/kg/hr (max 1,000 U/hr)

Titrate to aPTT/anti-Xa by local nomogram.

3. DOAC renal dosing (apixaban, rivaroxaban, edoxaban, dabigatran – AF and VTE)

US (FDA) and EU (EMA) labels differ, especially for CrCl < 30 and dialysis. Use CG with actual body weight per labels.

4. Pediatric salbutamol neb weight cut-off is inconsistent in this app: 20 kg (Tox/Pulm asthma) vs 25 kg (hyperkalemia, More tab)

Existing values kept as-is; owner to pick one cut-off (or an age-based rule) and harmonise both cards.

5. Adult asthma doses (salbutamol 5 mg neb back-to-back; ipratropium 0.5 mg q4–6h; prednisolone 40–50 mg ≥ 5 days or hydrocortisone 100 mg IV q6h; MgSO₄ 1.2–2 g IV over 20 min)

BTS/SIGN 2019 / GINA 2024; local protocol.

6. Pediatric asthma (ipratropium 250 mcg < 12 y / 500 mcg ≥ 12 y; prednisolone 1–2 mg/kg max 40 mg; MgSO₄ 40 mg/kg max 2 g; BTS HR/RR severity thresholds)

BTS 2019-based; PICU/local pediatric protocol governs.

7. COPD exacerbation (salbutamol 2.5–5 mg + ipratropium 0.5 mg neb; prednisone 40 mg × 5 days; antibiotic choice)

GOLD 2025; antibiotic choice by local resistance.

8. NIV starting settings (IPAP 15 → 20–30 cmH₂O, EPAP 4–5 cmH₂O) and BTS threshold (pH < 7.35 + $PaCO_2 > 6.5$ kPa after optimal therapy)

BTS/ICS 2016; device and unit protocols differ.

9. CAP empiric antibiotics (PCAP 2016 risk categories; IDSA/ATS 2019 regimens) and doses

Regimens and doses summarised – verify against the full PCAP 2016 document, local antibiogram and renal function.

10. Age-adjusted D-dimer (age × 10 µg/L FEU if > 50 y) and YEARS thresholds (1,000 / 500 ng/mL)

Assay-specific (FEU vs DDU units); validate with local lab.

11. PE anticoagulation (enoxaparin 1 mg/kg q12h; apixaban 10 mg BID × 7 d → 5 mg BID; rivaroxaban 15 mg BID × 21 d → 20 mg daily)

Renal function, weight extremes, cancer and pregnancy change choices.

12. PE thrombolysis: alteplase 100 mg over 2 h; accelerated 0.6 mg/kg over 15 min (max 50 mg); arrest 50 mg IV bolus

ESC 2019 / ERC; heparin handling during infusion differs (EU continue vs US pause); check contraindications.

13. Pneumothorax: needle aspiration (16–18 G, up to 2.5 L), needle decompression site/length (4th/5th ICS AAL–MAL or 2nd ICS MCL; ≥ 8 cm cannula adult)

BTS 2023 / ATLS 10th; site preference and device vary.

14. Lung-protective ventilation (Vt 6 mL/kg PBW, range 4–8; Pplat ≤ 30; driving pressure ≤ 15; proning ≥ 16 h/day if P/F < 150)

ARDSNet / PROSEVA; ICU protocol governs.

15. TXA for hemoptysis (nebulized 500 mg TID or IV 1 g)

Limited evidence; off-label.

16. Therapeutic thoracentesis volume limit (~1.5 L per session) and drain thresholds (pH < 7.2)

BTS 2023; stop earlier if chest pain/cough.

17. ROX index cut-offs (≥ 4.88 likely success; < 2.85 / 3.47 / 3.85 at 2 / 6 / 12 h high failure risk)

Derived in pneumonia-related hypoxemic failure (Roca 2019); not validated for all causes.

18. Asthma inline doses (adrenaline, aminophylline, ventilator settings)

Inline values from v1 / merged card, previously unflagged.

19. PE inline doses (enoxaparin, DOACs, alteplase 100 mg / 0.6 mg/kg / 50 mg arrest, fluid limit)

Inline values from v1 / merged card, previously unflagged.

Acute abdominal pain

- ☐ Migratory pain to RLQ (+1)
- ☐ Anorexia (+1)
- ☐ Nausea / vomiting (+1)
- ☐ RLQ tenderness (+2)
- ☐ Rebound tenderness (+1)
- ☐ Temperature $\geq 37.3^{\circ}\text{C}$ (+1)
- ☐ WBC $> 10 \times 10^9/\text{L}$ (+2)
- ☐ Neutrophils $> 75\%$ (+1)

0 Alvarado score △ VERIFY (0–10)

≤ 4 : appendicitis unlikely (not excluded)

- Always: pregnancy test in women of reproductive age, ECG in older patients (MI presents as epigastric pain), glucose, lipase, urinalysis
- **Elderly / immunosuppressed / diabetics**: blunted signs – low threshold for CT; high mortality
- **Pain out of proportion to exam** (AF, vascular disease): mesenteric ischemia → CT angiography; lactate rises late
- **Sudden severe pain + shock**: ruptured AAA (aorta card), perforated viscus (erect CXR/CT free air), ectopic (OB early pregnancy)
- **Bowel obstruction**: CT, NG decompression, IV fluids; strangulation (fever, peritonism, lactate) → surgery; sigmoid volvulus → endoscopic decompression
- **Incarcerated hernia**: gentle reduction with analgesia if $< 6\text{--}12\text{ h}$ and no signs of strangulation
- **Diverticulitis**: uncomplicated in immunocompetent patients may not need antibiotics (AGA 2021); abscess/perforation → CT, antibiotics, drainage
- Early analgesia does not mask surgical diagnoses

Intra-abdominal / biliary infection △ VERIFY

Community, mild–moderate

Ceftriaxone 2 g IV daily + metronidazole 500 mg IV q8h

Severe / healthcare-associated

Piperacillin–tazobactam 4.5 g IV q6h

Upper GI bleeding

- ☐ Pulse ≥ 100 (+1)
- ☐ Melena (+1)
- ☐ Syncope (+2)
- ☐ Hepatic disease (+2)
- ☐ Cardiac failure (+2)

0 Glasgow-Blatchford (0–23)

≤ 1 : very low risk – outpatient endoscopy may be appropriate △ VERIFY

- 1** Two large-bore IVs, crossmatch, crystalloid to perfusion; activate MTP if exsanguinating; protect airway in massive hematemesis/encephalopathy
- 2** Transfuse to Hb 7 g/dL (8 with cardiovascular disease) △ VERIFY; reverse anticoagulants (PCC / vitamin K); platelets if $< 50 \times 10^9/\text{L}$ and bleeding
- 3** Suspected varices (cirrhosis, stigmata): vasoactive drug + antibiotic **before** endoscopy; endoscopy within 12 h; balloon tamponade (Sengstaken/Minnesota) as a bridge
- 4** Non-variceal: endoscopy within 24 h after resuscitation; PPI; tranexamic acid **not** recommended (HALT-IT)

Proton-pump inhibitor – upper GI bleed △ VERIFY

IV

80 mg bolus, then 40 mg IV BID (or 8 mg/hr after endoscopic therapy)

Octreotide – variceal bleeding △ VERIFY

IV

50 mcg bolus, then 50 mcg/hr infusion $\times$ 2–5 days

Terlipressin (if available) △ VERIFY

IV

2 mg q4h $\times$ 24–48 h, then 1 mg q4h

Ischemia risk – avoid in coronary/peripheral vascular disease.

Ceftriaxone – cirrhosis with GI bleed △ VERIFY

IV

1 g daily (max 7 days)

Erythromycin – pre-endoscopy △ VERIFY

IV over 30 min

250 mg, 30–120 min before endoscopy

- Lower GI bleeding: most stop spontaneously; unstable → CT angiography then embolisation/colonoscopy; an upper source causes $\sim 10\text{--}15\%$ of brisk hematochezia (check BUN:Cr, consider NG aspirate/endoscopy)

Pancreas & biliary tree

Acute pancreatitis (revised Atlanta: 2 of pain, lipase $\geq 3\times$ ULN, imaging)

- 1 Moderate fluids**: lactated Ringer's – bolus 10 mL/kg only if hypovolemic, then 1.5 mL/kg/hr △ VERIFY
- 2 Analgesia** (opioids acceptable), early oral feeding as tolerated; no prophylactic antibiotics
- 3 Causes**: gallstones (ultrasound), alcohol, triglycerides $> 1,000\text{ mg/dL}$, hypercalcemia, drugs, post-ERCP; urgent ERCP only with cholangitis
- 4 Severity**: persistent organ failure $> 48\text{ h}$ = severe; SIRS, BUN rise and hematocrit rise predict poor outcome

Biliary

Condition	Key features	Action
Biliary colic	RUQ pain $< 6\text{ h}$, no fever, normal labs	Analgesia (NSAID), elective surgery
Acute cholecystitis	Pain $> 6\text{ h}$, fever, Murphy sign, US wall thickening/fluid	Antibiotics, early laparoscopic cholecystectomy ($< 72\text{ h}$)

Condition	Key features	Action
Acute cholangitis	Fever + jaundice + RUQ pain (Charcot) ± shock/confusion	Antibiotics + fluids; biliary drainage (ERCP) urgently if severe (Tokyo 2018)

Intra-abdominal / biliary infection

Community, mild-moderate

Ceftriaxone 2 g IV daily + metronidazole 500 mg IV q8h

Severe / healthcare-associated

Piperacillin-tazobactam 4.5 g IV q6h

Liver emergencies

- **Hepatic encephalopathy:** find the trigger (GI bleed, infection/SBP, constipation, hypokalemia, sedatives); lactulose ± rifaximin; protect airway if grade 3–4
- **SBP:** diagnostic paracentesis in every admitted cirrhotic with ascites (PMN $\geq$ 250/mm³); antibiotics + albumin
- **Acute liver failure** (INR $\geq$ 1.5 + encephalopathy, no prior cirrhosis): ICU, NAC for acetaminophen and some non-acetaminophen causes, glucose monitoring, early transplant-centre contact
- **Hepatitis A and E** (water-borne, common in PH), B (HBsAg high prevalence in PH) – check INR, bilirubin, glucose
- **Amoebic liver abscess** (PH): fever, RUQ pain, single right-lobe abscess – metronidazole; aspirate if large/left lobe/no response

Lactulose – hepatic encephalopathy

PO / NG

20 – 30 mL q1–2h until bowel movement, then 2–3 soft stools/day

Enema

300 mL lactulose + 700 mL water, retained 30–60 min

Albumin – SBP (day 1 / day 3)

IV

1.5 g/kg day 1, 1 g/kg day 3

Metronidazole – amoebic colitis / liver abscess

PO / IV

500 – 750 mg TID × 7–10 days, then luminal agent (paromomycin/diloxanide)

SBP antibiotics: cefotaxime 2 g IV q8h or ceftriaxone 2 g daily 

Vomiting, diarrhea & dehydration

Ondansetron

IV / PO / ODT

4 – 8 mg q8h

Metoclopramide

IV over $\geq$ 2 min

10 mg (akathisia: slow infusion; diphenhydramine)

Rehydration

Rehydration (WHO plans B and C)

Volumes

Plan B: 75 mL/kg ORS over 4 h

Plan C: 100 mL/kg Ringer's lactate

- Red flags: bloody diarrhea with fever (shigella/amoebiasis), rice-water stools with severe dehydration (cholera – notify DOH), prolonged fever with relative bradycardia (typhoid), recent antibiotics (C. difficile), immunosuppression
- Avoid loperamide in bloody/febrile diarrhea and in children
- Food poisoning within 1–6 h of eating (Staph aureus, Bacillus cereus "fried rice") – supportive
- Persistent vomiting: check glucose, electrolytes, pregnancy, DKA, bowel obstruction, raised ICP, cannabis hyperemesis (hot showers; haloperidol/capsaicin)

Azithromycin – cholera / shigella / typhoid

Cholera

1 g PO single dose

Shigella

500 mg PO daily × 3 days

Typhoid (uncomplicated)

1 g day 1, then 500 mg daily × 6 days

Metronidazole – amoebic colitis / liver abscess

PO / IV

500 – 750 mg TID × 7–10 days, then luminal agent (paromomycin/diloxanide)

Esophagus, anorectum & GI devices

Esophagus

- Food bolus: able to swallow saliva → observe briefly; drooling/complete obstruction → endoscopy within 6 h 
- Button battery or magnets: emergency (see Tox: button battery)
- Boerhaave (forceful vomiting → chest pain, subcutaneous emphysema): CT with oral contrast, NPO, broad antibiotics, surgery
- Caustic ingestion: no emesis/neutralisation; airway; endoscopy 12–24 h

Anorectal

- Thrombosed external hemorrhoid < 72 h: excision under local anesthesia; later → conservative (sitz baths, stool softeners)
- Anal fissure: fibre, sitz baths, topical nifedipine/GTN
- Perianal abscess: incision and drainage; perirectal/ischiorectal or diabetics/immunosuppressed → surgeon; Fournier gangrene = emergency

Devices & post-op

- Dislodged gastrostomy (PEG) tube: tract may close within hours – insert a Foley/replacement promptly if tract > 4–6 weeks old; confirm position with contrast if new tract
- Post-op fever: wound, urine, lungs, DVT/PE, anastomotic leak (tachycardia, peritonism days 3–7)

GI sources

Laine et al., ACG clinical guideline: upper GI and ulcer bleeding 2021 · Gralnek et al., ESGE NVUGIH 2021 · de Franchis et al., Baveno VII 2022 · Biggins et al., AASLD guidance on ascites, SBP, HRS 2021 · Vilstrup et al., AASLD/EASL hepatic encephalopathy 2014 · Tenner et al., ACG acute pancreatitis 2024 · de-Madaria et al., WATERFALL trial (NEJM 2022) · Tokyo Guidelines 2018 · Di Saverio et al., WSES Jerusalem appendicitis guidelines 2020 · Peery et al., AGA diverticulitis 2021 · WHO The treatment of diarrhoea (2005) · WHO typhoid

⚠ Items pending owner review in this section (14)

1. Ulcer bleeding PPI: pantoprazole/esomeprazole 80 mg IV bolus then 40 mg IV BID (or 8 mg/hr infusion × 72 h after endoscopic therapy)

ACG 2021 accepts intermittent high-dose; pre-endoscopy PPI downstages lesions but no mortality benefit.

2. Suspected variceal bleed: octreotide 50 mcg IV bolus then 50 mcg/hr × 2–5 days (or terlipressin 2 mg IV q4h); ceftriaxone 1 g IV daily (max 7 days); endoscopy ≤ 12 h; restrictive transfusion (Hb 7–8)

Baveno VII 2022.

3. Erythromycin 250 mg IV over 30 min, 30–120 min before endoscopy (improves visualisation)

ESGE/ACG; QT caution.

4. Glasgow-Blatchford score ≤ 1 (ACG: ≤ 1; ESGE: ≤ 1) → very low risk, outpatient management possible

Thresholds differ slightly between guidelines; some use 0 only.

5. Restrictive transfusion: Hb threshold 7 g/dL (8 g/dL with cardiovascular disease); aim 7–9 g/dL

ACG 2021 / AABB 2023.

6. Acute pancreatitis fluids: moderate goal-directed lactated Ringer's – 10 mL/kg bolus only if hypovolemic, then 1.5 mL/kg/hr, reassess at 12, 24, 48 h

WATERFALL 2022 / ACG 2024; aggressive fluids harmful.

7. Intra-abdominal infection: ceftriaxone 2 g IV daily + metronidazole 500 mg IV q8h; severe/healthcare-associated: piperacillin-tazobactam 4.5 g IV q6h

IDS/SIS 2010/2017, Tokyo 2018 for biliary; local antibiogram governs.

8. Hepatic encephalopathy: lactulose 20–30 mL (13–20 g) PO/NG q1–2h until bowel movement, then titrate to 2–3 soft stools/day (or enema 300 mL in 700 mL water); add rifaximin 550 mg BID

AASLD/EASL 2014.

9. SBP: ascitic PMN ≥ 250 cells/mm³ → cefotaxime 2 g IV q8h or ceftriaxone 2 g daily; albumin 1.5 g/kg day 1 and 1 g/kg day 3 if creatinine > 1 mg/dL, BUN > 30 or bilirubin > 4

AASLD 2021.

10. Antiemetics: ondansetron 4–8 mg IV/PO (peds 0.15 mg/kg, max 4 mg IV; ODT 2 mg 8–15 kg, 4 mg 15–30 kg, 8 mg > 30 kg); metoclopramide 10 mg IV over 2 min; promethazine not < 2 y

Common ED dosing; QT prolongation with ondansetron.

11. Dehydration (WHO): Plan A home ORS; Plan B 75 mL/kg ORS over 4 h; Plan C (severe) Ringer's lactate 100 mL/kg – infants 30 mL/kg in 1 h then 70 mL/kg in 5 h; older 30 mL/kg in 30 min then 70 mL/kg in 2.5 h; zinc 20 mg/day × 10–14 d (10 mg if < 6 months)

WHO 2005 Treatment of diarrhoea; cholera adults may need more.

12. Diarrhea antibiotics: cholera – azithromycin 1 g single dose (peds 20 mg/kg) or doxycycline 300 mg; shigellosis – azithromycin 500 mg × 3 d (peds 10 mg/kg/d); typhoid (uncomplicated) – azithromycin 1 g day 1 then 500 mg daily × 6 d (peds 20 mg/kg/d), severe – ceftriaxone 2 g IV daily (peds 50–80 mg/kg); amoebic dysentery – metronidazole 500–750 mg TID × 7–10 d

WHO / DOH PH; rising azithromycin and fluoroquinolone resistance – follow RITM/ARSP data.

13. Alvarado score: ≤ 4 appendicitis unlikely; 5–6 equivocal (imaging); ≥ 7 likely (surgical review)

WSES 2020 prefers AIR/AAS scores; Alvarado less accurate in women and children.

14. Esophageal food impaction: endoscopy within 24 h, within 6 h if complete obstruction (drooling) or sharp object/battery; glucagon not recommended

ESGE 2016 / ASGE 2011.

S10 • Renal & Genitourinary Disorders Merged

Uses weight, age, sex and height from the top of the page.

AKI staging – KDIGO 2012

Enter baseline + current creatinine and/or urine volume over hours (uses weight above).

Stage	Serum creatinine	Urine output
1	1.5–1.9 × baseline (within 7 d) <i>or</i> $\uparrow \geq 0.3$ mg/dL (≥ 26.5 μ mol/L) within 48 h	< 0.5 mL/kg/hr for 6–12 h △ VERIFY
2	2.0–2.9 × baseline	< 0.5 mL/kg/hr for ≥ 12 h △ VERIFY
3	≥ 3.0 × baseline <i>or</i> ≥ 4.0 mg/dL (353.6 μ mol/L) <i>or</i> RRT started <i>or</i> < 18 y: eGFR < 35	< 0.3 mL/kg/hr ≥ 24 h <i>or</i> anuria ≥ 12 h △ VERIFY

- Unknown baseline: use lowest value in last 3 months / repeat in 24–48 h; do not back-calculate in CKD
- Look for cause: volume, obstruction (bladder scan / ultrasound), sepsis, nephrotoxins, rhabdo (CK), GN (urine dip blood/protein)
- Stop/hold nephrotoxins and “sick-day” drugs (SADMANS: SGLT2i, ACEi, diuretics, metformin, ARB, NSAIDs, sulfonylureas)
- Creatinine-based CrCl/eGFR are **not valid** while creatinine is changing – assume GFR is lower than calculated in rising AKI

Creatinine clearance – Cockcroft–Gault

Enter age, weight, sex, creatinine

- **eGFR vs CrCl:** eGFR (CKD-EPI 2021) is indexed to 1.73 m^2 and is used for CKD staging. Most drug labels’ renal tables were built on Cockcroft–Gault CrCl (actual weight) – use CrCl for narrow-therapeutic-index drugs and DOACs. KDIGO 2024 accepts non-indexed eGFR (mL/min) for most other drugs
- Neither estimate is valid in AKI (non-steady state), extremes of muscle mass, amputation, pregnancy or dialysis – use clinical judgment and pharmacist
- Metformin and contrast thresholds use eGFR

Renal dose adjustments △ VERIFY

Enter age, weight, sex and creatinine (CrCl card) to highlight the matching band. Adult doses. **Every dose here is △ VERIFY – check the local formulary and pharmacist.** Give a full first (loading) dose of most antibiotics in sepsis; adjust maintenance.

Antibiotics

Piperacillin-tazobactam △ VERIFY

Usual: 4.5 g IV q6h (first dose always full, regardless of renal function)

> 40	No change
20–40	3.375 g q6h
< 20	2.25 g q6h
HD	2.25 g q8h + 0.75 g after each HD

For 3.375 g q6h regimen: 20–40 → 2.25 g q6h; < 20 → 2.25 g q8h; HD 2.25 g q12h. Vancomycin + pip-tazo: higher AKI signal than vancomycin + cefepime.

Cefepime △ VERIFY

Usual: 2 g IV q8h (severe / febrile neutropenia / Pseudomonas)

> 60	No change
30–60	2 g q12h
11–29	2 g q24h
< 11	1 g q24h
HD	1 g q24h (or 1 g day 1 then 500 mg q24h), after HD

Unadjusted dosing → encephalopathy, myoclonus, non-convulsive status. Consider EEG if confused on cefepime.

Meropenem △ VERIFY

Usual: 1 g IV q8h (2 g q8h for CNS infection)

> 50	No change
26–50	1 g q12h
10–25	500 mg q12h
< 10	500 mg q24h
HD	500 mg q24h, after HD

Vancomycin △ VERIFY

Usual: Loading 20–35 mg/kg actual body weight, max 3 g (25–30 mg/kg common in sepsis), regardless of renal function

> 50	15–20 mg/kg q8–12h, then AUC-guided
20–49	15–20 mg/kg q24h, then level/AUC-guided
< 20	Loading dose, then redose by levels
HD	Load 20–25 mg/kg, then 500–1,000 mg (or 5–10 mg/kg) after each HD; pre-HD levels

Target AUC₂₄ 400–600 mg·h/L (Bayesian software or 2 levels) for serious MRSA; trough-only targeting (15–20) no longer recommended. Infuse ≤ 1 g/hr (red-man / vancomycin infusion reaction). Pharmacy dosing service.

Trimethoprim-sulfamethoxazole △ VERIFY

Usual: Dose by TMP: 8–12 mg/kg/day (PCP 15–20 mg/kg/day) divided q6–12h; UTI 1 DS tab BID

> 30	No change
15–30	50% of usual dose
< 15	Not recommended (pharmacist if unavoidable)

Raises creatinine (blocks tubular secretion) without lowering true GFR. Hyperkalemia (amiloride-like) – avoid with ACEi/ARB/spironolactone where possible.

Acyclovir IV (A VERIFY)

Usual: 5–10 mg/kg IV q8h (HSV encephalitis 10 mg/kg)

> 50	Same dose q8h
25–50	Same dose q12h
10–25	Same dose q24h
< 10	50% dose q24h
HD	50% dose q24h, after HD

Hydrate; infuse over 1 h (crystal nephropathy). Neurotoxicity (confusion, tremor) in renal failure – dialyzable.

Levofloxacin (A VERIFY)

Usual: 750 mg PO/IV q24h

≥ 50	No change
20–49	750 mg q48h
10–19	750 mg once, then 500 mg q48h
HD	750 mg once, then 500 mg q48h (not removed by HD)

Ciprofloxacin (A VERIFY)

Usual: 500–750 mg PO q12h / 400 mg IV q8–12h

> 50	No change
30–50	PO 250–500 mg q12h
< 30	PO 250–500 mg q18h · IV 200–400 mg q18–24h
HD	PO 250–500 mg q24h, after HD

Moxifloxacin: no renal adjustment. All FQs: QT, tendinopathy, dysglycemia, delirium in elderly.

Aminoglycosides (extended-interval) (A VERIFY)

Usual: Gentamicin/tobramycin 5–7 mg/kg, amikacin 15–20 mg/kg (dosing weight: IBW, or adjusted BW if > 120% IBW)

≥ 60	q24h
40–59	q36h
20–39	q48h
< 20	Single dose, redose by level
HD	Single dose, redose by pre-HD level

Concentration-dependent: keep the mg/kg dose, extend the interval. A single ED dose is usually acceptable if indicated; avoid repeated dosing in AKI if alternatives exist. Levels (e.g. Hartford nomogram / 6–14 h level) with pharmacy.

Anticoagulants**Enoxaparin** (A VERIFY)

Usual: Treatment 1 mg/kg SC q12h · prophylaxis 40 mg SC daily

≥ 30	No change
15–29	Treatment 1 mg/kg q24h · prophylaxis 30 mg daily
< 15	Avoid – use UFH
HD	Avoid – use UFH

Anti-Xa monitoring in severe CKD, weight < 45 or > 150 kg, pregnancy.

Unfractionated heparin (preferred in severe CKD) (A VERIFY)

Usual: VTE: 80 U/kg bolus + 18 U/kg/hr · ACS: 60 U/kg (max 4,000 U) + 12 U/kg/hr (max 1,000 U/hr)

Any No renal adjustment – aPTT / anti-Xa nomogram

Preferred when CrCl < 30, on dialysis, high bleeding risk, or procedure/thrombolysis likely (short half-life, protamine-reversible).

Apixaban (A VERIFY)

Usual: AF 5 mg BID (2.5 mg BID if ≥ 2 of: age ≥ 80, weight ≤ 60 kg, creatinine ≥ 1.5 mg/dL / 133 μmol/L) · VTE 10 mg BID × 7 d → 5 mg BID

≥ 25	Dose by AF criteria; VTE no adjustment
15–24	AF per criteria (EU: 2.5 mg BID if CrCl 15–29); VTE limited data
< 15	EU: not recommended · US: allowed incl. HD (5 mg BID; 2.5 mg if age ≥ 80 or ≤ 60 kg)

Rivaroxaban (A VERIFY)

Usual: AF 20 mg daily with evening meal · VTE 15 mg BID × 21 d → 20 mg daily

> 50	No change
15–50	AF 15 mg daily · VTE: caution < 30
< 15	Avoid

Edoxaban (A VERIFY)

Usual: AF 60 mg daily · VTE 60 mg daily after ≥ 5 d parenteral

> 95	AF: US label – do not use (lower efficacy)
51–95	No change
15–50	30 mg daily (VTE also if ≤ 60 kg / P-gp inhibitor)
< 15	Not recommended

Dabigatran (≈ 80% renal) (A VERIFY)

Usual: AF 150 mg BID · VTE 150 mg BID after 5 d parenteral

> 50	No change (EU 110 mg BID if age ≥ 80)
30–50	No change (EU consider 110 mg BID)
15–29	US AF: 75 mg BID · EU: contraindicated · VTE: avoid
< 15	Avoid
HD	Avoid (dialyzable; reversal idarucizumab 5 g IV)

Other ED drugs

Gabapentin ⚠️ VERIFY

Usual: 900–3,600 mg/day in 3 doses

≥ 60	No change
30–59	400–1,400 mg/day in 2 doses
15–29	200–700 mg/day once daily
< 15	100–300 mg/day once daily
HD	Daily dose + 125–350 mg after each HD

Accumulation → sedation, myoclonus, encephalopathy, falls (dialyzable).

Pregabalin ⚠️ VERIFY

Usual: 150–600 mg/day in 2–3 doses

≥ 60	No change
30–60	75–300 mg/day in 2–3 doses
15–29	25–150 mg/day in 1–2 doses
< 15	25–75 mg once daily
HD	Daily dose + supplement after each HD

Opioids ⚠️ VERIFY

Usual: Choose agent by renal function

≥ 30	Usual agents, reduce dose / extend interval as CrCl falls
< 30	Avoid morphine (M3G/M6G accumulate), codeine, pethidine; tramadol q12h max 200 mg/day. Prefer fentanyl; cautious low-dose hydromorphone/oxycodone
HD	Fentanyl preferred; naloxone may need repeating

Metformin (uses eGFR) ⚠️ VERIFY

Usual: Hold in AKI, sepsis/shock, hypoxia, decompensated HF

eGFR ≥ 45	Continue
eGFR 30–44	Do not start; review, max ~1,000 mg/day
eGFR < 30	Contraindicated – stop

Iodinated contrast: stop at/before contrast if eGFR 30–60, liver disease, alcohol use, HF or intra-arterial contrast; restart after 48 h if renal function stable. MALA → see dialysis (AEIOU). Band highlight uses CrCl as a proxy.

NSAIDs

Usual: Avoid in AKI, eGFR < 30, volume depletion, hyperkalemia, HF, cirrhosis

Any CKD “Triple whammy” with ACEi/ARB + diuretic. Prefer paracetamol (acetaminophen); short topical NSAID courses if needed

Iodinated contrast & nephrotoxins ⚠️ VERIFY

Usual: Do not withhold indicated emergency contrast (stroke, PE, aortic, trauma) to wait for creatinine

eGFR ≥ 30	No prophylaxis needed (ACR–NKF 2020); 30–44 individualise
eGFR < 30 / AKI	Isotonic IV fluid prophylaxis if not on dialysis and no fluid contraindication; minimise dose; avoid repeat studies within 48–72 h
HD	No urgent post-contrast dialysis needed

Avoid combinations of nephrotoxins (aminoglycosides, vancomycin + pip-tazo, NSAIDs, amphotericin, IV acyclovir); check lithium/digoxin levels in AKI.

🔥 Urgent dialysis – AEIOU

Emergency RRT when life-threatening complications are **refractory to medical therapy**. Call nephrology / ICU early (line + machine take time). No benefit from pre-emptive start without an indication (AKIKI, STARTT-AKI).

- **A – Acidosis:** severe metabolic acidosis refractory to treatment (especially if bicarbonate limited by volume/Na load)
- **E – Electrolytes:** refractory hyperkalemia (e.g. $K \geq 6.5$ or ECG changes despite medical therapy), especially oligo-anuric AKI/ESKD; also severe refractory dysnatremia, hypercalcemia, tumour lysis
- **I – Intoxication** (dialyzable: small, water-soluble, low protein binding – **SLIME**: salicylates, lithium, isopropanol (rarely needed), methanol/metformin, ethylene glycol; also valproate, theophylline, barbiturates) ⚠️ VERIFY
 - Lithium: impaired kidney function + $Li > 4.0$ mmol/L, or ↓ consciousness, seizures, dysrhythmia (any level); suggested if $Li > 5.0$
 - Salicylate: level > 7.2 mmol/L (100 mg/dL), > 6.5 mmol/L (90 mg/dL) with impaired kidney function, altered mental status, new hypoxemia, or failure of standard therapy; suggested if $pH \leq 7.20$
 - Metformin (MALA): lactate > 20 mmol/L, $pH \leq 7.0$, shock, ↓ consciousness, or failure of standard therapy
 - Toxic alcohols (methanol, ethylene glycol): severe acidosis, coma/seizures, visual signs (methanol), AKI, or high levels – with fomepizole/ethanol; toxicology guides thresholds
- **O – Overload:** fluid overload / pulmonary edema refractory to diuretics (oligo-anuric), with hypoxemia
- **U – Uremia:** pericarditis (risk of tamponade – avoid anticoagulation), encephalopathy, seizures, uremic bleeding

Hemodynamically unstable patients: CRRT (ICU) rather than intermittent HD. Discuss toxin cases with the poison center (EXTRIP).

⚡ Hyperkalemia – stepwise

Mild 5.5–5.9 · moderate 6.0–6.4 · severe ≥ 6.5 mmol/L (UK Kidney Assoc.). **ECG changes = emergency regardless of level.** The **+** More tab has the compact dose card (open).

ECG: peaked T → flattened/absent P, long PR → wide QRS → sine wave, bradycardia, VF/asystole. Normal ECG does not exclude danger.

- 1 Exclude pseudohyperkalemia (hemolysed / delayed sample) – recheck on blood gas, but **do not delay** treatment if ECG changes. Continuous cardiac monitoring if $K \geq 6.5$ or ECG changes; IV access.
- 2 **Protect the heart** (ECG changes) ⚠️ VERIFY: **calcium gluconate 10% 30 mL** IV over 5–10 min, or **calcium chloride 10% 10 mL** (large/central vein). Repeat ECG after 5–10 min; repeat dose if changes persist. Effect lasts 30–60 min only. On digoxin: give slowly (over 20 min).

- 3 **Shift K⁺ in** ($K \geq 6.0$) (△ VERIFY): **10 units soluble insulin + 25 g glucose** (e.g. 50 mL 50%, 125 mL 20% or 250 mL 10%) IV over 15 min. Lowers K ~0.6–1.0 mmol/L in 30–60 min.
Glucose monitoring: check capillary glucose at 0, 15, 30, 60, 90, 120 min then hourly to 6 h. If pre-treatment glucose < 7 mmol/L (126 mg/dL): start 10% glucose 50 mL/hr for 5 h. Higher hypoglycemia risk: no diabetes, low baseline glucose, eGFR < 30, low weight.
- 4 **Salbutamol** 10–20 mg nebulised ($K \geq 6.5$) as an adjunct to insulin – never alone (△ VERIFY). Caution in tachyarrhythmia / ischemic heart disease (use 10 mg).
- 5 **Sodium bicarbonate** – not routine; consider only with significant metabolic acidosis, with senior/nephrology input (sodium and volume load) (△ VERIFY).
- 6 **Remove K⁺** (△ VERIFY): sodium zirconium cyclosilicate 10 g TDS up to 72 h ($K \geq 6.0$), or patiomer 8.4 g daily (slower onset); calcium/sodium polystyrene (e.g. Ca resonium 15 g TDS-QDS) if others unavailable (slow; GI necrosis risk). Loop diuretic if volume-replete and passing urine.
- 7 **Dialysis** if refractory, ESKD, oligo-anuric AKI, or K rising despite therapy – call nephrology early (see AEIOU).
- 8 **Monitor K** at 1, 2, 4, 6 and 24 h (rebound after shift therapy wears off at 4–6 h). Stop K-raising drugs (ACEi/ARB, MRA, K supplements, NSAIDs, trimethoprim, heparin); treat cause.

- 4 **Watch for aquaresis:** urine output > 100 mL/hr after volume repletion, stopping thiazide/desmopressin, or steroid replacement → Na can rise fast. Check Na every 2–4 h while correcting. (△ VERIFY)
- 5 **Overcorrection rescue** (△ VERIFY): stop saline; give electrolyte-free water – D5W 10 mL/kg over 1 h or 3 mL/kg/hr – plus **desmopressin 2 µg IV/SC** (not more often than q6–8h); re-lower Na below the 24-h limit; nephrology/ICU.

Correction tracker

Enter starting Na, current Na and hours since treatment start.

💧 Hypernatremia – free water deficit

Enter serum Na > 145 mmol/L.

- **Shock / hypovolemia first:** isotonic crystalloid until perfused, then free water
- **Chronic (> 48 h) or unknown:** lower by ≤ 10 –12 mmol/L per 24 h ($\approx \leq 0.5$ mmol/L/hr) – cerebral edema risk (△ VERIFY)
- **Acute (< 48 h, e.g. salt ingestion):** faster correction (~1 mmol/L/hr) toward 145 is acceptable (△ VERIFY)
- Recheck Na every 2–4 h initially; TBW fraction: 0.6 child/young man, 0.5 young woman/older man, 0.45 older woman (≥ 65 y)
- Consider diabetes insipidus (dilute urine, high output), osmotic diuresis (glucose, urea)

⚡ Hyperkalemia

Mild 5.5–5.9 · moderate 6.0–6.4 · severe ≥ 6.5 mmol/L (UK Kidney Assoc. 2023). ECG changes = emergency regardless of level.

Hyperkalemia – stabilize, shift, remove (△ VERIFY)

1 · **Calcium gluconate 10%** (if ECG changes / $K \geq 6.5$)
30 mL (3 g) IV over 5–10 min

Repeat if ECG changes persist (up to 3×). Or CaCl₂ 10% 10 mL via large/central vein.

2 · **Insulin + glucose** (△ VERIFY)

10 units regular insulin IV + 25 g glucose (50 mL D50)

If pre-treatment glucose < 126 mg/dL (7 mmol/L): add D10 50 mL/hr × 5 h. Glucose hourly × 6 h. Some use 5 U in renal failure.

3 · **Salbutamol**

10 – 20 mg nebulized

Adjunct to insulin; ~1/3 of ESKD patients do not respond.

4 · **Sodium bicarbonate 8.4%** (△ VERIFY)

50 – 100 mL (50–100 mEq) IV if metabolic acidosis

5 · **Remove K⁺**

Furosemide 40–80 mg IV (if volume-replete/urinating); binder (SZC 10 g TID, or Na polystyrene 15–30 g); dialysis if refractory/ESKD

Stepwise version (ECG, glucose monitoring, binders, dialysis):
stepwise card ↑

📖 Renal sources

KDIGO 2012 AKI guideline · KDIGO 2024 CKD guideline (drug dosing / eGFR) · Cockcroft & Gault 1976 · Spasovski et al., European clinical practice guideline on hyponatraemia 2014 (ESE/ESICM/ERA-EDTA) · Verbalis et al., US expert panel recommendations on hyponatremia 2013 · UK Kidney Association clinical practice guideline: treatment of acute hyperkalaemia in adults 2020 (2023 update) · Rybak et al., ASHP/IDSA/PIDS/SIDP vancomycin consensus 2020 · ACR-NKF consensus on IV iodinated contrast in kidney disease 2020 · EXTRIP workgroup recommendations (lithium 2015, salicylates 2015, metformin 2015, methanol 2015) · FDA product labeling (renal dosing tables). Renal dose ranges vary by reference (Sanford, Lexicomp, BNF, local antibiograms) – **always check the local formulary and pharmacist.**

▼ Hypokalemia

Mild 3.0–3.4 · moderate 2.5–2.9 · severe < 2.5 mmol/L or symptomatic (arrhythmia, U waves / ST depression / long QU, weakness, paralysis, rhabdomyolysis).

- 1 **Oral first** if $K \geq 3.0$ (or 2.5–3.0 and asymptomatic) and gut works: KCl 20–40 mmol PO 2–3×/day (△ VERIFY).
- 2 **IV potassium** – premixed bags only, never IV push or undiluted concentrate (△ VERIFY):
· **Peripheral:** ≤ 10 mmol/hr, concentration usually ≤ 40 mmol/L (premixed 10–20 mmol/100 mL bags via pump only per local policy)
· **Central line + continuous ECG:** up to 20 mmol/hr; faster only for life-threatening arrhythmia/paralysis in ICU
· Roughly 10 mmol raises serum K ~0.1 mmol/L (very variable – total deficit often 200–400 mmol)
- 3 **Check and replace magnesium** – hypokalemia is refractory until Mg is corrected: MgSO₄ 2 g (8 mmol) IV over 1–2 h (over 10–15 min if arrhythmia) (△ VERIFY).
- 4 Recheck K every 2–4 h during IV replacement; avoid glucose-containing carriers initially (insulin shifts K in). **Reduce dose/rate in renal impairment** (risk of overshoot).
- 5 Find the cause: GI/renal losses, diuretics, β -agonists/insulin, DKA treatment, refeeding, hypothermia, periodic paralysis.

💧 Hyponatremia

Treat by **symptoms**, not the number alone. Exclude hyperglycemia (corrected Na) and pseudohyponatremia. Severe symptoms: vomiting, cardiorespiratory distress, abnormal deep somnolence, seizures, GCS ≤ 8 . Moderate: nausea, confusion, headache.

- 1 **Severe symptoms: 3% NaCl 150 mL IV over 20 min** (European 2014) or **100 mL over 10 min** (US 2013) (△ VERIFY). Recheck Na after each bolus; repeat up to 3 boluses total until symptoms improve or Na has risen **5 mmol/L**. Then stop hypertonic saline and treat the cause.
- 2 **Moderately severe symptoms:** single 3% bolus (European: 150 mL over 20 min), aim $\uparrow$ ~5 mmol/L/24 h; senior/ICU input (△ VERIFY).
- 3 **Correction limits** (△ VERIFY): ≤ 10 mmol/L in the first 24 h and ≤ 8 mmol/L in every 24 h thereafter (≈ 18 mmol/L / 48 h). **High ODS risk** (Na ≤ 105 , hypokalemia, alcohol use disorder, malnutrition, advanced liver disease): aim 4–6, **never > 8 mmol/L in any 24 h**.

⚠ Items pending owner review in this section (35)

1. Hyperkalemia insulin 10 U + 25 g glucose (5 U in renal failure)

Insulin dose and glucose co-administration vary; hypoglycemia risk.

2. Sodium bicarbonate for hyperkalemia (adult 50–100 mEq if acidotic)

Evidence limited; local practice varies.

3. Pediatric hyperkalemia (Ca gluconate 60 mg/kg max 3 g; insulin 0.1 U/kg + dextrose 0.5 g/kg; salbutamol 2.5/5 mg; bicarbonate 1 mEq/kg)

Pediatric protocols vary.

4. All renal dose adjustments in the Renal tab (CrCl bands from FDA labels / common references)

Bands and doses differ between Sanford, Lexicomp, BNF and local formularies; check the local formulary and pharmacist before prescribing.

5. Cockcroft–Gault weight choice (actual vs ideal vs adjusted body weight)

Drug labels mostly used actual body weight; many pharmacists use IBW, or adjusted BW if > 120% IBW. Not valid when creatinine is unstable (AKI).

6. Piperacillin–tazobactam renal dosing (CrCl 20–40: 3.375 g q6h; < 20: 2.25 g q6h; HD 2.25 g q8h + 0.75 g post-HD, for the 4.5 g q6h regimen)

Label-based; extended-infusion regimens have different tables. First dose full.

7. Cefepime renal dosing (2 g q8h regimen: CrCl 30–60 2 g q12h; 11–29 2 g q24h; < 11 1 g q24h; HD 1 g q24h)

Neurotoxicity (encephalopathy, NCSE) if not adjusted; tables differ for 1–2 g q12h regimens.

8. Meropenem renal dosing (1 g q8h: CrCl 26–50 1 g q12h; 10–25 500 mg q12h; < 10 / HD 500 mg q24h)

Meningitis (2 g q8h) scales proportionally; check local protocol.

9. Vancomycin load 20–35 mg/kg actual body weight (max 3 g), then AUC/level-guided maintenance; HD 500–1,000 mg (or 5–10 mg/kg) after each session

ASHP/IDSA 2020 consensus; AUC 400–600 target for serious MRSA. Use pharmacy dosing service / local nomogram.

10. TMP-SMX: 50% of usual dose if CrCl 15–30; not recommended < 15

Label-based; some references allow reduced dose with monitoring in < 15 / HD. Hyperkalemia and creatinine rise.

11. IV acyclovir 5–10 mg/kg q8h; CrCl 25–50 q12h; 10–25 q24h; < 10 / HD 50% dose q24h

Dosing weight (IBW vs actual) varies; hydrate; neurotoxicity in renal failure.

12. Levofloxacin 750 mg q24h → CrCl 20–49 750 mg q48h; 10–19 / HD 750 mg then 500 mg q48h

Label-based.

13. Ciprofloxacin: CrCl 30–50 PO 250–500 mg q12h; < 30 PO 250–500 mg q18h / IV 200–400 mg q18–24h; HD 250–500 mg PO q24h after HD

Label-based; varies between references.

14. Aminoglycoside extended-interval dosing (gentamicin/tobramycin 5–7 mg/kg, amikacin 15–20 mg/kg; interval by CrCl / levels)

Hartford-type nomogram intervals; dosing weight (IBW/adjusted) and level targets are institution-specific.

15. Enoxaparin: treatment 1 mg/kg q12h → q24h if CrCl < 30; prophylaxis 40 mg → 30 mg daily; avoid in dialysis

Anti-Xa monitoring in severe CKD, obesity, low weight; local VTE protocol.

16. UFH: VTE 80 U/kg bolus + 18 U/kg/hr; ACS 60 U/kg (max 4,000 U) + 12 U/kg/hr (max 1,000 U/hr)

Titrate to aPTT/anti-Xa by local nomogram.

17. DOAC renal dosing (apixaban, rivaroxaban, edoxaban, dabigatran – AF and VTE)

US (FDA) and EU (EMA) labels differ, especially for CrCl < 30 and dialysis. Use CG with actual body weight per labels.

18. Gabapentin/pregabalin daily dose ranges by CrCl and post-HD supplements

Label-based ranges; titrate to effect and sedation.

19. Opioids in renal failure (avoid morphine/codeine/pethidine; tramadol CrCl < 30 max 200 mg/day q12h)

Choice (fentanyl vs hydromorphone vs oxycodone) varies by local palliative/renal guidance.

20. Metformin eGFR thresholds (30–44 do not start / reduce; < 30 contraindicated) and contrast hold

FDA 2016 / local diabetes protocol; uses eGFR, not CrCl.

21. Contrast-associated AKI prophylaxis (isotonic IV fluid if eGFR < 30 not on dialysis, or AKI)

ACR–NKF 2020 consensus; fluid regimen and thresholds vary locally.

22. EXTRIP dialysis thresholds (lithium, salicylate, metformin, methanol)

Summarised from EXTRIP; always discuss with toxicology/poison center and nephrology.

23. Adult hyperkalemia stepwise doses (Ca gluconate 30 mL 10% / CaCl₂ 10 mL 10%; salbutamol 10–20 mg; SZC 10 g TDS; patiromer 8.4 g; Ca resonium 15 g)

UK Kidney Association 2020/2023; binder availability and dosing vary.

24. IV potassium limits (peripheral ≤ 10 mmol/hr; ≤ 40 mmol/L; central with ECG monitoring ≤ 20 mmol/hr; oral 20–40 mmol 2–3×/day)

Concentration and rate limits are institution-specific; premixed bags only.

25. Pediatric IV potassium 0.5–1 mmol/kg (max 40 mmol) at ≤ 0.5 mmol/kg/hr

Pediatric protocol dependent.

26. Magnesium sulfate 2 g (8 mmol) IV for hypomagnesemia with hypokalemia

Rate (15 min if arrhythmia vs 1–2 h) and repeat dosing vary.

27. 3% NaCl for severe symptomatic hyponatremia: 150 mL over 20 min (Europe 2014) or 100 mL over 10 min (US 2013), up to 3 boluses; peds 2 mL/kg (max 100 mL)

Bolus volume, repeat count and pediatric dose differ between guidelines.

28. Sodium correction limits (≤ 10 mmol/L first 24 h, ≤ 8 mmol/L/24 h thereafter; high ODS risk ≤ 8 mmol/L in any 24 h)

European 2014 vs US 2013 numbers differ slightly; local protocol governs.

29. Overcorrection rescue: D5W 10 mL/kg over 1 h (or 3 mL/kg/hr) + desmopressin 2 µg IV/SC (not more often than q6–8h)

Expert-opinion regimens; involve nephrology/ICU.

30. Hypernatremia correction rate (chronic ≤ 10–12 mmol/L/day; acute < 48 h up to ~1 mmol/L/hr; peds ≤ 0.5 mmol/L/hr) and TBW fractions

Formula estimates (Adrogué–Madias) ignore ongoing losses – recheck Na q4–6h.

31. Pediatric salbutamol neb weight cut-off is inconsistent in this app: 20 kg (Tox/Pulm asthma) vs 25 kg (hyperkalemia, More tab)

Existing values kept as-is; owner to pick one cut-off (or an age-based rule) and harmonise both cards.

32. Pediatric doses are capped at an adult dose (engine-wide). Caps added where v1 had none: procainamide 1 g, bicarbonate 50 mEq, ketamine/propofol 200 mg, etomidate 30 mg, rocuronium 120 mg, succinylcholine 150 mg, fentanyl 100 mcg, phenobarbital 1.5 g, mannitol 100 g, fomepizole 1.5 g, methylene blue 200 mg, atropine (OP) 2 mg

For drugs dosed per kg in adults too, the cap is the adult dose at a ~100 kg reference weight – a policy choice for owner sign-off, not a guideline value.

33. Hyperkalemia quick card doses (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

34. KDIGO staging thresholds (creatinine, urine output mL/kg/hr)

Inline values from v1 / merged card, previously unflagged.

35. Hyponatremia aquaresis threshold (urine output > 100 mL/hr) and related inline numbers

Inline values from v1 / merged card, previously unflagged.

Maternal weight = weight at the top (Adult mode). Newborn doses use the separate newborn weight in the delivery card.

Early pregnancy: bleeding & ectopic

- 1 **Every person of reproductive age with abdominal pain, bleeding, syncope or shock: pregnancy test.** Positive test + no intrauterine pregnancy (IUP) on ultrasound = ectopic until proven otherwise.
- 2 **Unstable** (hypotension, peritonism, free fluid in Morison's pouch on FAST): 2 large-bore IVs, crossmatch, activate MTP, **emergency OB/surgery** – do not wait for formal imaging or hCG.
- 3 **Stable:** transvaginal ultrasound + quantitative β -hCG; no IUP with hCG above the local discriminatory zone (~1,500–3,500 IU/L) is highly suspicious; below it → repeat hCG in 48 h (expected rise $\geq$ ~35–50%) with clear return precautions.
- 4 **Rh(D)-negative** with bleeding, ectopic, miscarriage or trauma: anti-D immunoglobulin 300 mcg IM within 72 h ($\geq$ 12 weeks; may be omitted < 12 weeks per ACOG 2024) **△ VERIFY**
- 5 Heterotopic pregnancy is rare but more common with IVF. Methotrexate (50 mg/m² IM single-dose protocol) is an OB decision for stable, selected patients **△ VERIFY**.

Threatened/incomplete miscarriage: analgesia, anti-D if indicated, OB follow-up; heavy bleeding with open os → remove products from the os with ring forceps, uterotonic, OB.

Pre-eclampsia & eclampsia

Pre-eclampsia: new BP $\geq$ 140/90 after 20 weeks (or up to 6 weeks postpartum) + proteinuria or organ involvement (platelets < 100 × 10⁹/L, creatinine > 1.1 mg/dL or doubled, transaminases $\geq$ 2×, pulmonary edema, new headache or visual symptoms). **Severe features:** BP $\geq$ 160/110, organ involvement, persistent headache/visual change, RUQ/epigastric pain. **HELLP:** hemolysis, elevated liver enzymes, low platelets – BP may be only mildly raised. **Eclampsia:** seizure with pre-eclampsia (can occur postpartum, without prior hypertension).

- 1 **Eclamptic seizure:** left lateral, airway, O₂; **magnesium sulfate is the anticonvulsant** (benzodiazepine only if Mg delayed or seizures persist); then control BP; deliver once stable.

Magnesium sulfate – eclampsia **△ VERIFY**

Loading

4 g IV over 15–20 min

Zuspan; Pritchard adds 10 g IM (5 g each buttock). 4 g = 8 mL of 50%; dilute to 20 mL (20%). Some protocols load 6 g.

Maintenance

1 – 2 g/hr IV (or 5 g IM q4h)

Hold if reflexes absent, RR < 12, urine < 25–30 mL/hr. Antidote: Ca gluconate 1 g IV.

Magnesium toxicity → calcium gluconate **△ VERIFY**

IV over 10 min

1 g = 10 mL of 10%

Stop Mg. Loss of patellar reflexes → RR depression → arrest as levels rise.

Severe BP – pick ONE first-line agent **△ VERIFY**

Labetalol IV **△ VERIFY**

Severe BP

20 mg IV over 2 min

Recheck BP in 10 min → 40 mg → (10 min) 80 mg → (10 min) switch to hydralazine. Max cumulative 300 mg. Avoid: asthma, HF, HR < 60.

Hydralazine IV **△ VERIFY**

Severe BP

5 – 10 mg IV over 2 min

Recheck in 20 min → 10 mg → (20 min) switch to labetalol.

Nifedipine immediate-release PO **△ VERIFY**

Severe BP (no IV access or as first line)

10 mg PO

Recheck in 20 min → 20 mg → (20 min) 20 mg → switch to labetalol. Swallow whole.

- Target ~140–150 / 90–100 mmHg – avoid precipitous drops (placental perfusion); fetal monitoring if viable
- Avoid ACE inhibitors, ARBs, nitroprusside (except refractory, briefly); fluid-restrict (~80 mL/hr total) – pulmonary edema risk **△ VERIFY**
- Definitive treatment is delivery – urgent OB; transfer to a facility with OB/neonatal capability once stabilised

Postpartum hemorrhage

Definition **△ VERIFY**: $\geq$ 500 mL after vaginal birth (WHO) or $\geq$ 1,000 mL / any blood loss with signs of hypovolemia (ACOG). Visual estimates underestimate – weigh/measure. Causes: **4 Ts** – Tone ($\approx$ 70%), Trauma (tears, inversion, rupture), Tissue (retained placenta), Thrombin (coagulopathy, HELLP, AFE).

- 1 **Call for help**, two large-bore IVs, bloods + crossmatch, warm fluids, O₂, empty bladder (catheter), keep warm. **E-MOTIVE** bundle together: uterine **M**assage, **O**xytocics, **T**ranexamic acid, **I**V fluids, **E**xamination & escalation.
- 2 **Bimanual uterine compression** / aortic compression while drugs act; inspect for tears and retained placenta; check the placenta is complete.

Oxytocin – first-line uterotonic **△ VERIFY**

Bolus

10 IU IM, or slow IV over 1–2 min

Infusion

20 – 40 IU in 1,000 mL crystalloid, titrate to uterine tone

Water intoxication with prolonged high-dose infusion.

Tranexamic acid – PPH **△ VERIFY**

Within 3 h of birth

1 g IV over 10 min

= 10 mL of 100 mg/mL. Second 1 g if bleeding continues after 30 min or restarts within 24 h.

Ergometrine / methylergonovine **△ VERIFY**

IM (or slow IV)

0.2 mg

Contraindicated: hypertension, pre-eclampsia, cardiac disease.

Carboprost (15-methyl PGF₂ α) **△ VERIFY**

Deep IM (or intramyometrial by OB)

250 mcg q15–90 min, max 8 doses (2 mg)

Avoid in asthma.

Misoprostol **△ VERIFY**

Sublingual

800 mcg

If oxytocin unavailable; shivering/fever common.

- Refractory: intrauterine balloon tamponade, non-pneumatic anti-shock garment (NASG) for transfer, OR (compression sutures, embolisation, hysterectomy)
- Blood: early transfusion, **fibrinogen** (target > 2 g/L) and 1:1:1 MTP, calcium – see Trauma: MTP (△ VERIFY)
- Uterine inversion: stop oxytocin, immediately attempt manual replacement, then uterotonics; shock may be out of proportion to blood loss

Maternal cardiac arrest

Code timer not running. – this card then shows the 4-minute hysterotomy countdown.

- 1 **High-quality CPR** – standard hand position and ACLS drugs/energies (do not withhold defibrillation); call OB + neonatal team at once (△ VERIFY)
- 2 **Manual left uterine displacement** (push uterus up and left) if fundus at/above umbilicus (≥ 20 weeks); remove fetal monitors
- 3 Airway early by the most experienced operator (smaller ETT, aspiration risk); O₂ 100%; IV access **above the diaphragm**
- 4 **Resuscitative hysterotomy** (perimortem cesarean) if no ROSC by **4 minutes** – aim to deliver by ~5 min; do it at the bedside, do not move the patient
- 5 Pregnancy-specific causes – “**BEAU-CHOPS**”: Bleeding/DIC, Embolism (PE, amniotic fluid, air), Anesthetic complications (high block, LAST), Uterine atony, Cardiac disease (MI, dissection, peripartum cardiomyopathy), Hypertension/eclampsia, Other (Hs & Ts), Placenta abruptio/previa, Sepsis
- 6 On magnesium: stop it; calcium chloride 10% 10 mL or calcium gluconate 10% 30 mL IV (△ VERIFY). LAST → lipid emulsion (see Tox antidotes).

Arrest drug doses: Resus: arrest drugs · ACLS steps:

Emergency delivery & newborn

Imminent delivery in the ED

- Call OB + neonatal help; warm room; prepare newborn resuscitation area (warmer, towels, bag-mask with term & preterm masks, suction, cord clamps); oxytocin ready
- Control the head (gentle flexion), check for nuchal cord (slip over or somersault), deliver anterior then posterior shoulder with gentle traction; dry and place skin-to-skin
- Delay cord clamping ≥ 30–60 s if mother and baby are stable; oxytocin 10 IU IM after the baby is delivered (active management of the third stage) (△ VERIFY)

Shoulder dystocia (head delivered, “turtle sign”)

- 1 Call for help; note the time; **no fundal pressure**; stop maternal pushing while repositioning
- 2 **McRoberts** (hyperflex hips to abdomen) + **suprapubic pressure** (behind anterior shoulder, toward fetal face) – resolves most cases
- 3 Internal maneuvers: **deliver the posterior arm**, rotational maneuvers (Rubin II / Woods screw); consider episiotomy for access; then **all-fours** (Gaskin) and repeat
- 4 Head-to-body interval matters – neonatal team ready for resuscitation

Cord prolapse

- **Elevate the presenting part off the cord** (△ VERIFY) (**gloved hand in vagina or fill the bladder with 500–750 mL via catheter**). **knee-chest or exaggerated Sims position; minimal cord handling; emergency cesarean**

Breech

- “Hands off the breech” until the umbilicus delivers; support the body, keep the back anterior; flex the head (Mauriceau maneuver) – call OB

Newborn resuscitation (AHA/AAP 2020 · ILCOR 2023) (△ VERIFY)

- 1 **Term? Tone? Breathing/crying?** Yes → skin-to-skin, dry, observe. No → warm, position airway, clear secretions only if obstructing, dry, stimulate (~30 s)
- 2 Apneic/gasping or HR < 100 → **positive-pressure ventilation within 60 s** at 40–60 breaths/min, start in air (21%) for ≥ 35 weeks (21–30% if preterm); SpO₂ on right hand; ventilation corrective steps (MR SOPA) if chest not rising
- 3 HR < 60 after 30 s of **effective** ventilation → intubate/SGA, **chest compressions 3:1** (90 compressions + 30 breaths/min), FiO₂ 100%, UVC/IO
- 4 HR still < 60 → epinephrine; suspected blood loss → volume; check glucose

Newborn – epinephrine (0.1 mg/mL) (△ VERIFY)

IV / IO (preferred)

Enter newborn weight

0.01–0.03 mg/kg (0.1–0.3 mL/kg of 0.1 mg/mL)

Endotracheal (while obtaining access)

Enter newborn weight

0.05–0.1 mg/kg (0.5–1 mL/kg of 0.1 mg/mL)

Newborn – volume & glucose (△ VERIFY)

NS or O-negative blood (suspected blood loss)

Enter newborn weight

10 mL/kg over 5–10 min

Dextrose 10% (hypoglycemia)

Enter newborn weight

2 mL/kg D10

Enter newborn weight (estimate: term ≈ 3 kg; 32 weeks ≈ 1.7 kg) to compute newborn doses.

Other obstetric emergencies

Antepartum hemorrhage (> 20 weeks)

- **No digital vaginal examination until placenta previa is excluded** by ultrasound. Painless bright bleeding → previa; painful, tense uterus ± concealed bleeding, fetal distress → **abruption** (coagulopathy, check fibrinogen)
- Left lateral, two IVs, crossmatch, Kleihauer-Betke + anti-D if Rh-negative (△ VERIFY), continuous fetal monitoring if viable, urgent OB

Trauma in pregnancy

- Mother first – standard ATLS with **left uterine displacement** (or 15–30° left tilt on spinal board) after ~20 weeks; O-negative blood if needed; FAST is valid; indicated imaging should not be withheld
- Normal pregnancy vitals: HR +10–20, BP lower in 2nd trimester, mild respiratory alkalosis (PaCO₂ ~30 mmHg – a “normal” 40 is respiratory failure); blood volume ↑ ~45% masks hemorrhage
- ≥ 20–24 weeks: cardiotocography for ≥ 4–6 h even after minor trauma (abruption); Rh-negative → anti-D (△ VERIFY); consider intimate partner violence

Amniotic fluid embolism

- Sudden hypoxia + hypotension/arrest + DIC during labour or soon after delivery. Supportive: airway, ventilation, RV support (norepinephrine, inotropes), early massive transfusion with fibrinogen, delivery; hysterotomy if arrest. A-OK regimen (atropine 1 mg, ondansetron 8 mg, ketorolac 30 mg IV) is case-report level only (△ VERIFY)

VTE / PE in pregnancy

- D-dimer is less useful (pregnancy-adapted YEARS); leg compression ultrasound first if leg symptoms; CTPA or V/Q per local pathway. Treat with LMWH – enoxaparin 1 mg/kg SC q12h **△ VERIFY**; no DOACs or warfarin. See Pulmonary: PE

Hyperemesis gravidarum

- Ketonuria, weight loss, dehydration, hypokalemia; **thiamine 100 mg IV before dextrose** (Wernicke); IV fluids (NS/Hartmann's with K), antiemetics (metoclopramide 10 mg, ondansetron 4–8 mg, doxylamine-pyridoxine), VTE prophylaxis if admitted **△ VERIFY**

Peripartum cardiomyopathy & other

- New heart failure in the last month of pregnancy up to 5 months postpartum: BNP, echo, treat as acute HF (see Cardiovascular: AHF); avoid ACEi/ARB antepartum
- Postpartum (≤ 6 weeks) headache, seizures or hypertension → think pre-eclampsia, cerebral venous thrombosis, PRES, stroke

- Drugs best avoided in pregnancy: ACE inhibitors/ARBs, warfarin, statins, NSAIDs after ~20 weeks (oligohydramnios/ductus), tetracyclines, valproate; most standard resuscitation drugs are used as normal

Obstetric sources

ACOG Committee Opinion 767 (2019, reaffirmed) – acute-onset severe hypertension in pregnancy · ACOG Practice Bulletin 222 (2020) – gestational hypertension & pre-eclampsia · WHO recommendations on pre-eclampsia/eclampsia (2011, updated 2018/2021) · WHO 2023 consolidated PPH recommendations & E-MOTIVE bundle · WOMAN trial / WHO 2017 TXA recommendation · ACOG Practice Bulletin 183 (PPH) · AHA 2020 maternal cardiac arrest (and 2015 AHA scientific statement) · ERC 2021 special circumstances – pregnancy · AHA/AAP 2020 neonatal resuscitation (NRP 8th ed.) & ILCOR 2023 · ACOG Practice Bulletin 193 (tubal ectopic) · ACOG 2024 clinical practice update on Rh(D) immune globulin · RCOG Green-top 42 (shoulder dystocia) & 50 (cord prolapse) · DOH Philippines / POGS maternal care protocols. Tintinalli 9e Section 11 used only as a topic map.

△ Items pending owner review in this section (20)

1. Magnesium sulfate for eclampsia (Zuspan 4 g IV + 1–2 g/hr, or Pritchard 4 g IV + 10 g IM)

Regimen choice and loading dose (4 vs 6 g) follow local OB / DOH protocol.

2. Rh(D) immune globulin: 300 mcg IM within 72 h for Rh-negative bleeding/trauma ≥ 12 weeks; < 12 weeks may be omitted (ACOG 2024) or 50–120 mcg if given

Product sizes (50/120/300 mcg) and local policy (DOH/POGS) differ; Kleihauer–Betke for large fetomaternal hemorrhage.

3. Methotrexate for stable ectopic (50 mg/m² IM single-dose protocol) – OB decision only

Eligibility (hCG level, no cardiac activity, mass size) per ACOG PB 193; listed for awareness, not ED initiation.

4. Severe-range BP in pregnancy (≥ 160 systolic or ≥ 110 diastolic, persisting ≥ 15 min) → treat within 30–60 min; target ~140–150/90–100

ACOG CO 767; WHO/DOH targets similar; confirm local OB protocol.

5. Labetalol IV for severe pregnancy hypertension: 20 mg → 40 mg → 80 mg at 10-min intervals (max cumulative 300 mg), then switch agent

ACOG CO 767 algorithm; avoid in asthma, heart failure, bradycardia.

6. Hydralazine IV 5–10 mg over 2 min, repeat 10 mg at 20 min, then switch agent

ACOG CO 767; maternal hypotension → fetal distress; availability varies.

7. Nifedipine immediate-release 10 mg PO, then 20 mg at 20 min, then 20 mg at 20 min, then switch agent

ACOG CO 767; capsules swallowed, not sublingual.

8. Magnesium toxicity: stop infusion; calcium gluconate 10% 10 mL (1 g) IV over 10 min; monitor reflexes, RR ≥ 12, urine ≥ 25–30 mL/hr

Standard practice; reduce maintenance in renal impairment.

9. PPH definition/triggers: ≥ 500 mL vaginal (WHO) or ≥ 1,000 mL / any loss with instability (ACOG); obstetric shock index ≥ 0.9–1.0 = escalate

Definitions differ; E-MOTIVE uses calibrated drape ≥ 300 mL with abnormal signs or ≥ 500 mL.

10. Oxytocin 10 IU IM or slow IV, then infusion (e.g. 20–40 IU in 1 L crystalloid titrated to tone)

WHO 2023; infusion concentration and rate are protocol-specific; avoid rapid undiluted IV bolus (hypotension).

11. TXA for PPH: 1 g IV over 10 min within 3 h of birth; second 1 g if bleeding continues after 30 min or restarts within 24 h

WOMAN trial / WHO 2017; no benefit beyond 3 h.

12. Ergometrine/methylergonovine 0.2 mg IM (or slow IV), may repeat; contraindicated in hypertension/pre-eclampsia, heart disease

Product and repeat interval (q2–4h, max ~1 mg/day) vary.

13. Carboprost 250 mcg IM q15–90 min, max 8 doses (2 mg); avoid in asthma

Availability in PH limited; storage refrigerated.

14. Misoprostol 800 mcg sublingual for PPH treatment (if oxytocin unavailable/adjunct)

WHO; adjunct value with oxytocin is limited; fever/shivering common.

15. PPH blood products: early fibrinogen (target > 2 g/L), 1:1:1 MTP, calcium

Obstetric MTP packs and fibrinogen sources (cryo vs concentrate) are institution-specific.

16. Maternal arrest: manual left uterine displacement if fundus at/above umbilicus (~≥ 20 wk); hysterotomy decision at 4 min, delivery by ~5 min; standard

ACLS drugs/energies; Mg toxicity → calcium chloride 10% 10 mL / gluconate 10% 30 mL

AHA 2020 / ERC 2021; timing targets are aspirational – start preparing on arrival.

17. Newborn resuscitation doses: epinephrine IV/IO 0.01–0.03 mg/kg (0.1 mg/mL), ETT 0.05–0.1 mg/kg; volume 10 mL/kg; dextrose 10% 2 mL/kg; PPV 40–60/min; compressions 3:1 if HR < 60 after 30 s effective ventilation

AHA/AAP 2020 NRP & ILCOR 2023; check local neonatal protocol and drug strength.

18. Amniotic fluid embolism: supportive (A-OK regimen – atropine 1 mg, ondansetron 8 mg, ketorolac 30 mg IV – case-report evidence only)

Not guideline-endorsed; ketorolac contraindicated with bleeding/coagulopathy. Owner to decide whether to keep.

19. VTE in pregnancy: enoxaparin 1 mg/kg SC q12h (early-pregnancy weight); no DOACs/warfarin

RCOG GTG 37b / ASH 2018; anti-Xa monitoring at weight extremes and renal impairment.

20. Hyperemesis: thiamine 100 mg IV before dextrose; antiemetics (metoclopramide 10 mg, ondansetron 4–8 mg, doxylamine-pyridoxine)

RCOG GTG 69 (2024); first-trimester ondansetron small oral-cleft signal – shared decision.

S12 • 🧑🏻 Pediatrics – Pediatric tools Built

⚖️ Peds weight from age (APLS) △ VERIFY

Enter age at the top.

Age	Formula (kg)
0–12 months	$(0.5 \times \text{age in months}) + 4$
1–5 years	$(2 \times \text{age}) + 8$
6–12 years	$(3 \times \text{age}) + 7$

Prefer measured weight or a length-based tape. Obese/underweight children: formulas unreliable.

📈 Vital signs by age

Age	HR awake	RR	SBP	Hypotension
Neonate (0–28 d)	100–205	30–53	67–84	< 60
Infant (1–12 mo)	100–180	30–53	72–104	< 70

Age	HR awake	RR	SBP	Hypotension
Toddler (1–2 y)	98–140	22–37	86–106	< $70 + 2 \times \text{age}$
Preschool (3–5 y)	80–120	20–28	89–112	< $70 + 2 \times \text{age}$
School (6–11 y)	75–118	18–25	97–115	< $70 + 2 \times \text{age}$ (≤ 10 y), < 90
Adolescent (12–15 y)	60–100	12–20	110–131	< 90

Approximate ranges (PALS 2020). Values vary with sleep, fever, pain and anxiety – trend matters. Adult: HR 60–100, RR 12–20, SBP 90–140.

💧 Maintenance fluids (4-2-1)

Enter weight

4-2-1 rule (Holliday–Segar)

Pediatric doses appear throughout the app in Pediatric mode (capped at the adult dose). Full Pediatrics section coming soon.

△ Items pending owner review in this section (16)

1. APLS weight-from-age formulas

Derived from UK children; may overestimate weight in Filipino children – prefer measured or length-based weight.

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AHA/AAP 2020 NRP & ILCOR 2023; check local neonatal protocol and drug strength.

3. Dehydration (WHO): Plan A home ORS; Plan B 75 mL/kg ORS over 4 h; Plan C (severe) Ringer's lactate 100 mL/kg – infants 30 mL/kg in 1 h then 70 mL/kg in 5 h; older 30 mL/kg in 30 min then 70 mL/kg in 2.5 h; zinc 20 mg/day × 10–14 d (10 mg if < 6 months)

WHO 2005 Treatment of diarrhoea; cholera adults may need more.

4. Febrile infant empiric antibiotics: ≤ 28 d – ampicillin 50 mg/kg IV q8h (100 mg/kg if meningitis) + gentamicin 4 mg/kg IV q24h or cefotaxime/ceftazidime 50 mg/kg q8h; 29–60 d – ceftriaxone 50 mg/kg IV q24h (100 mg/kg if meningitis); acyclovir 20 mg/kg IV q8h if HSV risk

AAP 2021; neonatal intervals vary with postnatal age; avoid ceftriaxone in neonates on IV calcium/with hyperbilirubinemia.

5. PECARN febrile infant (29–60 d) low risk: normal UA, ANC ≤ 4090/μL, procalcitonin ≤ 0.5 ng/mL; AAP 2021 inflammatory marker thresholds: PCT > 0.5, ANC > 4000–5200, CRP > 2 mg/dL, T > 38.5 °C

Applies to well-appearing term infants only.

6. Croup: dexamethasone 0.15–0.6 mg/kg PO/IM once (max 16 mg; 0.15 mg/kg as effective for mild–moderate); nebulized epinephrine 1:1000 0.5 mL/kg (max 5 mL) for moderate–severe, observe 2–4 h after

Cochrane 2023; budesonide 2 mg neb alternative.

7. Kawasaki disease: IVIG 2 g/kg IV over 10–12 h + aspirin 30–50 mg/kg/day (moderate) or 80–100 mg/kg/day (high) divided q6h until afebrile 48 h, then 3–5 mg/kg/day

AHA 2017/2024; IVIG supply/cost in PH; consider dosing obese adolescents by IBW.

8. Alprostadil (PGE1) for suspected duct-dependent lesion: 0.05–0.1 mcg/kg/min IV (maintenance down to 0.01–0.05); apnea risk – airway ready

Neonatal formularies.

9. Pediatric UTI (> 2 mo): cefixime 8 mg/kg PO daily (max 400 mg) or cefalexin 50–100 mg/kg/day ÷ q6–8h × 7–14 d (febrile); IV ceftriaxone 50 mg/kg if unwell/vomiting

AAP 2011/2016; local E. coli resistance (PH ESBL common).

10. Pediatric CAP (non-severe, outpatient): amoxicillin 45 mg/kg BID (90 mg/kg/day, max 4 g/day) × 5 days; severe: ampicillin 50 mg/kg IV q6h (max 2 g) or ceftriaxone 50–100 mg/kg/day

IDSA 2011 / PIDSP-PAPP pCAP 2016 (PH guideline doses may differ – confirm); 5-day course per SCOUT-CAP.

11. Suspected metabolic crisis: stop protein, D10 with electrolytes at 1.5 × maintenance (glucose 8–10 mg/kg/min); ammonia urgently; specialist (metabolic) advice for scavengers/dialysis

BIMDG emergency protocols.

12. PH child protection: RA 7610 – physicians must report suspected child abuse (to DSWD/police) within 48 h; hospital WCPU; DSWD Bantay Bata 163 hotline

Confirm reporting timeframe and hotline number with hospital WCPU.

13. Neonatal jaundice: use AAP 2022 hour-specific phototherapy/exchange thresholds (gestational age, neurotoxicity risk factors); jaundice < 24 h of life, conjugated hyperbilirubinemia or ill infant = urgent

AAP 2022; thresholds require the nomogram tool.

14. AOM: amoxicillin 80–90 mg/kg/day ÷ BID (max 4 g/day) × 10 d if < 2 y or severe, 5–7 d if older; penicillin allergy (non-anaphylactic) cefdinir/cefuroxime; observation option for non-severe AOM ≥ 2 y

AAP 2013.

15. Kocher criteria (child with irritable hip): non-weight-bearing, fever > 38.5 °C, ESR > 40 mm/h, WBC > $12 \times 10^9/L$ – septic arthritis probability ~3% (1), 40% (2), 93% (3), 99% (4); CRP > 2 mg/dL often added

Kocher 1999 derivation; lower in validation cohorts.

16. Pediatric agitation: diphenhydramine 1 mg/kg (max 50 mg) PO/IM; midazolam 0.1–0.15 mg/kg IM (max 5 mg); olanzapine 2.5–5 mg (< 12 y) / 5–10 mg (≥ 12 y); haloperidol 0.05–0.1 mg/kg IM (max 5 mg) in ≥ 6 y

AACAP/BETA pediatric consensus 2019; specialist input.

Febrile infant & fever

Enter age in days △ VERIFY

- Applies to **well-appearing, term** infants; ill-appearing = sepsis workup and antibiotics at any age
- Fever = rectal $\geq 38.0^{\circ}\text{C}$; parental report of measured fever counts
- HSV risk (maternal lesions, seizures, vesicles, hepatitis, CSF pleocytosis, age < 21 d) → add acyclovir, HSV PCR
- Older children: look for focus; sepsis signs (tachycardia, cap refill > 3 s, altered mentation) → sepsis bundle
- PH: in febrile children ≥ 2 –3 days consider dengue (dengue), typhoid, leptospirosis after floods, measles (check vaccination)

Ampicillin – febrile neonate / meningitis △ VERIFY

IV q8h (≤ 28 d)

50 mg/kg, max 2,000 mg

100 mg/kg/dose if meningitis suspected.

Gentamicin – neonate (with ampicillin) △ VERIFY

IV q24h

4 mg/kg, max 400 mg

Levels if > 48 h; interval longer in preterm.

Ceftriaxone – infant 29–60 d / older child SBI △ VERIFY

IV q24h

50 mg/kg, max 2,000 mg

Meningitis: 100 mg/kg/day (max 4 g). Neonates ≤ 28 d: use cefotaxime/ceftazidime 50 mg/kg q8h instead.

Acyclovir – neonatal HSV risk △ VERIFY

IV q8h

20 mg/kg, max 800 mg

Ensure hydration; adjust for renal function.

Stridor, wheeze & pneumonia

Westley croup score

0 Westley (0–17)

Mild: dexamethasone, discharge with advice △ VERIFY

Dexamethasone – croup △ VERIFY

PO / IM once

0.15 – 0.6 mg/kg, max 16 mg

Nebulized epinephrine 1:1000 (1 mg/mL) – croup △ VERIFY

Nebulized

0.5 mL/kg, max 5 mL

Observe ≥ 2 –4 h for rebound.

Condition

Key points

Bronchiolitis (< 2 y)

Supportive: nasal suction, feeding support (NG), oxygen if $\text{SpO}_2 < 90$ –92 %, HFNC for work of breathing; salbutamol, steroids, antibiotics and routine CXR not recommended; high risk: < 6 –12 wk, prematurity, CHD

Epiglottitis / bacterial tracheitis

Toxic, drooling, tripod, muffled voice – keep calm with parent, no throat exam, airway expert in OR; ceftriaxone; Hib vaccination status

Foreign body aspiration

Sudden choking, unilateral wheeze; inspiratory/expiratory or decubitus films; rigid bronchoscopy. Button batteries in esophagus = emergency (tox)

Asthma (child)

See asthma card

Pneumonia

Tachypnea and chest indrawing; amoxicillin high dose outpatient; admit if hypoxemic, < 3 –6 mo, unable to feed, severe distress △ VERIFY

Amoxicillin – pediatric pneumonia / otitis △ VERIFY

PO BID

45 mg/kg, max 2,000 mg

Kawasaki disease

- ☐ Bilateral non-exudative conjunctival injection
- ☐ Lip/oral changes (red cracked lips, strawberry tongue)
- ☐ Polymorphous rash
- ☐ Extremity changes (erythema/edema, later peeling)
- ☐ Cervical lymph node ≥ 1.5 cm (usually unilateral)
- ☐ Fever ≥ 5 days

0 principal features (0–5)

Criteria not met – reassess daily if fever persists

△ VERIFY

- Infants < 6 months may have fever alone – low threshold for echo
- Treat within 10 days of fever onset (ideally ≤ 7) to reduce coronary aneurysms

- MIS-C: fever + multisystem involvement + recent SARS-CoV-2 – shock, myocarditis; IVIG + steroids with specialist input
- PH: Kawasaki disease is relatively common in East/Southeast Asian children

IVIG – Kawasaki disease (A VERIFY)

IV over 10–12 h

2 g/kg, max 140 g

Single infusion; consider IBW in obesity.

Aspirin – Kawasaki (acute phase) (A VERIFY)

PO per day (÷ q6h)

30 – 50 mg/kg, max 4,000 mg

Then 3–5 mg/kg/day once daily (max 81–325 mg) after 48 h afebrile.

PECARN head injury

< 2 years

- ☐ GCS < 15 / altered mental status / palpable skull fracture
- ☐ Occipital, parietal or temporal scalp hematoma
- ☐ LOC ≥ 5 s
- ☐ Severe mechanism
- ☐ Not acting normally per parent

≥ 2 years

- ☐ GCS < 15 / altered mental status / signs of basilar skull fracture
- ☐ Any LOC
- ☐ Vomiting
- ☐ Severe mechanism
- ☐ Severe headache

Very low risk (< 0.05 %): CT not recommended

Applies to GCS 14–15 within 24 h of injury. Severe mechanism: MVC with ejection/rollover/death, pedestrian or unhelmeted cyclist struck, fall > 0.9 m (< 2 y) or > 1.5 m (≥ 2 y), head struck by high-impact object. Consider abuse in infants (child maltreatment).

Neonatal emergencies & BRUE

Sick neonate – think broadly

- Sepsis (most common) – cultures and antibiotics early (see febrile infant)
- Cardiac: duct-dependent lesion presents day 3–14 with shock/cyanosis, weak femoral pulses, hepatomegaly → alprostadil (A VERIFY)
- Metabolic: hypoglycemia, inborn errors (vomiting, lethargy, acidosis, high ammonia), congenital adrenal hyperplasia (salt-wasting, low Na/high K) (A VERIFY)
- Surgical: bilious vomiting = malrotation with volvulus until proven otherwise → urgent surgery/contrast study; NEC; Hirschsprung
- Neurologic: seizures (subtle), non-accidental trauma; toxins/formula errors
- PH: check expanded newborn screening results (RA 9288 – includes G6PD deficiency, CAH, MSUD, galactosemia)

Alprostadil (PGE1) – duct-dependent lesion (A VERIFY)

Infusion

0.05 – 0.1 mcg/kg/min

Example dilution 10 mcg/mL. Apnea/hypotension – be ready to ventilate.

Jaundice

- Jaundice in first 24 h, rapidly rising, conjugated (pale stools/dark urine), or unwell infant → urgent evaluation
- Total serum bilirubin plotted on hour-specific AAP 2022 thresholds; risk factors lower thresholds (prematurity, hemolysis/G6PD deficiency, sepsis, albumin < 3) (A VERIFY)
- Feeding support; intensive phototherapy; exchange transfusion if at threshold or signs of encephalopathy

BRUE (brief resolved unexplained event)

- < 1 y; < 1 min; resolved; unexplained after history/exam: cyanosis/pallor, absent/irregular breathing, tone change, altered responsiveness
- **Lower risk:** age > 60 d, GA ≥ 32 wk (PCA ≥ 45 wk), first event, < 1 min, no CPR by trained provider; no concerning features → education, CPR training, brief observation with pulse oximetry, ECG optional
- Higher risk → admit/observe; consider abuse, infection (pertussis, RSV), seizures, cardiac (long QT), metabolic
- Sudden unexpected infant death: support family, medico-legal notification; safe-sleep counselling for siblings

Pediatric abdomen & GU

Condition	Key points
Intussusception	6 mo–3 y, intermittent colicky pain with drawing up of legs, lethargy, vomiting, "red currant jelly" stool (late); ultrasound (target sign) → air/hydrostatic enema; surgery if perforation/peritonitis
Pyloric stenosis	2–8 weeks, projectile non-bilious vomiting, hungry; hypochloremic hypokalemic metabolic alkalosis; ultrasound; correct fluids/electrolytes before surgery
Malrotation/volvulus	Bilious (green) vomiting in infant = surgical emergency
Appendicitis	Younger children perforate more often; Pediatric Appendicitis Score/ultrasound first, limit CT

Condition	Key points
Testicular torsion	Peak neonatal & adolescent; absent cremasteric reflex, high-riding testis; TWIST score; urology without delay (do not wait for ultrasound if high suspicion; 6-hour window)
Incarcerated inguinal hernia	Reduce with analgesia; surgery if irreducible/ischemic
Dehydration	Assess with clinical dehydration scale; ORS first (ORS plans); hypoglycemia check
UTI	Catheter or suprapubic sample in non-toilet-trained; febrile UTI in < 2 mo = admit/IV; renal ultrasound after first febrile UTI < 2 y △ VERIFY
Cefixime – pediatric UTI (oral) △ VERIFY	
PO once daily 8 mg/kg, max 400 mg	

Pediatric orthopedics & limp

Condition	Key points
Septic hip vs transient synovitis	Fever, non-weight-bearing, ESR/CRP, WBC → Kocher criteria (hot joint card); ultrasound for effusion; aspirate if suspicious
SCFE	10–16 y, obese; hip, thigh or knee pain; frog-leg lateral X-ray; non-weight-bearing, urgent orthopedics
Perthes	4–10 y, painless/mild limp; X-ray; orthopedics
Osteomyelitis	Bone tenderness, fever; MRI; S. aureus coverage

Condition	Key points
Toddler fracture	Spiral tibial fracture, may be occult; immobilize, repeat X-ray
Salter–Harris	Physis injuries (I–V): types III–V need orthopedics; tenderness over physis with normal X-ray = presume type I
Supracondylar humerus	Check radial pulse, AIN/radial/ulnar nerves; pulseless pink vs white hand (upper limb)
Leukemia/tumor	Night pain, systemic symptoms, abnormal CBC – do not miss

Child maltreatment

- Red flags: history inconsistent with injury/development ("those who don't bruise rarely bruise"), delay in seeking care, changing story, injury blamed on sibling
- **TEN-4-FACESp** bruising: Torso, Ears, Neck, Frenulum, Angle of jaw, Cheeks, Eyelids, Subconjunctivae in < 4 y; **any** bruise in an infant < 4–6 months; patterned bruises
- High-specificity fractures: classic metaphyseal lesions, posterior rib, scapula, spinous process, sternum; multiple fractures of different ages
- Abusive head trauma: irritability, vomiting, seizures, apnea, bulging fontanelle; subdural hematoma + retinal hemorrhage
- Work-up (< 2 y with concerns): skeletal survey, CT head (± MRI), ophthalmology, LFTs/lipase (abdominal injury), urinalysis, coagulation screen; screen siblings
- Neglect, sexual abuse (refer to WCPU – sexual assault), medical child abuse also occur
- **PH:** RA 7610 – report suspected abuse to DSWD/PNP; hospital Women and Children Protection Unit (WCPU); DSWD Bantay Bata 163 **△ VERIFY**
- Admit when safety is uncertain; document objectively with body map and photographs

Pediatric emergencies sources

Pantell et al., AAP CPG febrile infants 8–60 days (Pediatrics 2021) · Kuppermann et al., PECARN febrile infant rule (JAMA Pediatr 2019) · Kuppermann et al., PECARN head injury rule (Lancet 2009) · Ralston et al., AAP bronchiolitis guideline 2014 · Gates et al., Cochrane glucocorticoids for croup 2023; Westley score (1978) · McCrindle et al., AHA Kawasaki statement 2017 · Tieder et al., AAP BRUE guideline 2016 · Kemper et al., AAP hyperbilirubinemia 2022 · AAP UTI guideline 2011 (reaffirmed 2016) · Bradley et al., IDSA pediatric CAP 2011; PIDSP/PAPP pediatric CAP guideline 2016 · Kocher et al. (JBJS 1999) · Pierce et al., TEN-4-FACESp (JAMA Netw Open 2021) · Philippine RA 7610, RA 9288 (Newborn Screening Act) · BIMDG emergency guidelines. Tintinalli 9e Section 12 used only as a topic map.

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Neonatal formularies.

9. Pediatric UTI (> 2 mo): cefixime 8 mg/kg PO daily (max 400 mg) or cefalexin 50–100 mg/kg/day ÷ q6–8h × 7–14 d (febrile); IV ceftriaxone 50 mg/kg if unwell/vomiting

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Kocher 1999 derivation; lower in validation cohorts.

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AACAP/BETA pediatric consensus 2019; specialist input.

S13 • Infectious Diseases – Sepsis Built

Hour-1 bundle (SSC)

Time zero = triage / recognition. Checked items are time-stamped.

- ☐ Measure **lactate** (re-measure within 2–4 h if > 2 mmol/L)
- ☐ Obtain **blood cultures** before antibiotics (do not delay antibiotics)
- ☐ Give **broad-spectrum antibiotics** (within 1 h if shock/probable sepsis)
- ☐ Start **crystalloid 30 mL/kg** for hypotension or lactate ≥ 4 mmol/L △ VERIFY
- ☐ **Vasopressors** if hypotensive during/after fluids – target MAP ≥ 65 mmHg

qSOFA

- ☐ Respiratory rate ≥ 22 /min
- ☐ Altered mentation (GCS < 15)
- ☐ Systolic BP ≤ 100 mmHg

0 / 3 – < 2 does not exclude sepsis

SSC 2021 advises against qSOFA alone as a screening tool.

Fluid resuscitation

30 mL/kg crystalloid within 3 h △ VERIFY

Enter weight

Vasopressors & steroids

Norepinephrine (MAP ≥ 65)

Infusion △ VERIFY

0.05 – 0.5 mcg/kg/min

4 mg in 250 mL = 16 mcg/mL. Peripheral OK short-term via good proximal vein.

Vasopressin △ VERIFY

Add-on infusion

0.03 units/min (not weight-based)

Add when norepinephrine ≈ 0.25 –0.5 mcg/kg/min (SSC 2021).

Epinephrine

Add-on infusion △ VERIFY

0.05 – 0.5 mcg/kg/min

If MAP inadequate on norepinephrine $\pm$ vasopressin.

Hydrocortisone △ VERIFY

Vasopressor-dependent shock

200 mg/day IV (50 mg q6h)

If norepinephrine ≥ 0.25 mcg/kg/min for ≥ 4 h (SSC 2021).

△ Items pending owner review in this section (7)

1. Pediatric vasoactive infusion ranges (epinephrine/norepinephrine 0.05–1 mcg/kg/min; epi for bradycardia 0.1–1)

Titration ranges vary by PICU protocol; standard 16 mcg/mL bag assumed.

2. Sepsis 30 mL/kg crystalloid within 3 h

SSC 2021 weak recommendation; individualize in heart failure, ESKD, obesity (ideal body weight often used).

3. Pediatric septic shock fluids 10–20 mL/kg boluses, up to 40–60 mL/kg in hour 1

SSC 2020: only where intensive care is available; without ICU and no hypotension, avoid boluses (FEAST).

4. Adult norepinephrine range 0.05–0.5 mcg/kg/min

Many units titrate in mcg/min; ceiling and peripheral-line policies vary.

5. Pediatric hydrocortisone 2 mg/kg (max 100 mg) for catecholamine-resistant shock

Dose ranges 1–2 mg/kg up to 50 mg/m²/day in references.

6. Vasopressin 0.03 units/min add-on (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

7. Hydrocortisone 200 mg/day in vasopressor-dependent shock (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

 Dengue

- Home care if able to drink, passing urine every 4–6 h, no warning signs: oral fluids (ORS, juice, soup), paracetamol only (no NSAIDs/aspirin, no IM injections)
- Daily review (CBC) from day 3 until 48 h after defervescence; return immediately with warning signs

Recognise

- Phases: febrile (days 1–3), critical (around defervescence, days 3–7 – plasma leak, 24–48 h), recovery (fluid reabsorption – watch for overload)
- **Warning signs:** abdominal pain/tenderness, persistent vomiting, fluid accumulation, mucosal bleeding, lethargy/restlessness, liver > 2 cm, rising Hct with rapid platelet fall
- Shock may have normal BP with narrow pulse pressure (≤ 20 mmHg) and cold peripheries
- Tests: NS1 antigen (days 1–5), IgM (after day 5), CBC with Hct trend; tourniquet test supportive
- Prophylactic platelet transfusion is not recommended in stable patients
- Risk groups: infants, pregnancy, elderly, obesity, diabetes, renal/heart disease, living far from care
- PH: peak June–November; report to local epidemiology surveillance unit (RA 11332)

 Leptospirosis

- Suspect: fever + myalgia (calves), conjunctival suffusion, flood/wading exposure within 2–30 days
- Severe (Weil): jaundice, AKI (often non-oliguric, hypokalemic), pulmonary hemorrhage (high mortality), myocarditis, thrombocytopenia
- Tests: MAT/IgM, PCR early; do not wait – treat empirically
- Severe: admit, early dialysis for AKI, lung-protective ventilation for hemorrhage
- Prophylaxis after exposure: natural disaster card

Doxycycline – leptospirosis (mild) / rickettsia / scrub typhus **VERIFY**

PO

100 mg BID × 7 days**Penicillin G / ceftriaxone – severe leptospirosis** **VERIFY**

Penicillin G

1.5 million units IV q6h × 7 days

Ceftriaxone

1 – 2 g IV daily × 7 days **Malaria, measles, TB & tropical fever****Malaria**

- PH: mainly Palawan (and returning travelers/workers from Africa, Papua); malaria-free status in most provinces
- Test every febrile patient from an endemic area: thick/thin smear or RDT; repeat if negative and suspicion persists
- Severe: impaired consciousness, convulsions, shock, acidosis, hypoglycemia, severe anemia, jaundice, parasitemia > 5–10 %  **VERIFY**

Artemether-lumefantrine 20/120 mg – uncomplicated falciparum **VERIFY**

Tablets per dose (6 doses: 0, 8, 24, 36, 48, 60 h)

With fatty food/milk; repeat dose if vomited within 30 min**Artesunate IV – severe malaria** **VERIFY**

IV at 0, 12, 24 h then daily

2.4 mg/kg

Switch to full oral ACT course once tolerating

Measles

- Fever, cough, coryza, conjunctivitis, then rash from face down; Koplik spots
- Airborne isolation; complications – pneumonia, diarrhea, encephalitis, keratitis; vitamin A for all cases  **VERIFY**
- PH outbreaks follow low vaccination coverage – check MMR/MR status, notify

Vitamin A – measles **VERIFY**

PO daily × 2 days

Enter age**Influenza**

- Antiviral for hospitalized, severe, or high-risk patients (age < 5 or ≥ 65, pregnancy, chronic disease) – best within 48 h but still useful later in severe disease  **VERIFY**

Oseltamivir – influenza (5 days) **VERIFY**

PO

75 mg BID**TB & others**

- PH has one of the highest TB burdens: cough ≥ 2 weeks, weight loss, hemoptysis → mask/isolate, Xpert MTB/RIF, refer to TB-DOTS  **VERIFY**
- Typhoid (GI card), scrub typhus/rickettsia (eschar – doxycycline), melioidosis (diabetes, rice farmers – ceftazidime/meropenem)
- Returning traveler: where, when, activities, vaccines, prophylaxis; malaria first in fever from endemic zones

 HIV, STIs & PEP

- PH: rapidly growing HIV epidemic (young men who have sex with men, PWID); offer testing to everyone with STI, TB, pneumonia, unexplained weight loss/fever  **VERIFY**
- Advanced HIV (CD4 < 200): PCP (hypoxemia, LDH, diffuse infiltrates – TMP-SMX + steroids), cryptococcal meningitis (headache, high opening pressure – CrAg), toxoplasmosis (ring-enhancing lesions), TB (often extrapulmonary), CMV
- IRIS after ART start; drug interactions with ART (rifampicin, statins, antiepileptics)
- **STIs:** treat partners, test for HIV/syphilis, abstain 7 days; syphilis is rising in PH (congenital syphilis screening in pregnancy)  **VERIFY**
- **Occupational exposure** (needlestick/splash): wash with soap and water (do not squeeze/scrub with bleach), assess source (HIV, HBV, HCV testing with consent), start HIV PEP within 72 h (ideally ≤ 2 h) if significant exposure (PEP regimen), HBV vaccine ± HBIG if non-immune, baseline and follow-up testing

STI treatment **VERIFY**

Gonorrhea

Ceftriaxone 500 mg IM once

Chlamydia

Doxycycline 100 mg PO BID × 7 days (pregnancy: azithromycin 1 g once)

Early syphilis

Benzathine penicillin G 2.4 million units IM once

♥ Infective endocarditis

- Fever + new murmur, embolic phenomena, prosthetic valve, PWID (PWID card), or rheumatic heart disease (still common in PH)
- Three sets of blood cultures from separate sites before antibiotics (unless septic shock – take 2 sets quickly and treat)
- Echo: TTE first, TEE for prosthetic valves/high suspicion
- Major criteria: typical organism in blood cultures, imaging evidence (vegetation, abscess, new dehiscence); minor: predisposition, fever, vascular/immunologic phenomena
- Urgent surgery: heart failure from valve dysfunction, uncontrolled infection/abscess, large vegetation with embolism **△ VERIFY**

Empiric endocarditis (native valve) **△ VERIFY**

Vancomycin IV load

25 – 30 mg/kg, max 3,000 mg

+ Ceftriaxone

2 g IV daily

🔗 Other infections

- Sepsis bundle

- Skin & soft-tissue infections
- Tetanus prophylaxis & treatment
- Rabies post-exposure prophylaxis
- Gastroenteritis, typhoid & cholera
- Meningitis & encephalitis
- Febrile infant
- Leptospirosis prophylaxis after floods

📖 Infectious diseases sources

WHO Dengue: guidelines for diagnosis, treatment, prevention and control (2009) · WHO clinical management of arboviral diseases (2025) · DOH Revised Dengue Clinical Case Management Guidelines (2011) · DOH/PSMID/PPS Leptospirosis CPG (2010) · WHO Guidelines for malaria (2024); DOH Malaria Control and Elimination Program · WHO vitamin A for measles · CDC influenza antiviral guidance · CDC STI Treatment Guidelines 2021 · Fowler et al., Duke-ISCVID criteria (Clin Infect Dis 2023); ESC endocarditis 2023 · DOH NTP Manual of Procedures 6th ed. · Philippine RA 11166, RA 11332 · NIH/IDSA adult OI guidelines. Tintinalli 9e Section 13 used only as a topic map.

△ Items pending owner review in this section (15)

1. Leptospirosis post-flood prophylaxis (DOH): low risk – doxycycline 200 mg single dose within 24–72 h of exposure; moderate risk (wounds, repeated wading, swallowed water) – 200 mg once daily for 3–5 days; high risk (continuous exposure) – 200 mg once weekly until exposure ends

DOH categories recalled; pregnant women/children: azithromycin or amoxicillin per DOH – confirm current DOH memorandum.

2. Rabies PEP (WHO 2018, DOH): Category II → vaccine; Category III → vaccine + rabies immunoglobulin. Vaccine: intradermal 2-site on days 0, 3, 7 (PH ABTC regimen) or IM 1 dose on days 0, 3, 7 and 14–28; previously vaccinated → 2 doses (days 0, 3), no RIG

Schedules per WHO 2018 / DOH NRPC; confirm current DOH memorandum and product (PCECV/PVRV) ID volume (0.1 mL per site).

3. Rabies immunoglobulin: HRIG 20 IU/kg or ERIG 40 IU/kg, infiltrated into and around all wounds as far as anatomically feasible (no need to inject the remainder IM); given up to day 7 after first vaccine dose

WHO 2018; ERIG skin testing not required by WHO but some PH protocols still perform it.

4. Diarrhea antibiotics: cholera – azithromycin 1 g single dose (peds 20 mg/kg) or doxycycline 300 mg; shigellosis – azithromycin 500 mg × 3 d (peds 10 mg/kg/d); typhoid (uncomplicated) – azithromycin 1 g day 1 then 500 mg daily × 6 d (peds 20 mg/kg/d), severe – ceftriaxone 2 g IV daily (peds 50–80 mg/kg); amoebic dysentery – metronidazole 500–750 mg TID × 7–10 d

WHO / DOH PH; rising azithromycin and fluoroquinolone resistance – follow RITM/ARSP data.

5. Dengue fluids (WHO 2009/DOH): warning signs without shock – isotonic crystalloid 5–7 mL/kg/h × 1–2 h, then 3–5 mL/kg/h × 2–4 h, then 2–3 mL/kg/h, taper over 24–48 h guided by Hct/vitals/urine; compensated shock – 5–10 mL/kg/h (adult) or 10–20 mL/kg/h (child) × 1 h; hypotensive shock – 20 mL/kg over 15–30 min, repeat/colloid if no response; use IBW in obesity

WHO 2009 / DOH 2011; WHO 2025 arboviral guideline wording differs slightly – confirm.

6. Leptospirosis treatment: mild – doxycycline 100 mg PO BID × 7 d (children 2 mg/kg/dose BID, or amoxicillin); moderate–severe – penicillin G 1.5 MU IV q6h or ceftriaxone 1–2 g IV daily × 7 d

DOH/PSMID 2010 leptospirosis CPG.

7. Malaria: uncomplicated P. falciparum – artemether-lumefantrine (20/120 mg) weight-band 6-dose regimen over 3 days with fatty food; severe – IV artesunate 2.4 mg/kg (3 mg/kg if < 20 kg) at 0, 12, 24 h then daily; primaquine (single low dose 0.25 mg/kg for falciparum; G6PD testing before radical cure of vivax)

WHO 2024; DOH Malaria Control Program (Palawan is main endemic area) – confirm drug access.

8. Measles vitamin A: < 6 mo 50 000 IU, 6–11 mo 100 000 IU, ≥ 12 mo 200 000 IU PO daily × 2 days (3rd dose 2–4 weeks later if eye signs)

WHO.

9. Oseltamivir × 5 days: adult 75 mg BID; child ≥ 1 y – ≤ 15 kg 30 mg, 15–23 kg 45 mg, 23–40 kg 60 mg, > 40 kg 75 mg BID; < 1 y 3 mg/kg BID

CDC/AAP; renal adjustment.

10. STIs: gonorrhea ceftriaxone 500 mg IM once (1 g if ≥ 150 kg); chlamydia doxycycline 100 mg BID × 7 d (azithromycin 1 g in pregnancy); early syphilis benzathine penicillin G 2.4 MU IM once (late: weekly × 3); PID ceftriaxone + doxycycline + metronidazole × 14 d

CDC 2021; PH benzathine penicillin shortages occur.

11. Endocarditis empiric (native valve, after cultures): vancomycin 25–30 mg/kg load + ceftriaxone 2 g IV daily (or + gentamicin); prosthetic valve: vancomycin + gentamicin + rifampicin (specialist)

ESC 2023 / AHA 2015.

12. TB: airborne isolation for suspected pulmonary TB; Xpert MTB/RIF first test (NTP); do not start empiric single-drug therapy in ED

NTP MOP 6th ed 2020.

13. PH HIV: RA 11166 – voluntary, confidential testing with counselling (minors 15–17 may consent); link to DOH treatment hub; PCP: TMP-SMX 15–20 mg/kg/day (TMP) + q6–8h + prednisone if PaO₂ < 70 mmHg

RA 11166 IRR; IDSA/NIH OI guidelines.

14. Bacterial meningitis (adult): dexamethasone 10 mg IV q6h × 4 d (start with/before first antibiotic); ceftriaxone 2 g IV q12h + vancomycin 15–20 mg/kg q8–12h; + ampicillin 2 g IV q4h if > 50 y/immunocompromised/pregnant; acyclovir 10 mg/kg IV q8h if encephalitis possible. Contacts of meningococcal disease: ciprofloxacin 500 mg PO once (or rifampicin 600 mg BID × 2 d; ceftriaxone 250 mg IM in pregnancy). Peds: ceftriaxone 50 mg/kg q12h (100 mg/kg/day, max 4 g), vancomycin 15 mg/kg q6h, dexamethasone 0.15 mg/kg q6h

IDSA 2004 / ESCMID 2016 / NICE NG240; PH pneumococcal resistance data limited.

15. HIV PEP ≤ 72 h: tenofovir DF 300 mg + lamivudine 300 mg (or emtricitabine 200 mg) + dolutegravir 50 mg daily × 28 days; children: weight-band dosing per DOH/WHO

WHO 2024 / CDC nPEP; PH supply via DOH treatment hubs – confirm local protocol and starter packs.

 Glasgow Coma Scale

Tap one option in each section. Intubated: record V as "T".

Eye opening (E)**Verbal (V)****Motor (M)** **NIH Stroke Scale**

UN (untestable, e.g. amputation, joint fusion) scores 0 in the total – document why.

 Stroke time windows **△ VERIFY**

Enter last-known-well time to see remaining windows. Wake-up stroke: MRI DWI–FLAIR mismatch may allow lysis.

Alteplase (tPA) 1 mg/mL – 10% bolus, 90% over 60 min

0.9 mg/kg, max 90 mg **△ VERIFY**

Tenecteplase (TNK) – stroke dose **△ VERIFY**

0.25 mg/kg, max 25 mg

Stroke dose differs from the STEMI weight-band dose.

- Glucose first (treat if < 60 mg/dL); non-contrast CT ± CTA
- BP < 185/110 before lysis; < 180/105 for 24 h after
- No antithrombotics for 24 h after lysis; neuro checks q15 min × 2 h

 Status epilepticus ladder

① 0–5 min • Stabilize: ABC, O₂, monitor, time seizure, IV/IO. Check glucose – if < 60 mg/dL: thiamine 100 mg IV then dextrose 25 g (50 mL D50). **△ VERIFY**

5–20 min • Benzodiazepine (choose ONE; may repeat once)**Lorazepam IV**

IV (may repeat once after 5–10 min) **△ VERIFY**

0.1 mg/kg, max 4 mg

Midazolam IM

IM, single dose **△ VERIFY**

> 40 kg: 10 mg • 13–40 kg: 5 mg • < 13 kg: 0.2 mg/kg

Diazepam **△ VERIFY**

IV (may repeat once)

0.15 – 0.2 mg/kg, max 10 mg

Rectal **△ VERIFY**

0.5 mg/kg (2–5 y) • 0.3 mg/kg (6–11 y) • 0.2 mg/kg (≥ 12 y); max 20 mg

Midazolam intranasal / buccal **△ VERIFY****Intranasal**

0.2 mg/kg, max 10 mg

Undiluted 5 mg/mL with atomizer, split nostrils (≤ 1 mL each).

Buccal

0.3 mg/kg, max 10 mg

20–40 min • Second line (choose ONE)**Levetiracetam** **△ VERIFY**

IV over 5–15 min

60 mg/kg, max 4,500 mg

500 mg/5 mL vials = 100 mg/mL; dilute in 100 mL NS.

Fosphenytoin **△ VERIFY**

IV

20 mg PE/kg, max 1,500 mg PE

Rate ≤ 150 mg PE/min. Cardiac monitoring; hypotension/arrhythmia.

Valproate **△ VERIFY**

IV over 10 min

40 mg/kg, max 3,000 mg

Avoid: pregnancy, liver disease, < 2 y / suspected mitochondrial disease, pancreatitis.

Phenytoin (if no fosphenytoin) **△ VERIFY**

IV

20 mg/kg, max 1,500 mg

Rate ≤ 50 mg/min; NS only; cardiac monitor. Avoid in toxin-induced seizures.

Phenobarbital **△ VERIFY**

IV over 10–20 min

15 – 20 mg/kg

Rate ≤ 50–100 mg/min. Respiratory depression/hypotension. Vial strength varies – compute mL from the vial.

ESETT: levetiracetam, fosphenytoin and valproate are similarly effective (~50%).

> 40 min • Refractory – intubate, ICU, continuous EEG**Refractory SE (intubated, EEG)** **△ VERIFY**

Midazolam load

0.2 mg/kg

Midazolam infusion

0.05 – 2 mg/kg/hr

Propofol load

1 – 2 mg/kg

Propofol infusion

20 – 200 mcg/kg/min

PRIS risk with prolonged high doses (esp. children).

Pregnancy ≥ 20 wk / postpartum**Magnesium sulfate – eclampsia** **△ VERIFY**

Loading

4 g IV over 15–20 min

Zuspan; Pritchard adds 10 g IM (5 g each buttock). 4 g = 8 mL of 50%; dilute to 20 mL (20%). Some protocols load 6 g.

Maintenance

1 – 2 g/hr IV (or 5 g IM q4h)

Hold if reflexes absent, RR < 12, urine < 25–30 mL/hr. Antidote: Ca gluconate 1 g IV.

 Headache & Ottawa SAH rule

Ottawa SAH rule: alert patients ≥ 15 y with new severe non-traumatic headache peaking within 1 h.

- ☐ Age ≥ 40
- ☐ Neck pain or stiffness
- ☐ Witnessed loss of consciousness
- ☐ Onset during exertion
- ☐ Thunderclap (instantly peaking) headache
- ☐ Limited neck flexion on exam

Rule negative – SAH unlikely (sensitivity ~100 %); still consider other red flags

Red flags

- Sudden/thunderclap, worst ever, exertional/positional, new after age 50, fever/neck stiffness, focal deficit, papilledema, immunosuppression, cancer, pregnancy/postpartum (preeclampsia – OB HTN, CVST), anticoagulation, head injury
- Older adults: giant cell arteritis (jaw claudication, visual symptoms, ESR/CRP) → steroids same day to protect vision
- Household clusters/winter/charcoal or generator use → carbon monoxide (CO)

Migraine / primary headache

- Antiemetic dopamine antagonist $\pm$ NSAID, IV fluids if dehydrated; dexamethasone reduces 72-h recurrence; avoid opioids and butalbital
- Occipital nerve block or sphenopalatine block for refractory cases
- Cluster: high-flow oxygen 12–15 L/min via non-rebreather $\times$ 15 min **(Δ VERIFY)**

Metoclopramide / prochlorperazine – migraine **(Δ VERIFY)**

Metoclopramide IV

10 mg slow IV

Prochlorperazine IV

10 mg (with diphenhydramine 25 mg)

NSAID – headache **(Δ VERIFY)**

Ketorolac IV/IM

15 – 30 mg (15 mg if ≥ 65 y)

Dexamethasone – migraine recurrence prevention **(Δ VERIFY)**

IV

10 mg once

ICH & SAH

- Airway if GCS ≤ 8 or deteriorating; head of bed 30°; treat fever, glucose 140–180 mg/dL
- **ICH**: lower SBP to ~ 140 mmHg within 2 h, smoothly (avoid < 130 and big swings) **(Δ VERIFY)**
- Reverse anticoagulants immediately: anticoagulant reversal card; platelet transfusion not helpful for antiplatelet-associated ICH unless surgery
- Neurosurgery: cerebellar hematoma > 3 cm / brainstem compression / hydrocephalus; spot sign = expansion risk
- **aSAH**: Hunt–Hess/WFNS grade; BP < 160 until aneurysm secured; nimodipine; seizure treatment if seizures (no routine prophylaxis); EVD for hydrocephalus
- PH: hypertensive ICH very common – check adherence to antihypertensives; transfer early to neurosurgical centre

Nimodipine – aneurysmal SAH **(Δ VERIFY)**

PO / NG

60 mg q4h $\times$ 21 days (30 mg q2h if hypotension)

Vertigo (HINTS)

Pattern	Approach
Triggered episodic (seconds, positional)	BPPV: Dix-Hallpike (torsional upbeat nystagmus) → Epley maneuver; supine roll test for horizontal canal
Spontaneous episodic (minutes–hours)	Vestibular migraine, Ménière; TIA (posterior circulation – ABCD ² , imaging)
Acute vestibular syndrome (continuous, with nystagmus)	HINTS by a trained examiner: <i>central</i> if head impulse normal, direction-changing nystagmus, or skew deviation present (any one) → MRI (CT misses posterior strokes); plus gait (truncal ataxia = central)
Toxic/metabolic	Drugs (anticonvulsants, alcohol), CO, hypoglycemia

- Hearing loss with acute vertigo can be labyrinthine infarction (AICA) – treat as central
- Vestibular suppressants only briefly; vestibular rehabilitation **(Δ VERIFY)**

Vestibular suppressant (short term) **(Δ VERIFY)**

Meclizine PO

25 mg q8h PRN (≤ 3 days)

Dimenhydrinate PO/IV

50 mg q6h PRN

Altered mental status & coma

- 1 ABCs, glucose at the bedside, temperature, SpO₂/CO₂
- 2 Treat reversible: glucose if low, thiamine if at risk, naloxone if opioid toxidrome **(Δ VERIFY)** (naloxone, hypoglycemia)
- 3 Look for: meningism, focal signs, pupils, toxidrome (toxidromes), trauma, needle marks, medical-alert IDs
- 4 Mnemonic AEIOU-TIPS: Alcohol, Epilepsy/Encephalopathy/Electrolytes, Insulin, Opioids/Oxygen, Uremia; Trauma/Temperature, Infection, Psychiatric/Poisoning, Stroke/Shock
- 5 CT head; labs (Na, Ca, renal/liver, ammonia, VBG, tox screen, cultures); LP if infection possible; EEG for non-convulsive status (subtle twitching, unexplained persistent coma after seizure)
- 6 PH: consider cerebral malaria (Palawan), Japanese encephalitis, TB meningitis, rabies, dengue encephalopathy

Acute neuromuscular weakness

Condition	Key points
Guillain-Barré	Ascending weakness, areflexia, post-infection (Campylobacter, dengue, Zika); autonomic instability; serial FVC/NIF (VERIFY) ; IVIG or plasma exchange
Myasthenic crisis	Fatigable ptosis/diplopia, bulbar and respiratory weakness; triggers – infection, drugs (aminoglycosides, fluoroquinolones, macrolides, magnesium, β -blockers); avoid succinylcholine resistance/rocuronium sensitivity – use reduced rocuronium; IVIG/PLEX
Spinal cord compression	Back pain + bilateral leg weakness, sensory level, retention – MRI whole spine urgently; malignant: dexamethasone 10 mg IV (VERIFY) + oncology/radiotherapy (oncologic emergencies); epidural abscess
Bell palsy	Lower motor neuron facial palsy involving forehead; exclude stroke (forehead sparing), Ramsay Hunt (vesicles), otitis/parotid, Lyme; steroids ≤ 72 h, eye care (VERIFY)
Hypokalemic periodic paralysis	Sudden weakness after carbohydrate load/exercise; thyrotoxic form in Asian men – check K and TFTs; cautious K replacement (rebound hyperkalemia)

Condition

Key points

Botulism / tetanus	Descending paralysis (botulism – home-canned food, wound); tetanus – tetanus card
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Prednisolone – Bell palsy [\(VERIFY\)](#)

PO

50 mg daily $\times$ 10 days (or 60 mg $\times$ 5 days then taper)

Meningitis & encephalitis

- 1 Suspected bacterial meningitis: blood cultures then **antibiotics within 1 h** – do not delay for CT/LP
- 2 CT before LP if: immunocompromised, focal deficit, papilledema, new seizure, GCS $< 10-12$, history of CNS lesion
- 3 Dexamethasone before or with the first antibiotic dose (benefit mainly pneumococcal; stop if not bacterial) [\(VERIFY\)](#)
- 4 Encephalitis (confusion, seizures, behaviour change, fever): acyclovir empirically; HSV PCR, MRI, EEG
- 5 CSF: bacterial – high neutrophils, low glucose ratio (< 0.4), high protein; viral – lymphocytes, normal glucose; TB – lymphocytes, very high protein, low glucose
- 6 Droplet precautions for meningococcal disease; chemoprophylaxis for close contacts (ciprofloxacin 500 mg once / rifampicin) [\(VERIFY\)](#)
- 7 PH: TB meningitis (subacute, cranial nerve palsies), Japanese encephalitis, eosinophilic meningitis from raw snails/kuhol (Angiostrongylus), neurocysticercosis, rabies

Dexamethasone – bacterial meningitis [\(VERIFY\)](#)

IV

10 mg q6h $\times$ 4 days, first dose before/with antibiotic

Ceftriaxone – bacterial meningitis [\(VERIFY\)](#)

IV

2 g q12h

Vancomycin – meningitis [\(VERIFY\)](#)

IV q8–12h

15 – 20 mg/kg, max 2,000 mg

Load 25–30 mg/kg in critically ill; levels.

Ampicillin – Listeria cover [\(VERIFY\)](#)

IV

2 g q4h (if > 50 y, immunocompromised, pregnant, alcohol)

Acyclovir – HSV encephalitis [\(VERIFY\)](#)

IV q8h

10 mg/kg, max 1,000 mg

Use IBW in obesity; hydrate; renal adjust.

Neurology (extended) sources

Perry et al., Ottawa SAH rule (JAMA 2013) and CT within 6 h (BMJ 2011) · ACEP clinical policy: acute headache 2019 · AHS acute migraine treatment (Orr 2016) · Greenberg et al., AHA/ASA spontaneous ICH guideline 2022 · Hoh et al., AHA/ASA aSAH guideline 2023 · Kattah et al., HINTS (Stroke 2009); Edlow et al., GRACE-3 (Acad Emerg Med 2023) · van de Beek et al., ESCMID bacterial meningitis 2016; Tunkel et al., IDSA 2004; NICE NG240 (2024) · Tunkel et al., IDSA encephalitis 2008 · van Doorn et al., EAN/PNS GBS guideline 2023 · Sanders et al., MGFA consensus 2016/2020 · Baugh et al., AAO-HNS Bell palsy 2013. Tintinalli 9e Section 14 used only as a topic map.

⚠ Items pending owner review in this section (26)

1. Stroke time windows & eligibility (IV lytic ≤ 4.5 h; EVT ≤ 6 h, 6–24 h with imaging)

Follow your stroke team / imaging criteria; local capability (CTA/CTP, EVT) varies.

2. Tenecteplase for stroke 0.25 mg/kg (max 25 mg), 5 mg/mL

Product availability/registration and reconstitution differ (25 mg vs 50 mg vial).

3. Lorazepam IV 0.1 mg/kg (max 4 mg), 2 mg/mL assumed

Lorazepam injection may be unavailable/4 mg/mL locally – diazepam or midazolam often used instead.

4. IM midazolam weight bands (10 mg > 40 kg, 5 mg 13–40 kg; 0.2 mg/kg < 13 kg)

The < 13 kg 0.2 mg/kg value is extrapolated (RAMPART/AES only cover ≥ 13 kg).

5. Intranasal 0.2 mg/kg / buccal 0.3 mg/kg midazolam (max 10 mg)

Volumes per nostril and buccal dosing differ between protocols.

6. Rectal diazepam age-based dosing (0.5 / 0.3 / 0.2 mg/kg, max 20 mg)

Product-label banding; < 2 y not labeled. Uses IV solution 5 mg/mL rectally if no gel.

7. Phenytoin 20 mg/kg (cap 1500 mg), ≤ 50 mg/min (peds ≤ 1 mg/kg/min)

Cap and infusion-rate limits vary by reference; fosphenytoin often unavailable locally.

8. Phenobarbital 15–20 mg/kg

Maximum single dose and vial strength (30/60/65/130 mg/mL) vary – no mL computed.

9. Refractory status epilepticus infusions (midazolam 0.2 mg/kg then 0.05–2 mg/kg/hr; propofol 1–2 mg/kg then 20–200 mcg/kg/min)

ICU/neurology protocol dependent; EEG guided.

10. Magnesium sulfate for eclampsia (Zuspan 4 g IV + 1–2 g/hr, or Pritchard 4 g IV + 10 g IM)

Regimen choice and loading dose (4 vs 6 g) follow local OB / DOH protocol.

11. Pediatric doses are capped at an adult dose (engine-wide). Caps added where v1 had none: procainamide 1 g, bicarbonate 50 mEq, ketamine/propofol 200 mg, etomidate 30 mg, rocuronium 120 mg, succinylcholine 150 mg, fentanyl 100 mcg, phenobarbital 1.5 g, mannitol 100 g, fomepizole 1.5 g, methylene blue 200 mg, atropine (OP) 2 mg

For drugs dosed per kg in adults too, the cap is the adult dose at a ~100 kg reference weight – a policy choice for owner sign-off, not a guideline value.

12. Diazepam rectal/IV for seizures (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

13. Levetiracetam 60 mg/kg (max 4.5 g) – status epilepticus (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

14. Fosphenytoin 20 mg PE/kg (max 1.5 g PE) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

15. Valproate 40 mg/kg (max 3 g) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

16. Status epilepticus timeline doses (thiamine, dextrose, benzodiazepines, second-line agents)

Inline values from v1 / merged card, previously unflagged.

17. Alteplase 0.9 mg/kg (max 90 mg), 10% bolus – stroke card

Inline values from v1 / merged card, previously unflagged.

18. Migraine (adult): metoclopramide 10 mg IV or prochlorperazine 10 mg IV (+ diphenhydramine 25 mg for akathisia); ketorolac 15–30 mg IV; dexamethasone 10 mg IV to reduce recurrence; avoid opioids; peds: metoclopramide 0.1–0.15 mg/kg (max 10 mg), ibuprofen 10 mg/kg (max 400–600 mg)

ACEP 2019 / AHS 2016.

19. Spontaneous ICH: target SBP ~140 mmHg (avoid < 130), start within 2 h, smooth (labetalol 10–20 mg IV / nicardipine 5 mg/h titrate); reverse anticoagulation; aSAH: nimodipine 60 mg PO q4h $\times 21$ d, BP < 160 until aneurysm secured

AHA 2022 ICH / 2023 aSAH.

20. Bacterial meningitis (adult): dexamethasone 10 mg IV q6h $\times 4$ d (start with/before first antibiotic); ceftriaxone 2 g IV q12h + vancomycin 15–20 mg/kg q8–12h; + ampicillin 2 g IV q4h if > 50 y/immunocompromised/pregnant; acyclovir 10 mg/kg IV q8h if encephalitis possible. Contacts of meningococcal disease: ciprofloxacin 500 mg PO once (or rifampicin 600 mg BID $\times 2$ d; ceftriaxone 250 mg IM in pregnancy). Peds: ceftriaxone 50 mg/kg q12h (100 mg/kg/day, max 4 g), vancomycin 15 mg/kg q6h, dexamethasone 0.15 mg/kg q6h

IDSA 2004 / ESCMID 2016 / NICE NG240; PH pneumococcal resistance data limited.

21. Vertigo symptomatic relief (short course only): meclizine 25 mg PO, dimenhydrinate 50 mg PO/IV, diazepam 2–5 mg; vestibular neuritis: prednisone 50–60 mg $\times 5$ d (evidence limited)

GRACE-3 2023.

22. Bell palsy: prednisolone 50 mg $\times 10$ d or 60 mg $\times 5$ d then taper, within 72 h; add valacyclovir 1 g TID $\times 7$ d only in severe palsy (optional); eye protection

AAN/AAO-HNS 2013.

23. GBS/myasthenia respiratory failure thresholds ("20/30/40"): FVC < 20 mL/kg, NIF weaker than -30 cmH₂O, PEF/MEP < 40 cmH₂O $\rightarrow$ intubation/ICU; GBS: IVIG 0.4 g/kg/day $\times 5$ d or plasma exchange

EAN/PNS 2023; MGFA.

24. Malignant spinal cord compression: dexamethasone 10 mg IV load then 16 mg/day (some use 8–16 mg); MRI whole spine within 24 h

NICE NG234 (2023).

25. Coma "cocktail" – give based on findings: glucose (adult D50 25 g / child D10 2–5 mL/kg) if low, thiamine 100–500 mg IV before/with glucose if at risk, naloxone 0.04–0.4 mg IV titrated if opioid toxidrome

Standard practice.

26. Thiamine: prophylaxis 100–250 mg IV/IM daily; suspected Wernicke encephalopathy 500 mg IV TID $\times 3$ days then 250 mg daily

RCP/EFNS; give before or with glucose (do not delay glucose).

 Poisoned patient – first steps

- 1 ABCs first** – airway, breathing, circulation; glucose (point-of-care) in every altered patient; 12-lead ECG (QRS, QTc, R in aVR); temperature.
- 2 History:** what, how much (mg/kg), when, why (intentional?), co-ingestants, pill bottles/photos; staggered or sustained-release?
- 3 Look for a toxidrome** (next card) and an antidote; check paracetamol and salicylate levels in every intentional overdose.
- 4 Decontaminate selectively** △ VERIFY: activated charcoal **50 g** only if within ~1 h of a dangerous, charcoal-adsorbable ingestion and the airway is protected. Not for alcohols, iron, lithium, metals, caustics, hydrocarbons. Whole-bowel irrigation (PEG 1.5–2 L/hr) for iron, lithium, sustained-release drugs, body packers. **No induced emesis.**
- 5 Enhanced elimination:** urinary alkalinization (salicylate); hemodialysis for salicylate, lithium, toxic alcohols, metformin, valproate (EXTRIP – see Renal: AEIOU).
- 6 Call the poison center early** △ VERIFY: UP-PGH National Poison Management & Control Center (NPMCC) **(02) 8524-1078** · Globe **0966-718-9904**. Suicide risk assessment before disposition.

Observe asymptomatic ingestions 4–6 h (≥ 12–24 h for sustained-release, sulfonylureas, methadone, button batteries, paraquat). Hypoglycemia, hyperthermia and seizures kill early – treat them before chasing levels.

 Toxidromes

Toxidrome	Vitals	Mental status	Pupils	Skin	Clues	Typical agents
Sympathomimetic	↑ HR, ↑ BP, ↑ T	Agitated, paranoid	Dilated	Sweaty	Tremor, seizures, rhabdo	Methamphetamine (shabu), cocaine, caffeine excess
Anticholinergic	↑ HR, ↑ T	Delirium, picking, mumbling	Dilated, sluggish	Dry, flushed	Urinary retention, ↓ bowel sounds	Antihistamines, TCAs, Datura (talampanay)

Toxidrome	Vitals	Mental status	Pupils	Skin	Clues	Typical agents
Cholinergic	↓ or ↑ HR	Confused → coma	Pinpoint	Wet (sweat, saliva)	Bronchospasm, vomiting, diarrhea, fasciculations → weakness	Organophosphate / carbamate pesticides
Opioid	↓ RR, ↓ HR	Depressed	Pinpoint	Normal / cool	Hyperventilation; tramadol → seizures	Morphine, tramadol, fentanyl, nalbuphine
Sedative-hypnotic	Near-normal; mild ↓ RR	Depressed	Normal	Normal	Ataxia, slurred speech	Benzodiazepines, alcohol, Z-drugs
Serotonin syndrome	↑ HR, ↑ T, labile BP	Agitated	Dilated	Sweaty	Clonus (ocular/inducibles), hyperreflexia, legs > arms; onset hours	SSRI / SNRI + tramadol, linezolid, MAOI, dextrometorphan

Toxic rome	Vital s	Ment al statu s	Pupil s	Skin	Clues	Typic al agen ts
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NMS	↑ T, labile BP	Alter ed, mute	Norm al	Swea ty	Lead- pipe rigidi ty, ↓ reflex es; onset days	Antip sycho tics, meto clopr amid e, abru pt levod opa stop
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Sweaty vs dry skin separates sympathomimetic from anticholinergic. Clonus = serotonin; rigidity with slow onset = NMS. Mixed ingestions blur the picture.

Acetaminophen / paracetamol

Enter level and time for a single acute ingestion with a known time

[\(VERIFY\)](#)

- **Start NAC without waiting for a level** [\(VERIFY\)](#) if: > 8 h since ingestion with a potentially toxic dose (≥ 150 mg/kg or ≥ 10 g), unknown time, staggered ingestion, or any liver injury (raised ALT)
- Toxic dose [\(VERIFY\)](#) $\approx \geq 150$ mg/kg (≥ 200 mg/kg or 10 g in 24 h for repeated supratherapeutic use)
- Stop NAC when paracetamol undetectable and ALT normal/falling with INR < 2 (per local protocol). Liver unit referral: INR rising, pH < 7.3, creatinine rising, encephalopathy

Acetaminophen → N-acetylcysteine (IV 21-h) [\(VERIFY\)](#)

Three-bag regimen

150 mg/kg over 1 h → 50 mg/kg over 4 h → 100 mg/kg over 16 h

Dosing weight capped at 100 kg

NAC – two-bag IV regimen (alternative) [\(VERIFY\)](#)

200 mg/kg over 4 h → 100 mg/kg over 16 h

Bag 1: 200 mg/kg over 4 h · Bag 2: 100 mg/kg over 16 h
(dosing weight capped at 100 kg)

Salicylates

Early: tinnitus, vomiting, **tachypnea** (respiratory alkalosis) → mixed respiratory alkalosis + high anion gap metabolic acidosis → confusion, hyperthermia, pulmonary/cerebral edema. Repeat levels q2h until clearly falling (enteric-coated / bezoars absorb late).

- 1 Activated charcoal if alert and early; fluids for volume depletion (avoid overload)
- 2 **Urinary alkalization** for symptomatic patients or rising levels:

Sodium bicarbonate – salicylate alkalization [\(VERIFY\)](#)

Bolus (8.4%)

1 – 2 mEq/kg

Then infusion; repeat bolus if serum pH falls.

Infusion: 150 mEq NaHCO₃ in 1 L D5W

1.5–2 × maintenance rate (4-2-1)

Target urine pH 7.5–8.0, serum pH ≤ 7.55

- **Altered mental status = neuroglycopenia:** give dextrose even if serum glucose normal [\(VERIFY\)](#)
- **Avoid intubation if possible; if unavoidable: bicarbonate bolus first, then match pre-intubation minute ventilation (high RR)** [\(VERIFY\)](#)
- **Hemodialysis** (EXTRIP) [\(VERIFY\)](#): level > 100 mg/dL (7.2 mmol/L), > 90 mg/dL with impaired kidneys, altered mental status, new hypoxemia, pH ≤ 7.20 , or failure of standard therapy – call nephrology early (see AEIOU)

Naloxone

Naloxone [\(VERIFY\)](#)

Apneic / arrest / overdose

0.4 – 2 mg IV/IO/IM, repeat q2–3 min

0.4 mg/mL ampule. Reconsider diagnosis if no response after 10 mg.

Opioid-dependent, still breathing

0.04 – 0.1 mg IV, titrate q2–3 min

0.4 mg diluted to 10 mL = 0.04 mg/mL. Goal RR ≥ 10 , not full arousal.

Intranasal

4 mg spray; repeat in 2–3 min other nostril

Infusion $\frac{2}{3}$ of the effective bolus dose per hour [\(VERIFY\)](#).

TCA & sodium-channel blockers

TCAs, diphenhydramine (large), cocaine, propranolol, flecainide, carbamazepine. Danger signs: **QRS > 100 ms**, terminal R in aVR > 3 mm, right-axis deviation of terminal QRS, hypotension, seizures

[\(VERIFY\)](#)

TCA / Na-channel blocker (QRS > 100 ms) → bicarbonate

[\(VERIFY\)](#)

Sodium bicarbonate 8.4%

1 – 2 mEq/kg

Repeat until QRS narrows; target pH 7.45–7.55.

- Repeat boluses until QRS narrows / BP improves; target arterial pH 7.45–7.55; watch K and Na [\(VERIFY\)](#)
- Seizures: benzodiazepines (avoid phenytoin). Hypotension: fluids then norepinephrine. Refractory: hypertonic saline, lipid emulsion (see antidotes), ECMO
- Avoid physostigmine, class IA/IC antiarrhythmics; amiodarone only with expert advice. Lidocaine for refractory VT (expert)
- Asymptomatic with normal ECG at 6 h → medically clear (psychiatric assessment)

Cardiotoxic drugs (β-blocker, CCB, digoxin)

β-blocker / calcium-channel blocker

Bradycardia + hypotension ± hyperglycemia (CCB) or hypoglycemia (β-blocker; children). Atropine often fails. Early high-dose insulin is the key therapy; add vasopressors (norepinephrine/epinephrine), pacing, lipid or ECMO for refractory shock (AHA 2023).

β-blocker / Ca-channel blocker → calcium, HIET, glucagon

[\(VERIFY\)](#)

Calcium

CaCl₂ 10% 10–20 mL (1–2 g) or Ca gluconate 10% 30–60 mL; repeat q10–20 min PRN

High-dose insulin (HIET)

1 U/kg bolus, then 1–10 U/kg/hr + dextrose 0.5 g/kg/hr

Glucagon

3 – 5 mg IV over 1–2 min (≈ 50 mcg/kg), then 2 – 5 mg/hr

Digoxin

Vomiting, visual change, confusion; any dysrhythmia (classic: atrial tachycardia with block, bidirectional VT, regularized AF). Acute toxicity → hyperkalemia; chronic → often hypokalemia/renal impairment. Treat hypokalemia and hypomagnesemia; avoid calcium only if Fab is immediately available (old “stone heart” concern is largely refuted).
Fab indications (△ VERIFY): life-threatening dysrhythmia, $K > 5-5.5$ mmol/L in acute poisoning, end-organ hypoperfusion, or very high level. Without Fab: atropine/pacing for bradycardia, lidocaine for ventricular dysrhythmias, avoid cardioversion unless pulseless.

Digoxin → digoxin immune Fab (△ VERIFY)

Vials

Acute, unknown amount: 10 vials · Chronic: 3–6 vials adult / 1–2 vials child

Known level: vials = level (ng/mL) × weight (kg) / 100. Each 40 mg vial binds ≈ 0.5 mg.

Toxic alcohols

Enter measured osmolality, Na, glucose and BUN (mg/dL; urea mmol/L × 2.8 = BUN mg/dL) (△ VERIFY).

- **Methanol** (adulterated lambanog / bootleg spirits, outbreaks): delayed 12–24 h, blurred/“snowfield” vision, high anion gap acidosis. **Ethylene glycol** (antifreeze, brake fluid): intoxication → acidosis, AKI, oxalate crystals, hypocalcemia. **Isopropanol**: ketosis without acidosis – supportive
- Treat on suspicion (history + acidosis or osmolar gap) – do not wait for levels (rarely available)

Methanol / ethylene glycol → fomepizole (△ VERIFY)

Loading dose

15 mg/kg

Then 10 mg/kg q12h × 4, then 15 mg/kg q12h; extra dosing during dialysis. Ethanol if unavailable.

Ethanol (if no fomepizole) (△ VERIFY)

10% ethanol IV load (over 30–60 min)

8 mL/kg

= 0.8 g/kg. Central line preferred (hyperosmolar).

10% IV maintenance

0.8–2 mL/kg/hr; double during HD

Oral 40% spirits (e.g. 80-proof) load

2 mL/kg

Then ≈ 0.2–0.4 mL/kg/hr; lambanog strength varies – check label.

- Correct acidosis with bicarbonate ($pH < 7.3$) – less formic acid enters the brain/eye
- Adjuncts (△ VERIFY): methanol → folinic/folic acid 50 mg IV q6h; ethylene glycol → thiamine 100 mg + pyridoxine 50 mg IV q6h
- **Hemodialysis**: severe acidosis, visual signs, coma/seizures, AKI, or high level – removes both alcohol and toxic acids (EXTRIP)

Pesticides

Organophosphate / carbamate (cholinergic crisis)

Decontaminate (remove clothes, wash skin; staff PPE). Killers: bronchorrhea, bronchospasm, respiratory muscle weakness. Atropinize fast, by doubling, to a dry chest – not to pupil size. Avoid succinylcholine (prolonged paralysis) – use rocuronium. Watch for intermediate syndrome (proximal/neck/respiratory weakness) at 24–96 h.

Organophosphate / carbamate → atropine + pralidoxime

(△ VERIFY)

Atropine

1 – 2 mg IV, double every 5 min until atropinized

Targets: clear chest, HR > 80, SBP > 80, dry axillae, pupils not pinpoint. Then infusion 10–20% of total loading dose per hour.

Pralidoxime

30 mg/kg (max 2 g) over 30 min, then 8 mg/kg/hr

Paraquat (e.g. gramoxone) – very high lethality

- Burning mouth/throat ulcers, vomiting → AKI, liver injury, progressive lung fibrosis
- Decontaminate early: activated charcoal or Fuller’s earth 1 g/kg (△ VERIFY); **avoid supplemental O₂ unless SpO₂ < 88–90%** (oxygen worsens lung injury)
- Urine dithionite test (blue = positive) supports exposure; early hemoperfusion/immunosuppression are unproven – NPMCC + palliative discussion

Glyphosate-surfactant · rodenticides

- Glyphosate: GI corrosion, hypotension, acidosis, hyperkalemia, AKI – supportive; dialysis for refractory acidosis/hyperkalemia
- Superwarfarin rat poison (brodifacoum): INR at 24–48 h; bleeding → vitamin K (high doses, weeks–months) + PCC/FFP. Zinc/aluminium phosphide fumigants: refractory shock, phosphine gas risk – supportive, protect staff

Agitated / hyperthermic poisoned patient

- 1 **Temperature > 40 °C = emergency**: rapid external cooling (ice-water immersion / ice packs + evaporative); paralysis + intubation if rigid/agitated hyperthermia persists (antipyretics do not work)
- 2 **Benzodiazepines first** for agitation, tachycardia, hypertension and seizures; minimise physical restraint (rhabdo, sudden death)

Benzodiazepine – toxic agitation / seizures (△ VERIFY)

Diazepam IV

5 – 10 mg q5–10 min, titrate to calm

Midazolam

2.5 – 5 mg IV, or 5 – 10 mg IM; repeat PRN

- **Sympathomimetic** (shabu, cocaine): treat chest pain as ACS (aspirin, nitrates, benzodiazepines); hypertension → benzodiazepine, then phentolamine or nitroglycerin; cocaine wide-QRS → bicarbonate (TCA card). Avoid pure β-blockade early (unopposed α concern – evidence mixed)
- **Serotonin syndrome** (Hunter): stop serotonergic drugs, benzodiazepines, cool; cyproheptadine for moderate cases; severe ($T > 41$ °C, rigidity) → paralyse and ventilate
- **NMS**: stop the antipsychotic, cool, fluids, benzodiazepines; ICU for rigidity/hyperthermia; check CK
- **Anticholinergic delirium**: benzodiazepines; physostigmine only with toxicology advice and a narrow QRS (see antidotes)

Cyproheptadine – serotonin syndrome (△ VERIFY)

PO / crushed via NG

12 mg, then 2 mg q2h while symptomatic

Maintenance 8 mg q6h. No IV form.

NMS / malignant hyperthermia-like states ⚠️ VERIFY

Dantrolene IV

1 – 2.5 mg/kg

Repeat to max 10 mg/kg/day. Adjunct only – cool and sedate first.

Bromocriptine (NMS, adult)

2.5 mg PO/NG q8h



Other common poisonings

Iron

- Toxicity by ⚠️ VERIFY **elemental iron**: < 20 mg/kg usually benign; 20–60 mg/kg GI toxicity; > 60 mg/kg potentially lethal. Vomiting/diarrhea → latent phase → shock, acidosis, liver failure
- Abdominal X-ray (tablets radio-opaque); WBI if tablets beyond pylorus; charcoal ineffective; serum iron at 4–6 h; deferoxamine for shock, acidosis, altered status, or high levels

Iron → deferoxamine ⚠️ VERIFY

IV infusion

15 mg/kg/hr IV

Lithium ⚠️ VERIFY

- Acute: GI symptoms first, neurotoxicity later; chronic (often AKI/dehydration, new ACEi/thiazide/NSAID): tremor, ataxia, confusion, seizures at lower levels
- No charcoal; WBI for sustained-release; isotonic fluids to restore urine output; avoid nephrotoxins; dialysis per EXTRIP (AEIOU)

Methemoglobinemia

- Cyanosis/“chocolate” blood with SpO₂ stuck ~85% not responding to O₂; dapsone, topical benzocaine/prilocaine, nitrites, aniline dyes, some herbal remedies. Co-oximetry Methb; treat if symptomatic or > 20–30%

Methemoglobinemia → methylene blue 1% ⚠️ VERIFY

IV over 5 min

1 – 2 mg/kg

Repeat 1 mg/kg after 30–60 min if needed (max ~7 mg/kg). G6PD deficiency (common in the Philippines): ineffective/hemolysis – avoid; serotonergic drugs: serotonin syndrome.

Caustics & button batteries

- Caustics (lye, toilet/drain cleaners, soda): no emesis, no neutralisation, no charcoal; airway assessment (stridor, drooling → early intubation by expert); endoscopy within 12–24 h if symptomatic or intentional
- **Button battery**: X-ray neck–abdomen immediately; **esophageal battery = endoscopic removal within 2 h**; meanwhile, for age ≥ 1 y and ≤ 12 h:

Button battery – mucosal protection while arranging removal ⚠️ VERIFY

Honey (age ≥ 1 y, ≤ 12 h since ingestion)

10 mL PO q10 min, up to 6 doses

Sucralfate (in hospital)

1 g = 10 mL PO q10 min × 3

Hydrocarbons (kerosene, gasoline, lamp oil)

- Main danger is **aspiration pneumonitis** (cough, choking, tachypnea) – no emesis, no charcoal, no lavage
- Asymptomatic: observe 6 h + CXR; symptomatic: O₂, admit; antibiotics/steroids not routine. Huffing: sudden sniffing death (catecholamine-triggered VF) – avoid adrenergic stimulation



Common antidotes

Acetaminophen → N-acetylcysteine (IV 21-h) ⚠️ VERIFY

Three-bag regimen

150 mg/kg over 1 h → 50 mg/kg over 4 h → 100 mg/kg over 16 h

Dosing weight capped at 100 kg

Organophosphate / carbamate → atropine + pralidoxime

⚠️ VERIFY

Atropine

1 – 2 mg IV, double every 5 min until atropinized

Targets: clear chest, HR > 80, SBP > 80, dry axillae, pupils not pinpoint. Then infusion 10–20% of total loading dose per hour.

Pralidoxime

30 mg/kg (max 2 g) over 30 min, then 8 mg/kg/hr

β-blocker / Ca-channel blocker → calcium, HIET, glucagon

⚠️ VERIFY

Calcium

CaCl₂ 10% 10–20 mL (1–2 g) or Ca gluconate 10% 30–60 mL; repeat q10–20 min PRN

High-dose insulin (HIET)

1 U/kg bolus, then 1–10 U/kg/hr + dextrose 0.5 g/kg/hr

Glucagon

3 – 5 mg IV over 1–2 min (≈ 50 mcg/kg), then 2 – 5 mg/hr

TCA / Na-channel blocker (QRS > 100 ms) → bicarbonate

⚠️ VERIFY

Sodium bicarbonate 8.4%

1 – 2 mEq/kg

Repeat until QRS narrows; target pH 7.45–7.55.

Benzodiazepine → flumazenil (rarely indicated) ⚠️ VERIFY

IV

0.2 mg over 30 s, then 0.2 mg q1 min; max 1 mg

Avoid in unknown/mixed overdose, chronic benzo use, seizure history, TCA – may precipitate seizures. Re-sedation likely.

Methemoglobinemia → methylene blue 1% ⚠️ VERIFY

IV over 5 min

1 – 2 mg/kg

Repeat 1 mg/kg after 30–60 min if needed (max ~7 mg/kg). G6PD deficiency (common in the Philippines): ineffective/hemolysis – avoid; serotonergic drugs: serotonin syndrome.

Cyanide (smoke inhalation) → hydroxocobalamin ⚠️ VERIFY

IV over 15 min

5 g (may repeat 5 g)

5 g vial in 200 mL NS = 25 mg/mL.

Methanol / ethylene glycol → fomepizole ⚠️ VERIFY

Loading dose

15 mg/kg

Then 10 mg/kg q12h × 4, then 15 mg/kg q12h; extra dosing during dialysis. Ethanol if unavailable.

Digoxin → digoxin immune Fab ⚠ VERIFY

Vials

Acute, unknown amount: 10 vials · Chronic: 3–6 vials adult / 1–2 vials child

Known level: vials = level (ng/mL) × weight (kg) / 100. Each 40 mg vial binds ≈ 0.5 mg.

Iron → deferoxamine ⚠ VERIFY

IV infusion

15 mg/kg/hr IV

Heparin → protamine ⚠ VERIFY

Slow IV ≤ 5 mg/min

1 mg per 100 units heparin given in last 2–3 h; max 50 mg

10 mg/mL. Enoxaparin: 1 mg per 1 mg enoxaparin (within 8 h).

Warfarin (major bleed) → vitamin K + PCC ⚠ VERIFY

Vitamin K

10 mg slow IV over 20–30 min

4-factor PCC

INR 2–<4: 25 IU/kg · 4–6: 35 IU/kg · > 6: 50 IU/kg (weight cap 100 kg)

Isoniazid (seizures) → pyridoxine ⚠ VERIFY

IV 0.5–1 g/min until seizures stop

1 g per g ingested

Unknown: 70 mg/kg, max 5 g

Sulfonylurea → octreotide (+ dextrose) ⚠ VERIFY

SC/IV

50 mcg q6h

Observe ≥ 24 h.

Anticholinergic delirium → physostigmine ⚠ VERIFY

Slow IV over 5 min

0.5 – 2 mg (e.g. 0.5 mg q5 min)

Avoid if QRS widening/TCA, bradycardia, AV block. Atropine ready.

Local anesthetic toxicity → lipid emulsion 20% ⚠ VERIFY

ASRA regimen

Bolus 1.5 mL/kg over 2–3 min

Infusion 0.25 mL/kg/min

Upper limit ≈ 12 mL/kg

Activated charcoal ⚠ VERIFY

PO/NG

50 g

Within ~1 h of an adsorbable toxin; airway protected. Not for alcohols, iron, lithium, caustics, hydrocarbons.

Snake envenomation (e.g. Philippine cobra) ⚠ VERIFY

Antivenom

Per product insert / poison center

Immobilize limb, airway/ventilation support for neurotoxicity; neostigmine + atropine trial per protocol.

Contact your regional poison center (e.g. UP-PGH National Poison Management & Control Center) early.

 Toxicology sources

AACT/EAPCCT position papers on GI decontamination (single-dose activated charcoal 2005, whole-bowel irrigation 2015 update) · Chiew et al., Australian & NZ guideline on paracetamol poisoning 2019 (2-bag regimen) · Dart et al., US expert consensus on acetaminophen poisoning 2023 · UK MHRA/CHM paracetamol 100-line (2012) · EXTRIP workgroup (salicylate, lithium, toxic alcohol, metformin) · AHA 2023 focused update on management of patients in cardiac arrest or life-threatening toxicity due to poisoning · Eddleston et al. / WHO guidance on organophosphate poisoning · Boyer & Shannon, serotonin syndrome (Hunter criteria) · National Capital Poison Center button-battery triage & treatment guideline (2018 honey/sucralfate) · AHA 2020 ACLS (toxicology-related arrest) · UP-PGH National Poison Management & Control Center (NPMCC). Tintinalli 9e Section 15 used only as a topic map.

⚠ Items pending owner review in this section (38)

1. Glucagon (anaphylaxis on β-blocker / β-blocker overdose): adult 1–5 mg, peds 20–30 mcg/kg (max 1 mg); tox 50 mcg/kg

Doses differ between allergy and toxicology references.

2. Pediatric naloxone for therapeutic-opioid respiratory depression 1–5 mcg/kg

Titration dose ranges vary.

3. Naloxone infusion = ⅓ of effective bolus dose per hour

Empiric rule; titrate.

4. N-acetylcysteine IV 21-h regimen (150 / 50 / 100 mg/kg; weight cap 100 kg)

Some centers use 2-bag (200 + 100 mg/kg) or SNAP regimens; diluent volumes for children differ.

5. Organophosphate: atropine doubling regimen and pralidoxime (adult 30 mg/kg max 2 g then 8 mg/kg/hr; peds 25–50 mg/kg then 10–20 mg/kg/hr)

Regimens (WHO/Eddleston) and pralidoxime availability vary.

6. β-blocker / CCB toxicity: high-dose insulin (1 U/kg then 1–10 U/kg/hr), calcium repeats, glucagon

Toxicology-consult regimens; titration ceilings vary.

7. Pediatric hydroxocobalamin 70 mg/kg (max 5 g)

Pediatric labeling limited.

8. Fomepizole 15 mg/kg then 10 mg/kg q12h

Often unavailable locally – ethanol protocol needed.

9. Digoxin Fab vial numbers

Product-specific (DigiFab 40 mg vial); availability limited.

10. Deferoxamine 15 mg/kg/hr (iron)

Daily maximum (commonly ~6–8 g/24 h) differs between references.

11. Warfarin reversal: 4F-PCC 25–50 IU/kg by INR; pediatric vitamin K

PCC product/availability varies; pediatric vitamin K dose not given – per hematology/poison center.

12. Isoniazid: pyridoxine 1 g per g ingested, or 70 mg/kg (max 5 g)

Pyridoxine IV stock usually insufficient – check pharmacy.

13. Octreotide for sulfonylurea hypoglycemia (adult 50 mcg q6h; peds 1–1.5 mcg/kg max 50 mcg)

Dosing varies by reference.

14. Physostigmine (adult 0.5–2 mg; peds 0.02 mg/kg max 0.5 mg)

Toxicology consultation recommended; often not stocked.

15. Snake envenomation (e.g. Philippine cobra) – antivenom and neostigmine trial

No doses given: follow product insert and poison center/DOH protocol.

16. Pediatric calcium (CaCl₂ 10% 20 mg/kg max 2 g; gluconate 10% 60 mg/kg max 3 g)

Maximum single doses differ (some cap CaCl₂ at 1 g).

17. Pediatric doses are capped at an adult dose (engine-wide). Caps added where v1 had none: procainamide 1 g, bicarbonate 50 mEq, ketamine/propofol 200 mg, etomidate 30 mg, rocuronium 120 mg, succinylcholine 150 mg, fentanyl 100 mcg, phenobarbital 1.5 g, mannitol 100 g, fomepizole 1.5 g, methylene blue 200 mg, atropine (OP) 2 mg

For drugs dosed per kg in adults too, the cap is the adult dose at a ~100 kg reference weight – a policy choice for owner sign-off, not a guideline value.

18. Antidote doses carried over from v1 that had no review flag (bicarbonate for Na-channel blockade, flumazenil, methylene blue, adult hydroxocobalamin, protamine, lipid emulsion, charcoal, naloxone adult)

Flagged for completeness in the Toxicology sample section; values unchanged from v1.

19. Salicylate hemodialysis (EXTRIP 2015): level > 100 mg/dL (7.2 mmol/L), > 90 mg/dL with impaired kidney function, altered mental status, new hypoxemia, or pH ≤ 7.20 despite treatment

EXTRIP thresholds as recalled; confirm units (mg/dL vs mmol/L) with the lab.

20. Iron toxicity by elemental iron: < 20 mg/kg usually benign, 20–60 mg/kg GI toxicity, > 60 mg/kg potentially lethal; serum iron > 500 mcg/dL at 4–6 h = severe

Traditional thresholds; poor correlation with outcome – clinical status and anion gap acidosis matter more.

21. UP-PGH NPMCC hotline (02) 8524-1078 / Globe 0966-718-9904 (24 h)

Numbers taken from NPMCC public pages (2025–26); confirm and add your regional poison center / toxicology referral number.

22. Decontamination thresholds (charcoal 1 g/kg max 50 g within ~1 h; WBI with PEG 1.5–2 L/hr adult, 25 mL/kg/hr child, for iron, lithium, sustained-release drugs, body packers)

AACT/EAPCCT position papers – evidence weak; airway must be protected; owner to confirm local practice.

23. Paracetamol nomogram: treatment line 150 mg/L (≈ 1,000 μmol/L) at 4 h halving every 4 h; UK uses 100 mg/L line; staggered/unknown-time ingestion → treat

US/ANZ use the 150 line; UK treats on the 100 line. Choose the institutional standard.

24. NAC two-bag regimen (200 mg/kg over 4 h then 100 mg/kg over 16 h), weight cap 100 kg here (ANZ uses 110 kg); massive ingestion may need higher-dose NAC

ANZ 2019 consensus; US 2023 consensus allows either regimen. Weight cap and high-dose rules vary.

25. Salicylate alkalization: NaHCO₃ 1–2 mEq/kg bolus then 150 mEq/L in D5W at 1.5–2× maintenance; urine pH 7.5–8.0; blood pH ≤ 7.55; keep K ≥ 4

Common toxicology regimen (ACMT/EXTRIP context) – infusion recipes differ; hypokalemia blocks urinary alkalization.

26. Salicylate: avoid intubation if possible; if intubated, bicarbonate bolus first and match the pre-intubation minute ventilation (often RR 30+); dextrose for neuroglycopenia despite normal glucose

Expert practice, low-quality evidence.

27. Na-channel blocker toxicity thresholds (QRS > 100 ms, R in aVR > 3 mm) and bicarbonate 1–2 mEq/kg boluses to QRS narrowing, target pH 7.45–7.55; hypertonic saline / lipid if refractory

AHA 2023 toxicology update and tox-center practice; maximum cumulative bicarbonate is clinical.

28. Digoxin Fab indications (life-threatening dysrhythmia, K > 5–5.5 mmol/L in acute toxicity, end-organ hypoperfusion, level > 10–15 ng/mL acute) and empiric vial numbers

Thresholds differ between references; Fab availability in the Philippines is limited.

29. Osmolar gap = measured – (2Na + glucose/18 + BUN/2.8 + ethanol/3.7) with mg/dL units; gap > 10 suspicious

Ethanol divisor 3.7 vs 4.6 and the normal range (–14 to +10) vary; a normal gap does not exclude late toxic alcohol poisoning.

30. Ethanol as alcohol dehydrogenase blocker (10% IV: 8 mL/kg load, 0.8 mL/kg/hr non-drinker / 1.6–2 mL/kg/hr chronic drinker, doubled on HD; oral 40% spirits ≈ 2 mL/kg load); target serum ethanol 100–150 mg/dL

Only if fomepizole unavailable (usual in PH); regimens vary – levels hourly; hypoglycemia in children.

31. Toxic alcohol adjuncts (methanol: folinic/folic acid 50 mg IV q6h; ethylene glycol: thiamine 100 mg + pyridoxine 50 mg IV q6h)

Low-quality evidence; dosing intervals vary.

32. Paraquat: avoid supplemental O₂ unless SpO₂ < 88–90%; early charcoal or Fuller’s earth 1 g/kg; immunosuppression (cyclophosphamide + methylprednisolone) not routinely recommended

Evidence conflicting; discuss every case with NPMCC.

33. Benzodiazepine for toxic agitation/seizures (diazepam 5–10 mg IV q5–10 min; midazolam 2.5–5 mg IV / 5–10 mg IM; peds diazepam 0.1–0.2 mg/kg max 10 mg, midazolam IM 0.1–0.2 mg/kg max 10 mg)

Titration to effect; large cumulative doses may be needed in sympathomimetic toxicity – airway readiness.

34. Cyproheptadine for serotonin syndrome (12 mg PO/NG, then 2 mg q2h while symptomatic; maintenance 8 mg q6h)

Oral only; evidence limited; pediatric dosing not established here.

35. NMS / severe hyperthermia adjuncts (dantrolene 1–2.5 mg/kg IV, max 10 mg/kg/day; bromocriptine 2.5 mg PO q8h)

Weak evidence; cooling and benzodiazepines are first line.

36. Button battery: honey 10 mL q10 min up to 6 doses (age ≥ 1 y, ≤ 12 h, swallowing OK) pre-endoscopy; sucralfate 1 g (10 mL) q10 min × 3 in hospital; esophageal battery → endoscopic removal within 2 h

NCPC 2018 guideline based on animal data; must not delay removal.

37. Lithium: no charcoal; WBI for sustained-release; isotonic fluids to normal urine output; EXTRIP dialysis criteria (see Renal: AEIOU)

Fluid volume/rate depends on cardiac and renal function.

38. UP-PGH National Poison Management and Control Center: (02) 8524-1078 · Globe 0966-718-9904

Numbers from public NPMCC listings – confirm before printing for use.

❄️ Hypothermia

Enter core temperature (low-reading thermometer; esophageal if intubated).

- 1 Handle gently and keep horizontal; remove wet clothes, insulate (vapour barrier); warm humidified O₂; check glucose
- 2 Check for a pulse for up to **1 minute** (bradycardia, vasoconstriction); POCUS/capnography help. Low BP + organised rhythm = do not start CPR if any signs of life
- 3 **Arrest:** standard CPR rate; < 30 °C: no adrenaline, max 3 shocks; 30–35 °C: double the interval; mechanical CPR for transport; **ECLS (VA-ECMO)** rewarming **△ VERIFY**
- 4 Prognosis: "not dead until warm and dead" – but stop if obvious lethal injury, body frozen solid, airway packed with snow/ice and asystole, or K⁺ > 12 mmol/L (HOPE score better) **△ VERIFY**
- 5 Rewarming afterdrop/collapse: watch for hypovolemia, arrhythmia; consider sepsis, hypoglycemia, hypothyroidism, adrenal insufficiency, drugs/alcohol in "urban" hypothermia (PH: elderly, near-drowning, flood victims)

🔥 Heat illness

Heat stroke = core > 40 °C + CNS dysfunction (confusion, seizure, coma). **Heat exhaustion** = no significant CNS dysfunction, core usually < 40 °C. Classic (elderly, chronic illness, hot humid PH summer) vs exertional (athletes, soldiers, workers).

- 1 **Cool first, transport second.** Measure rectal temperature. **Cold/ice-water immersion** up to the neck (1–15 °C), stir water, keep airway above water – target ≥ 0.15 °C/min **△ VERIFY**
- 2 No tub: tarp-assisted cooling, rotating ice-water-soaked towels, ice packs + evaporative (mist + fan) – slower
- 3 **Stop at –38.6–39 °C** to avoid overshoot hypothermia **△ VERIFY**; continue to monitor
- 4 No antipyretics (acetaminophen/NSAIDs ineffective, harmful to liver/kidney); dantrolene not recommended; benzodiazepine for shivering/seizures
- 5 Check glucose, Na, K, CK, creatinine, LFTs, coagulation (DIC) – rhabdomyolysis, AKI, liver failure often peak at 24–72 h

Exercise-associated hyponatremia (mimic)

Endurance event + excess water, confusion/seizure with normal-ish temperature: check Na before giving fluids. Do not give hypotonic fluids.

3% NaCl – symptomatic exercise-associated hyponatremia

△ VERIFY

IV bolus

100 mL over ~1 min; repeat up to 2 more times at 10-min intervals

- Heat cramps/syncope/edema: rest, shade, oral electrolyte solution; return to activity gradually
- Drug contributors: anticholinergics, sympathomimetics, antipsychotics (NMS), serotonergic (see Tox: hyperthermic syndromes)

🌊 Drowning

- 1 Hypoxia is the cause of arrest: **5 initial rescue breaths**, then CPR (30:2); early airway and O₂ **△ VERIFY**
- 2 Do not attempt to drain water (no abdominal thrusts); vomiting is common – suction, lateral position
- 3 Spinal precautions only if injury mechanism (diving, surf, waterslide) or signs

- 4 Hypoxemia: O₂, early CPAP/NIV; intubate if failing – lung-protective ventilation with PEEP; bronchospasm → salbutamol
- 5 Assess for hypothermia (see above), trauma, intoxication, seizure or arrhythmia (long QT) as the cause
- 6 **Asymptomatic** (normal SpO₂, no cough, normal chest exam) → observe ≥ 4–6 h then discharge; any symptom → admit; prophylactic antibiotics/steroids not indicated (consider in grossly contaminated/flood water) **△ VERIFY**

⚡ Lightning & electrical injury

PH household supply is **220 V, 60 Hz** (AC: tetanic "can't let go"). High voltage ≥ 1,000 V (power lines). Skin burns underestimate deep muscle injury.

- 1 **Scene safety** – isolate power first. **Lightning mass casualty: reverse triage** – resuscitate those apparently dead first (respiratory arrest; survivors breathing usually do well); prolonged ventilation may be needed
- 2 ECG for all; cardiac monitoring 24 h if loss of consciousness, abnormal ECG/arrhythmia, chest pain, high voltage, known heart disease **△ VERIFY**. Asymptomatic household-voltage injury with normal ECG → discharge
- 3 Look for: compartment syndrome, fractures/dislocations (posterior shoulder), spinal injury from fall, cataracts, TM rupture (lightning), keraunoparalysis (transient limb paralysis), fern-pattern skin (Lichtenberg figures)
- 4 CK, myoglobinuria, creatinine, K: **urine output 1–1.5 mL/kg/hr** if myoglobinuria **△ VERIFY**; Parkland (burns card) underestimates fluid needs
- 5 Pregnancy: fetal monitoring – fetal injury possible even with minor maternal injury. Oral commissure burns in children (biting cords): delayed labial artery bleeding at 5–14 days

🏔️ Altitude illness

Rare in PH ED practice but relevant for climbers (Mt Apo 2,954 m, Pulag 2,926 m) and travellers abroad. Typically > 2,500 m; onset 6–12 h (AMS), 2–5 days (HAPE).

- **AMS:** headache + GI upset, fatigue, dizziness → stop ascent, analgesia, antiemetic; acetazolamide; descend if worsening
- **HACE:** ataxia, altered mental status in AMS → **descend** (≥ 300–1,000 m), O₂, dexamethasone, portable hyperbaric bag
- **HAPE:** dyspnea at rest, cough, ↓ exercise tolerance, crackles, SpO₂ low for altitude → **descend + O₂** (SpO₂ > 90%); nifedipine if descent/O₂ unavailable; no diuretics

Acetazolamide **△ VERIFY**

Prevention

125 mg PO BID (start day before ascent)

AMS treatment

250 mg PO BID

Dexamethasone – AMS / HACE **△ VERIFY**

Moderate–severe AMS

4 mg PO/IM/IV q6h

HACE

8 mg IV/IM/PO once, then 4 mg q6h

Nifedipine ER – HAPE (if descent/O₂ unavailable) **△ VERIFY**

PO

30 mg ER q12h (or 20 mg ER q8h)

Diving (brief)

- Symptoms within 24 h of diving (joint pain, neuro deficit, rash, dyspnea, confusion): **100% O₂**, supine, fluids, no flying – contact a hyperbaric facility / DAN; arterial gas embolism = sudden neuro deficit within 10 min of surfacing

△ VERIFY

Snakebite

Medically important PH snakes: Philippine cobra (*Naja philippinensis*, Luzon/Mindoro – neurotoxic, spits), Samar cobra (*N. samarensis*, Visayas/Mindanao), king cobra (*Ophiophagus hannah*), kraits (rare), pit vipers (*Trimeresurus/Parias flavomaculatus*, Wagler's *Tropidolaemus* – mostly local swelling/pain, occasionally coagulopathy), sea snakes (*Hydrophis*, *Laticauda* – myotoxic/neurotoxic).

First aid

- Reassure, immobilise the whole limb (splint), keep still, remove rings/watches, transport now
- **No tourniquet**, cutting, suction, ice, electric shock or herbal "tandok"; pressure-pad/bandage for suspected elapid (cobra) bite only if trained and it doesn't delay transport **△ VERIFY**
- Spitting cobra eye exposure: irrigate copiously with water/saline, fluorescein, topical antibiotic; no antivenom drops

Assess (repeat hourly for 24 h – asymptomatic ≥ 24 h before discharge)

- **Neurotoxic:** ptosis (earliest), diplopia, dysphagia, drooling, dysarthria, neck flexor weakness, single-breath count, paradoxical breathing → prepare to intubate
- **Hemotoxic:** 20WBCT, bleeding gums/sites, hematuria · **Myotoxic** (sea snake): myalgia, dark urine, ↑ CK, ↑ K · Local: swelling progression (mark edge hourly), blistering, necrosis; true compartment syndrome is rare – measure pressure before fasciotomy

20-minute whole blood clotting test **△ VERIFY**

1–2 mL venous blood → new, clean, dry **glass** tube → leave 20 min → tip once.

Antivenom

- **Indications:** any neurotoxic sign, shock, spontaneous systemic bleeding or unclotted 20WBCT, rapidly progressive swelling (> half the limb in 48 h / bite on digit), myoglobinuria/hyperkalemia (sea snake)
- **PCAV** (RITM, monovalent *N. philippinensis*) – antivenom for pit vipers and sea snakes is generally **not** locally produced; ask NPMCC for availability **△ VERIFY**
- Same dose for children (venom dose, not weight); give in an area with epinephrine ready – see anaphylaxis

Purified Cobra Antivenom (PCAV, RITM) – *Naja philippinensis* **△ VERIFY**

IV infusion (same dose adults & children)

5 ampoules (severe neurotoxicity) in 0.9% NaCl over ~30–60 min

Reconstitute per insert. Epinephrine IM ready for reactions (stop infusion, treat, then restart slowly). Reassess at 1–2 h; repeat per protocol/NPMCC.

Neostigmine trial (cobra neurotoxicity)

Neostigmine trial – step 1: atropine **△ VERIFY**

IV

0.6 mg

Neostigmine trial – step 2: neostigmine **△ VERIFY**

IM

1.5 – 2.0 mg

Reassess ptosis, single-breath count, neck flexion at 10–20 min.

Positive → 0.5 mg IV/SC q30 min + atropine 0.6 mg q8h.

- Tetanus prophylaxis; antibiotics only for infected wounds/necrosis; analgesia (avoid NSAIDs if coagulopathic)
- **UP-PGH NPMCC (24 h):** (02) 8524-1078 · Globe 0966-718-9904 **△ VERIFY**

Spiders, scorpions, stings

- **Spiders:** most PH spider bites cause only local pain/redness. Widow-type (*Latrodectus*) envenoming → regional pain, sweating, muscle cramps, hypertension: analgesia (opioids), benzodiazepines; widow antivenom generally unavailable locally. "Necrotic spider bite" is usually infection (MRSA) – treat as cellulitis/abscess **△ VERIFY**
- **Scorpions** (PH species): local pain, paresthesia – ice, analgesia, local anesthetic infiltration/digital block; systemic toxicity rare (watch children: hypertension, agitation, hypersalivation)
- **Centipedes** (*Scolopendra*): intense local pain, swelling – hot water immersion (40–45 °C), analgesia, tetanus; rare rhabdomyolysis/AKI
- **Bees, wasps, hornets:** remove stings quickly by any method (speed matters more than scraping); local reaction → cold compress, antihistamine; any systemic features → IM epinephrine; **mass envenomation** (> 50 stings adult, fewer in children; hornets) → admit for rhabdomyolysis, hemolysis, AKI **△ VERIFY**
- Fire ants, caterpillars (lepidopterism): local care, antihistamine, remove hairs with tape

Epinephrine IM – anaphylaxis **△ VERIFY**

IM anterolateral thigh

0.01 mg/kg, max 0.5 mg

1 mg/mL (1:1,000). Repeat every 5–15 min if no improvement.

Marine envenomation

- 1 **Box jellyfish** (*Chironex/Chiropsalmus*; fatalities in PH waters): get out of water, call for help, **CPR if needed**. Flood the sting with **vinegar for ≥ 30 s** (stops undischarged nematocysts), then pick off tentacles; then hot water (~45 °C, 20 min) or ice for pain **△ VERIFY**
- 2 **Never** rub, apply fresh water, urine, alcohol or sand (triggers discharge). **Bluebottle/Portuguese man-o'-war** (*Physalia*): no vinegar – rinse with seawater, remove tentacles, hot water
- 3 Irukandji-like syndrome (delayed severe back/abdominal pain, sweating, hypertension, pulmonary edema): opioids, antihypertensive (phentolamine/GTN), ICU. CSL box jellyfish antivenom is not routinely available in PH

Venomous spines (stingray, stonefish, scorpionfish, catfish, sea urchin)

- **Hot water immersion 40–45 °C for 30–90 min** (test with the rescuer's hand – avoid scalds), analgesia/regional block **△ VERIFY**
- X-ray/ultrasound for retained spines; explore and irrigate; stingray wounds to chest/abdomen → treat as penetrating trauma
- Tetanus; antibiotics for deep wounds/high-risk hosts with **Vibrio cover** (doxycycline + ceftazidime or ciprofloxacin) **△ VERIFY**; *V. vulnificus* (liver disease, seafood/sea-water wound) → necrotising infection, sepsis

Seafood poisoning

- **Scombroid** (tuna, mackerel, "galunggong"): flushing, headache, urticaria within 1 h → H1 + H2 antihistamines; not an allergy
- **Tetrodotoxin** (pufferfish – *butete*; some crabs): perioral numbness → ascending paralysis, respiratory failure within hours → airway and ventilation support; no antidote; good recovery if ventilated

- **Paralytic shellfish poisoning** (red tide – check BFAR shellfish bulletin): similar to tetrodotoxin → supportive ventilation; cooking does not destroy the toxin
- Ciguatera (reef fish): GI then neuro symptoms, temperature reversal – supportive

Carbon monoxide

PH context: generators and charcoal grills indoors during typhoon power outages, vehicles in closed garages, fires (with **cyanide**). Symptoms are non-specific (headache, nausea, dizziness – "flu" in a household); SpO₂ is falsely normal – measure COHb (co-oximetry, venous OK). COHb correlates poorly with severity.

- 1 **100% O₂** via non-rebreather at 15 L/min (or intubate) until symptom-free and COHb < 3–5% (≥ 6 h typical) **△ VERIFY**
- 2 **Hyperbaric O₂ referral criteria:** loss of consciousness/syncope, neurological deficit or altered mental status, COHb > 25% (> 15% in pregnancy), myocardial ischemia, severe metabolic acidosis **△ VERIFY**
- 3 ECG + troponin (myocardial injury predicts mortality); pregnancy: fetal Hb binds CO more – treat longer, low HBOT threshold
- 4 Smoke inhalation with lactate > 8–10 mmol/L, shock or arrest → empiric **hydroxocobalamin** (Tox antidotes)
- 5 Delayed neurological sequelae (2–40 days): counsel and arrange follow-up; check other household members and pets; report the source

Frostbite

Rare in PH except returning travellers/climbers and industrial cold (freezers, liquid nitrogen, LPG/refrigerant burns – treat refrigerant as frostbite).

- 1 Treat hypothermia first; avoid thaw-refreeze (keep frozen until definitive rewarming is guaranteed)
- 2 **Rapid rewarming** in circulating water **37–39 °C** for 15–30 min until tissue is soft and red/purple; opioid analgesia (painful) **△ VERIFY**
- 3 Do not rub; drain clear blisters (or aspirate), leave hemorrhagic blisters intact; aloe vera, elevate, splint; tetanus
- 4 **Ibuprofen** 12 mg/kg/day in 2 doses **△ VERIFY**
- 5 Severe (proximal phalanges or more) within 24 h: imaging (bone scan/angiography) and **thrombolysis** (tPA) or **iloprost** (within 48–72 h) at an experienced centre; amputation decisions delayed weeks **△ VERIFY**

Ibuprofen – frostbite **△ VERIFY**

PO

400 – 600 mg q12h (12 mg/kg/day, max 2,400 mg/day)

Environmental sources

Lott et al., ERC 2021 guidelines – cardiac arrest in special circumstances (and ERC 2025 update) · Dow et al., WMS clinical practice guidelines for out-of-hospital evaluation and treatment of accidental hypothermia (2019 update) · Paal et al., Accidental hypothermia 2022 update (Swiss staging, HOPE score) · Lipman et al., WMS heat illness guidelines 2019/2024 · Roberts et al., ACSM expert consensus on exertional heat illness 2023 · Bierens et al., ILCOR/ERC drowning 2021–2023 · Luks et al., WMS acute altitude illness guidelines 2019/2024 · McIntosh et al., WMS frostbite guidelines 2019/2024 · WHO Guidelines for the management of snakebites, 2nd ed (SEARO 2016) · WHO snakebite envenoming strategy 2019 · DOH/RITM PAV guidance and UP-PGH National Poison Management and Control Center (NPMCC) · Isbister et al. (box jellyfish, vinegar) and Australian Resuscitation Council guideline 9.4.5 (2021) · Hampson et al., UHMS CO poisoning guidance 2012 · ACEP clinical policy: CO poisoning 2017 · CDC Vibrio vulnificus guidance · BFAR shellfish bulletins. Tintinalli 9e Section 16 used only as a topic map.

⚠ Items pending owner review in this section (20)

1. Pediatric hydroxocobalamin 70 mg/kg (max 5 g)

Pediatric labeling limited.

2. Snake envenomation (e.g. Philippine cobra) – antivenom and neostigmine trial

No doses given: follow product insert and poison center/DOH protocol.

3. Burn fluid formula: Parkland 4 mL/kg/%TBSA vs ATLS 10th (2 adult / 3 child / 4 electrical)

Starting rate only – titrate to urine output; local burn unit may prefer a different formula.

4. Hypothermia staging: mild 35–32 °C (conscious, shivering), moderate < 32–28 °C (impaired consciousness, shivering stops), severe < 28 °C (unconscious, arrest risk; high risk < 24 °C)

Swiss/WMS staging relies on clinical signs; temperature bands are approximate (2022 update moved to risk-of-arrest wording).

5. Hypothermic arrest (ERC): < 30 °C withhold adrenaline and limit to 3 shocks until > 30 °C; 30–35 °C double adrenaline interval (6–10 min); ECLS rewarming; K⁺ > 12 mmol/L (HOPE score preferred) suggests futility

ERC 2021 (confirm 2025 update); AHA has a less restrictive stance on epinephrine.

6. Exertional heat stroke: cold/ice-water immersion (1–15 °C) to a cooling rate ≥ 0.15 °C/min; stop at ~38.6–39 °C core (rectal); "cool first, transport second" ACSM 2023 / WMS 2019 differ slightly on the endpoint (38.6 vs 39 °C).

7. Exercise-associated hyponatremia with neuro symptoms: 3% NaCl 100 mL IV bolus, repeat up to × 2 at 10-min intervals (pediatric 2 mL/kg, max 100 mL) WMS EAH 2019; pediatric extrapolation is not guideline-derived.

8. Drowning: 5 initial rescue breaths before compressions; asymptomatic patients observe ≥ 4–6 h (pulse oximetry, auscultation) before discharge

ERC/ILCOR; observation duration varies 4–8 h in sources.

9. Electrical injury: urine output target 1–1.5 mL/kg/hr if myoglobinuria (else 0.5–1); asymptomatic low-voltage (≤ 220–240 V household) exposure with normal ECG may be discharged; monitor 24 h if LOC, abnormal ECG, high voltage, chest pain

Expert consensus/observational data; Parkland underestimates deep tissue injury.

10. Altitude: acetazolamide prevention 125 mg BID (peds 1.25 mg/kg BID, max 125), treatment 250 mg BID (peds 2.5 mg/kg BID, max 250); dexamethasone AMS 4 mg q6h, HACE 8 mg then 4 mg q6h (peds 0.15 mg/kg q6h, max 4 mg); HAPE nifedipine ER 30 mg q12h

WMS 2019/2024. Descent and O₂ are definitive.

11. Philippine cobra antivenom (PCAV, RITM): start 5 ampoules IV for severe neurotoxic envenoming, reassess at 1–2 h and repeat per protocol; same dose for children; no routine skin test

Starting dose per RITM/DOH guidance (Eastern Visayas data: 1–4 vials often sufficed); confirm current product insert and NPMCC advice.

12. Neostigmine trial (WHO SEARO 2016): atropine 0.6 mg IV (peds 50 mcg/kg) then neostigmine 1.5–2.0 mg IM (peds 50–100 mcg/kg); positive = clear improvement in ptosis/strength in 10–20 min → continue 0.5 mg q30 min with atropine

Pediatric neostigmine dose recalled from WHO SEARO – check (some sources give 0.04 mg/kg). Mainly helps post-synaptic (cobra) neurotoxicity, not kraits.

13. 20-minute whole blood clotting test: 1–2 mL venous blood in a new, clean, dry GLASS tube, leave undisturbed 20 min, tip once – unclotted = coagulopathy (antivenom indication, repeat 6 h after antivenom)

WHO SEARO 2016; plastic tubes or washed glass give false positives.

14. UP-PGH National Poison Management and Control Center: (02) 8524-1078 · Globe 0966-718-9904

Numbers from public NPMCC listings – confirm before printing for use.

15. Box jellyfish: vinegar (4–6% acetic acid) ≥ 30 s on the sting, then remove tentacles, then hot water immersion (~45 °C, 20 min) or ice for pain; stingray/stonefish/catfish/urchin: hot water 40–45 °C for 30–90 min

ARC 2021 and Isbister/Australian toxinology sources; vinegar NOT advised for bluebottle (Physalia) in ARC – hot water instead.

16. Marine wound infection (Vibrio): doxycycline 100 mg BID + ceftazidime 2 g q8h (or ciprofloxacin); peds: avoid doxycycline < 8 y for long courses – TMP-SMX + aminoglycoside

CDC/IDSA SSTI 2014. Local susceptibility unknown.

17. CO: 100% O₂ via non-rebreather (COHb half-life ~300 min air → ~75 min O₂ → ~20–30 min HBOT); HBOT referral if LOC/syncope, neuro deficit, altered mental status, COHb > 25% (> 15% pregnancy), myocardial ischemia, severe acidosis

UHMS criteria; ACEP 2017 Level B (HBOT or high-flow normobaric for moderate-severe). Very few HBOT chambers in PH.

18. Frostbite: rapid rewarming in 37–39 °C water for 15–30 min (only if refreezing will not occur); ibuprofen 12 mg/kg/day in 2 doses (max 2,400 mg/day); tPA within 24 h for severe injury with no contraindication (e.g. alteplase 0.15 mg/kg bolus then 0.15 mg/kg/hr × 6 h, max 100 mg – protocols vary); iloprost 0.5–2 ng/kg/min IV × 6 h daily for 5–8 days (within 48–72 h)

WMS 2019/2024; thrombolysis and iloprost only in experienced centres with imaging.

19. Mass Hymenoptera envenomation: > 50 stings (adult; fewer in children) → admit for rhabdomyolysis, hemolysis, AKI monitoring

Threshold is approximate; hornet (Vespa) stings are more toxic per sting.

20. IM epinephrine 0.01 mg/kg (1 mg/mL) – max 0.5 mg adult, 0.3 mg child; repeat q5–15 min

v1 values (WAO/EAACI/ASCI); reused here for sting anaphylaxis and now flagged everywhere it appears.

📊 Hyperglycemic crisis calculator

- ☐ Known diabetes (DKA glucose criterion met regardless of glucose)

Enter labs to calculate anion gap, corrected Na, effective osmolality and classify DKA vs HHS (2024 consensus criteria).

DKA: glucose ≥ 200 mg/dL (11.1 mmol/L) or known diabetes $\cdot \beta\text{-OHB} \geq 3.0$ mmol/L (urine ketones $\geq 2+$ if no blood test) $\cdot \text{pH} < 7.3$ and/or $\text{HCO}_3^- < 18$. HHS: glucose ≥ 600 mg/dL (33.3 mmol/L) $\cdot$ effective osmolality > 300 mOsm/kg $\cdot \beta\text{-OHB} < 3.0$ $\cdot \text{pH} \geq 7.3$ $\cdot \text{HCO}_3^- \geq 15$

⚠️ VERIFY

🦋 DKA quick card

Calculator: anion gap, osmolality, DKA vs HHS †

Diagnosis (ADA 2024 / ISPAD 2022): glucose ≥ 200 mg/dL (11.1 mmol/L) or known diabetes $\cdot \beta\text{-hydroxybutyrate} \geq 3$ mmol/L (or urine ketones $\geq 2+$) $\cdot \text{pH} < 7.3$ and/or $\text{HCO}_3^- < 18$ mmol/L.

- 1 **Fluids:** 0.9% NaCl or balanced crystalloid 15–20 mL/kg in hour 1 ($\approx 1\text{--}1.5$ L), then 250–500 mL/hr by hemodynamics/Na **⚠️ VERIFY**
 - 2 **Potassium** **⚠️ VERIFY**: $\text{K} < 3.5 \rightarrow$ hold insulin, replace K first $\cdot 3.5\text{--}5.0 \rightarrow$ add 20–40 mEq K per L of fluid $\cdot > 5.0 \rightarrow$ no K, recheck q2h **⚠️ VERIFY**
 - 3 **Insulin** (regular IV, after $\text{K} \geq 3.5$): 0.1 U/kg/hr; no bolus needed **⚠️ VERIFY**
 - 4 Glucose < 250 mg/dL (13.9 mmol/L): add D5–D10, reduce insulin to 0.05 U/kg/hr; keep glucose 150–200 mg/dL until resolved **⚠️ VERIFY**
 - 5 Bicarbonate only if $\text{pH} < 7.0$ (per protocol). Check glucose hourly, electrolytes/VBG q2–4h **⚠️ VERIFY**
 - 6 Resolution: $\text{pH} \geq 7.3$, $\text{HCO}_3^- \geq 18$ (or ketones < 0.6 mmol/L). Give basal SC insulin 1–2 h before stopping the drip **⚠️ VERIFY**
- **Euglycemic DKA** (glucose < 250 mg/dL): SGLT2 inhibitors, pregnancy, starvation, low-carb diet – start dextrose with insulin from the outset; stop the SGLT2 inhibitor **⚠️ VERIFY**
 - **Mild DKA** ($\text{pH} 7.25\text{--}7.30$, alert, tolerating fluids): some protocols allow SC rapid-acting insulin in a monitored ward setting **⚠️ VERIFY**
 - **Alcoholic / starvation ketoacidosis:** glucose low-normal, high $\beta\text{-OHB}$, $\pm$ lactate – dextrose-containing fluids + thiamine; insulin rarely needed **⚠️ VERIFY**

💧 Hyperosmolar hyperglycemic state

Older adults with type 2 diabetes, days of polyuria/poor intake, infection or stroke as precipitant. Fluid deficit is huge (~100–220 mL/kg); mortality is higher than DKA – from the precipitant, thrombosis and over-rapid osmolar shifts.

- 1 **Fluids first** **⚠️ VERIFY**: 0.9% NaCl (or balanced crystalloid) 1–1.5 L in hour 1, then ~250–500 mL/hr guided by perfusion, urine output and osmolality; replace ~half the deficit in 12 h, the rest over the next 12 h
- 2 **Monitor hourly** glucose; osmolality, Na, K, urea at 0, 1, 2, 4, 6 h. Target falls: osmolality 3–8 mOsm/kg/hr $\cdot$ glucose $\leq 90\text{--}120$ mg/dL/hr (≤ 5 mmol/L/hr) $\cdot$ Na ≤ 10 mmol/L in 24 h **⚠️ VERIFY**
- 3 **Insulin** only when glucose stops falling with fluids alone: fixed-rate 0.05 U/kg/hr; start at once if significant ketonemia (mixed DKA/HHS) **⚠️ VERIFY**
- 4 Add dextrose (D5) when glucose < 300 mg/dL (16.7 mmol/L); keep glucose 200–300 mg/dL in the first 24 h
- 5 Potassium as for DKA (see calculator); **VTE prophylaxis** (LMWH) unless contraindicated; look for MI, stroke, sepsis, rhabdomyolysis; foot check

🦋 Hypoglycemia

Levels (ADA 2025): level 1 < 70 mg/dL (3.9 mmol/L) $\cdot$ level 2 < 54 mg/dL (3.0 mmol/L) $\cdot$ level 3 = altered mental/physical function

needing assistance. Neonates/children: see ISPAD/neonatal thresholds.

- 1 **Alert and can swallow:** 15–20 g fast glucose (e.g. 4 glucose tablets, $\frac{3}{4}$ cup juice, 3 tsp sugar in water); recheck in 15 min, repeat until > 70 mg/dL, then a complex-carbohydrate snack/meal **⚠️ VERIFY**
- 2 **Altered / unable to swallow:** IV dextrose (below); no IV $\rightarrow$ glucagon IM/SC or nasal. Thiamine 100 mg IV with (not before, if it delays) dextrose in malnutrition/alcohol use **⚠️ VERIFY**

IV dextrose – ISPAD-style pediatric dosing **⚠️ VERIFY**

Adult

25 g IV = 50 mL D50, or 100–250 mL D10 (less phlebitis)

Then oral carbohydrate or D10 infusion; recheck in 15 min.

Dextrose – hypoglycemia **⚠️ VERIFY**

Glucose < 70 mg/dL (3.9 mmol/L)

25 g IV = 50 mL D50

Or 250 mL D10. Thiamine 100 mg IV first if malnourished/alcohol use.

Glucagon – hypoglycemia (no IV access) **⚠️ VERIFY**

IM / SC

1 mg

Recovery in 10–15 min; vomiting common. Nasal glucagon 3 mg if available.

- **Sulfonylurea** (gliclazide, glibenclamide) or long-acting insulin: recurrent hypoglycemia – admit, D10 infusion, frequent glucose; octreotide for sulfonylurea-induced recurrence (see Tox antidotes); observe ≥ 24 h (longer for modified-release)
- Intentional insulin overdose: hypokalemia, prolonged effect – infusion + K monitoring; consider local excision of a large depot only in exceptional cases
- Non-diabetic hypoglycemia: sepsis, liver failure, adrenal insufficiency, alcohol, malaria/quinine, insulinoma – draw a sample (glucose, insulin, C-peptide, $\beta\text{-OHB}$, cortisol) before treatment if feasible
- Discharge only if cause understood, long-acting agent excluded, eating, responsible adult present, follow-up arranged

🔥 Thyroid storm

0 Burch–Wartofsky point scale

< 25: storm unlikely. A clinical diagnosis – do not wait for TSH/FT4 (send them).

- 1 Supportive: cooling, acetaminophen (avoid aspirin – displaces T4), fluids, treat precipitant (infection, missed antithyroid drugs, iodine contrast, delivery, trauma), ICU
- 2 Order matters: **β -blocker $\rightarrow$ thionamide $\rightarrow$ iodine ≥ 1 h later $\rightarrow$ steroid** **⚠️ VERIFY**

 β -blockade **⚠️ VERIFY**

Propranolol PO/NG

60 – 80 mg q4h

Also blocks $\text{T}_4 \rightarrow \text{T}_3$ conversion at high dose. Avoid/caution in decompensated HF – use esmolol.

Esmolol IV load (over 1 min)

250 – 500 mcg/kg

Then 50–100 mcg/kg/min infusion; titrate HR.

Thionamide (block synthesis) **VERIFY**

Propylthiouracil (preferred in storm, 1st trimester)

500 – 1,000 mg PO/NG load, then 250 mg q4h

or Methimazole

60 – 80 mg/day PO/NG (e.g. 20 mg q4–6h)

Iodine (block release) – ≥ 1 h AFTER thionamide **VERIFY**

SSKI

5 drops (0.25 mL ≈ 250 mg) PO q6h

or Lugol solution

8 – 10 drops PO q6–8h

Hydrocortisone (storm / relative adrenal insufficiency)

 **VERIFY**

IV

300 mg load, then 100 mg q8h

- Cholestyramine 4 g PO q6h reduces enterohepatic recycling  **VERIFY**
- AF with storm: rate control as above; anticoagulate per CHA₂DS₂-VA; cardioversion often fails until euthyroid
- Refractory: plasmapheresis, urgent thyroidectomy (specialist centre)

Myxedema coma

Elderly woman, winter or infection, sedatives/amiodarone, stopped levothyroxine: hypothermia, bradycardia, hypoventilation (↑ CO₂), hyponatremia, hypoglycemia, coma, pericardial effusion. High mortality – ICU.

- 1 Airway/ventilation (hypercapnic failure – low threshold for intubation); passive rewarming (active rewarming → vasodilation/collapse)
- 2 **Hydrocortisone first**, then IV thyroid hormone  **VERIFY**

Hydrocortisone – give BEFORE thyroid hormone **VERIFY**

IV

100 mg q8h until adrenal insufficiency excluded

Thyroid hormone – myxedema coma **VERIFY**

Levothyroxine IV load

200 – 400 mcg (lower end if small, elderly, cardiac disease)

Levothyroxine IV daily

1.6 mcg/kg/day × 0.75 IV

± Liothyronine (T3) IV

5 – 20 mcg load, then 2.5 – 10 mcg q8h

Avoid high doses in elderly/cardiac patients.

- Avoid hypotonic fluids; treat hyponatremia cautiously (see Renal: hyponatremia); dextrose for hypoglycemia; look for sepsis
- Drug clearance is slow – reduce sedative and opioid doses

Adrenal crisis

Suspect in: known Addison's / hypopituitarism / congenital adrenal hyperplasia, **long-term steroid use** (incl. stopping abruptly, or sick-day without dose increase), shock unresponsive to fluids/pressors, hyponatremia + hyperkalemia, hypoglycemia, unexplained vomiting/abdominal pain. Many carry a steroid emergency card.

- 1 **Give hydrocortisone immediately** – never delay for tests (draw cortisol/ACTH with first bloods if no delay)  **VERIFY**

Hydrocortisone – adrenal crisis **VERIFY**

Immediate

100 mg IV (or IM if no access)

Do not wait for cortisol – draw a sample first only if it causes no delay.

Then

200 mg / 24 h: 50 mg IV/IM q6h or continuous infusion

Taper to oral double-dose once stable and eating.

- 1 **Fluids:** 1 L 0.9% NaCl (or D5NS if hypoglycemic) in the first hour, then by response (often 3–4 L/24 h)  **VERIFY**
- 2 Treat precipitant (infection, GI loss, surgery); monitor glucose, Na, K; ICU if shock persists. Fludrocortisone is unnecessary while hydrocortisone ≥ 50 mg/day  **VERIFY**
- 3 Dexamethasone does not cover mineralocorticoid needs – use hydrocortisone unless a diagnostic test must be done very soon
- 4 Before discharge: **sick-day rules** (double oral dose with fever/illness; IM hydrocortisone if vomiting), emergency card, injection kit education

Endocrine sources

Umpierrez et al., Hyperglycemic crises in adults with diabetes – consensus report (ADA/EASD/JBDS/AACE/DTS), Diabetes Care 2024 · JBDS-IP DKA (2023) and HHS (2022) guidelines · ISPAD 2022 DKA & hypoglycemia chapters · ADA Standards of Care 2025 (hypoglycemia classification) · Ross et al., ATA guidelines for hyperthyroidism and other causes of thyrotoxicosis 2016 · Burch & Wartofsky 1993 (score) · Jonklaas et al., ATA guidelines for hypothyroidism 2014 (myxedema coma) · Bornstein et al., Endocrine Society primary adrenal insufficiency guideline 2016 · Endocrine Society/ESE glucocorticoid-induced adrenal insufficiency guideline 2024 · Society for Endocrinology adrenal crisis emergency guidance (2020). Tintinalli 9e Section 17 used only as a topic map.

⚠ Items pending owner review in this section (11)

1. Octreotide for sulfonyleurea hypoglycemia (adult 50 mcg q6h; peds 1–1.5 mcg/kg max 50 mcg)

Dosing varies by reference.

2. DKA potassium replacement thresholds (< 3.5 hold insulin; 3.5–5 add 20–40 mEq/L)

Local DKA protocol / concentration limits for peripheral lines.

3. Pediatric DKA fluids (deficit 5–10% over 24–48 h + maintenance), insulin 0.05–0.1 U/kg/hr, cerebral edema doses

ISPAD 2022 based – local pediatric DKA protocol governs.

4. Adult DKA: fluids 15–20 mL/kg ($\approx$ 1–1.5 L) hour 1; fixed-rate IV insulin 0.1 U/kg/hr without bolus; add dextrose and halve insulin when glucose < 250 mg/dL

2024 consensus allows 0.1 U/kg/hr (or 0.05–0.1 with bolus variants); JBDS keeps 0.1 U/kg/hr and adds glucose at < 14 mmol/L. Local protocol governs.

5. Hyperglycemic crisis criteria (2024 consensus): DKA = glucose $\geq$ 200 mg/dL or known diabetes + β -OHB $\geq$ 3.0 mmol/L + pH < 7.3 and/or HCO_3^- < 18; HHS = glucose $\geq$ 600 mg/dL + effective osmolality > 300 (or total > 320) + β -OHB < 3.0, pH $\geq$ 7.3, $\text{HCO}_3^- \geq$ 15

Thresholds recalled from the 2024 consensus; confirm against the published table. Severity bins use the older ADA 2009 cut-offs.

6. HHS: 0.9% NaCl 1–1.5 L hour 1; aim osmolality fall 3–8 mOsm/kg/hr, glucose fall $\leq$ 90–120 mg/dL/hr (5 mmol/L/hr), Na fall $\leq$ 10 mmol/L/24 h; insulin 0.05 U/kg/hr only once glucose stops falling with fluids (start early if β -OHB > 1.0–3.0)

JBDS-IP HHS 2022 numbers; 2024 consensus is less prescriptive. VTE prophylaxis per local policy.

7. Hypoglycemia: oral 15–20 g glucose if alert; IV dextrose adult 25 g (D50 50 mL, or D10 100–250 mL), peds 0.2–0.5 g/kg as D10 2–5 mL/kg; glucagon 1 mg IM/SC (0.5 mg if < 25 kg), nasal 3 mg ($\geq$ 4 y)

ADA 2025 / ISPAD 2022; v1 peds dextrose card uses 0.5–1 g/kg – owner to harmonise. Glucagon less effective in alcohol use, starvation, sulfonyleurea toxicity.

8. Thyroid storm: propranolol 60–80 mg PO q4h (or esmolol 250–500 mcg/kg load + 50–100 mcg/kg/min); PTU 500–1,000 mg load then 250 mg q4h, or methimazole 60–80 mg/day; iodine (SSKI 5 drops / 250 mg q6h or Lugol 8–10 drops q6–8h) $\geq$ 1 h after thionamide; hydrocortisone 300 mg IV load then 100 mg q8h; cholestyramine 4 g q6h

ATA 2016; pediatric dosing not included (pediatric endocrinology). PTU preferred in first trimester and in storm; hepatotoxicity.

9. Myxedema coma: levothyroxine 200–400 mcg IV load, then 1.6 mcg/kg/day $\times$ 0.75 IV daily (lower in elderly/cardiac disease); $\pm$ liothyronine 5–20 mcg IV then 2.5–10 mcg q8h; hydrocortisone 100 mg IV q8h before thyroid hormone

ATA 2014; IV levothyroxine/liothyronine availability in PH limited – NG levothyroxine used in practice.

10. Adrenal crisis: hydrocortisone 100 mg IV/IM then 200 mg/24 h (50 mg q6h or infusion); peds 50–100 mg/m² ($\approx$ 25 mg < 3 y, 50 mg 3–12 y, 100 mg $\geq$ 12 y); fluids 1 L 0.9% NaCl in hour 1 (peds 20 mL/kg)

Endocrine Society 2016 PAI / 2024 GI-AI; age bands are an approximation of 50–100 mg/m².

11. Dextrose doses reused from v1 in the Hypoglycemia card (adult 25 g D50; peds 0.5–1 g/kg D10/D25; neonate 0.2 g/kg)

Unflagged in v1; peds range differs from ISPAD (0.2–0.5 g/kg) – see en-hypo.

 Anemia & transfusion

- Restrictive thresholds: Hb 7 g/dL for most stable inpatients, 8 g/dL with cardiovascular disease/orthopedic surgery; symptoms and bleeding override numbers ([△ VERIFY](#))
- Platelets: < 10 prophylactic, < 50 bleeding or procedure, < 100 intracranial/ocular bleeding ([△ VERIFY](#))
- Chronic severe anemia (e.g. thalassemia, common in PH; nutritional): transfuse slowly, diuretic between units if heart failure; send hemoglobin electrophoresis/iron studies **before** transfusion
- Polycythemia (Hct > 55–60%) with symptoms: phlebotomy; secondary causes (COPD, OSA, testosterone, high altitude)

Packed red cells ([△ VERIFY](#))

IV

1 unit, then reassess (≈ +1 g/dL)**Fresh frozen plasma** ([△ VERIFY](#))

IV

10 – 15 mL/kg, max 1,200 mL

≈ 4 units adult; check volume overload.

Cryoprecipitate ([△ VERIFY](#))

IV

1 unit per 10 kg**Transfusion reactions – stop the transfusion first**

Reaction	Clues	Action
Acute hemolytic (ABO)	Fever, back pain, hypotension, hemoglobinuria, DIC	Stop, fluids, recheck identity, DAT, notify blood bank
Anaphylaxis	Hypotension, wheeze, angioedema (IgA deficiency)	IM epinephrine (anaphylaxis)
TRALI	Hypoxia, bilateral infiltrates within 6 h, often hypotension	O ₂ /ventilation; no diuretic
TACO	Hypoxia, hypertension, raised JVP	Diuretic, slow rates, sit up
Febrile non-hemolytic	Fever/chills only	Antipyretic; exclude hemolysis/sepsis
Allergic (urticaria only)	Hives	Antihistamine; may resume slowly

 Anticoagulant reversal

- Life-threatening or critical-site bleed (intracranial, airway, GI with shock, compartment): reverse **now**, resuscitate, source control
- Warfarin: vitamin K 10 mg IV + 4F-PCC by INR (antidotes card); FFP only if no PCC ([△ VERIFY](#))
- Heparin / enoxaparin: protamine (antidotes)
- Thrombolytic-associated bleed: stop infusion, cryoprecipitate (fibrinogen), tranexamic acid
- Time since last DOAC dose and renal function decide drug levels (dabigatran especially renally cleared – dialyzable)

Idarucizumab – dabigatran reversal ([△ VERIFY](#))

IV

5 g (2 × 2.5 g vials) as bolus or two 5–10 min infusions**4-factor PCC – factor Xa inhibitor bleed** ([△ VERIFY](#))

IV

50 IU/kg (or fixed 2,000 IU) **Hemophilia & vWD****Factor VIII dose** ([△ VERIFY](#))

Enter weight

units = weight × rise (40%) × 0.5

- 1 **Treat first, image later:** give factor immediately for head injury, neck/throat bleeds, iliopsoas bleed, or any suspected serious bleed – even before CT
- 2 Ask the patient/family: they often know their dose and carry factor; inhibitor patients need bypassing agents (rFVIIa / aPCC) – call hematology
- 3 No IM injections, no NSAIDs; acetaminophen for pain; RICE for joint bleeds
- 4 PH: factor concentrate access is limited – cryoprecipitate (contains FVIII, vWF, fibrinogen) or FFP (FIX) if no factor; Philippine Hemophilia Foundation / NCHH for support

Desmopressin (DDAVP) – mild hemophilia A / vWD type 1 / uremic platelets ([△ VERIFY](#))

IV/SC over 30 min

0.3 mcg/kg, max 20 mcg

Restrict fluids 24 h; check Na. Not < 2 y.

 Hemolysis, TTP/HUS, DIC, HIT**G6PD deficiency (common in PH)**

- Acute hemolysis after oxidant drug, infection or fava beans/mothballs: pallor, dark urine, jaundice; supportive, transfuse if severe ([△ VERIFY](#))
- Methylene blue for methemoglobinemia is contraindicated – use ascorbic acid/exchange transfusion via toxicology

Thrombotic microangiopathy (schistocytes + low platelets)

- **TTP** (ADAMTS13 < 10%): PLASMIC score ≥ 6 → urgent plasma exchange, steroids; avoid platelet transfusion unless life-threatening bleed
- **HUS** (often post-diarrheal E. coli in children): supportive, dialysis; no antibiotics/antimotility for STEC
- **DIC**: treat the cause (sepsis, obstetric, trauma, APL); replace platelets/fibrinogen if bleeding (ISTH score)
- **HIT** (platelet fall > 50% 5–10 days after heparin; 4Ts score): stop all heparin, start non-heparin anticoagulant (argatroban, fondaparinux, DOAC); no platelet transfusion

Sickle cell disease (uncommon in PH)

- Pain crisis: analgesia within 60 min, fluids if dehydrated; acute chest syndrome (fever, chest pain, new infiltrate) → O₂, antibiotics, transfusion; stroke → exchange transfusion; fever = emergency (functional asplenia)

Oncologic emergencies

Febrile neutropenia

- 1 Antibiotics within 60 min of arrival – do not wait for counts if recent chemotherapy (△ VERIFY)
- 2 Cultures (peripheral + each line lumen), lactate; MASCC ≥ 21 / CISNE low risk → possible oral outpatient therapy

Febrile neutropenia – empiric antibiotics (within 60 min)

(△ VERIFY)

IV

Cefepime 2 g q8h or piperacillin–tazobactam 4.5 g q6h (meropenem 1 g q8h if ESBL risk)

Tumour lysis syndrome

- High uric acid, K, phosphate; low Ca; AKI – after chemo for high-grade lymphoma/leukemia or spontaneously. Hydration, no K in fluids, treat hyperkalemia (card), rasburicase (not with G6PD deficiency) (△ VERIFY)

Rasburicase – tumour lysis (check G6PD first) (△ VERIFY)

IV over 30 min

0.1 – 0.2 mg/kg, max 15 mg

Single dose often sufficient; fixed 3–6 mg used in adults.

Hypercalcemia of malignancy

- Confusion, polyuria, constipation, short QT: saline rehydration, then bisphosphonate; calcitonin for rapid effect (△ VERIFY)

Zoledronic acid – hypercalcemia of malignancy (△ VERIFY)

IV over ≥ 15 min

4 mg (reduce if CrCl 30–60; avoid < 30)

Calcitonin (△ VERIFY)

SC/IM q12h

4 IU/kg

Rapid but short effect (tachyphylaxis in 48 h).

Spinal cord compression

- Back pain + weakness/sensory level/sphincter loss in cancer: dexamethasone, MRI whole spine within 24 h, radiotherapy/surgery (△ VERIFY)

Dexamethasone – cord compression (△ VERIFY)

IV

10 mg, then 4 mg q6h

- **SVC syndrome:** facial swelling, distended neck veins – elevate head, CT; stridor = airway emergency; tissue diagnosis before steroids if lymphoma suspected
- **Hyperviscosity** (Waldenström, myeloma) / **leukostasis** (WBC > 100 × 10⁹/L): neuro, visual, respiratory symptoms → plasmapheresis/leukapheresis, hydration; avoid transfusing RBCs before leukoreduction

Heme/Onc sources

Carson et al., AABB international guidelines for RBC transfusion (JAMA 2023) · Kaufman et al., AABB platelet transfusion 2015 · Srivastava et al., WFH guidelines for the management of hemophilia, 3rd ed (2020) · Connell/James et al., ASH ISTH NHF WFH vWD guidelines 2021 · Tomaselli et al., 2020 ACC ECDP on bleeding with oral anticoagulants · Freifeld et al., IDSA febrile neutropenia 2010; Taplitz et al., ASCO/IDSA 2018 · Cairo & Bishop TLS classification; Jones et al., BSH TLS 2015 · Shane et al., Endocrine Society hypercalcemia of malignancy 2023 · NICE NG234 metastatic spinal cord compression 2023 · Zheng et al., ISTH TTP guidelines 2020 · Cuker et al., ASH HIT 2018 · PH Newborn Screening Program (RA 9288). Tintinalli 9e Section 18 used only as a topic map.

⚠ Items pending owner review in this section (10)

1. Restrictive transfusion: Hb threshold 7 g/dL (8 g/dL with cardiovascular disease); aim 7–9 g/dL

ACG 2021 / AABB 2023.

2. Transfusion: 1 unit PRBC raises Hb ~1 g/dL in adults; children 10–15 mL/kg raises Hb ~2–3 g/dL; platelets: prophylactic $< 10 \times 10^9/L$, procedure/bleeding < 50 , CNS/eye < 100 ; FFP 10–15 mL/kg; cryoprecipitate 1 unit per 10 kg (fibrinogen < 1.5 g/L)

AABB 2015/2023, BSH; local blood bank products (PH: PRC volumes ~ 250 mL) differ.

3. DOAC reversal (life-threatening bleed): dabigatran → idarucizumab 5 g IV; apixaban/rivaroxaban → andexanet alfa (low/high dose) or 4-factor PCC 50 IU/kg (or fixed 2,000 IU)

ACC 2020 ECDP / ASH; andexanet rarely available in PH – PCC is the practical option.

4. Factor replacement: FVIII units = weight × desired rise (%) × 0.5; FIX units = weight × desired rise (%) × 1 (plasma-derived; recombinant ~1.2–1.4); life-threatening bleed target 80–100%

WFH 2020; PH factor supply limited – cryoprecipitate (FVIII) / FFP (FIX) as fallback.

5. Desmopressin 0.3 mcg/kg IV/SC (max 20 mcg) over 30 min for mild hemophilia A / type 1 vWD (fluid restriction, hyponatremia; avoid < 2 y)

WFH/ASH vWD 2021.

6. Febrile neutropenia ($T \geq 38.3$ °C once or ≥ 38.0 °C for 1 h, ANC < 500): antipseudomonal β -lactam within 60 min – cefepime 2 g IV q8h or piperacillin-tazobactam 4.5 g IV q6h (peds cefepime 50 mg/kg q8h, max 2 g); add vancomycin for line infection, skin/soft tissue, MRSA, hemodynamic instability

IDSA 2010 / ASCO-IDSA 2018; local resistance (ESBL in PH) may require a carbapenem.

7. Tumour lysis: IV fluids 2–3 L/m²/day (urine output 80–100 mL/m²/hr); rasburicase 0.1–0.2 mg/kg IV (contraindicated in G6PD deficiency); allopurinol 100 mg/m² q8h (max 800 mg/day) for prevention

Cairo-Bishop / BSH 2015; check G6PD (common in PH) before rasburicase.

8. Hypercalcemia of malignancy: 0.9% NaCl 200–300 mL/hr titrated to urine output 100–150 mL/hr; zoledronic acid 4 mg IV over 15 min (adjust for CrCl); calcitonin 4 IU/kg SC/IM q12h for 24–48 h

Endocrine Society 2023 hypercalcemia guideline.

9. Malignant spinal cord compression: dexamethasone 10 mg IV then 4 mg q6h (16 mg/day; some use 16 mg/day without load); MRI whole spine within 24 h

NICE 2023 / ESMO; high-dose 96 mg loading no longer favoured.

10. G6PD deficiency (PH newborn screening, prevalence ~ 2–3%): avoid oxidants – primaquine, dapsone, nitrofurantoin, sulfonamides/co-trimoxazole (high dose), methylene blue, rasburicase, naphthalene (mothballs), fava beans

WHO/G6PD Deficiency Foundation lists; risk depends on variant and dose.

 Eye emergencies

Always record visual acuity (each eye, with glasses/pinhole) – it is the vital sign of the eye.

- 1 **Chemical injury** (alkali worst – cement, lye, drain cleaner): irrigate **before** anything else; topical anesthetic, eyelid speculum/Morgan lens; check pH (△ VERIFY)
- 2 **Sudden painless monocular vision loss** = CRAO until proven otherwise: stroke pathway, urgent ophthalmology/stroke team (thrombolysis within 4.5 h in some centres), check ESR/CRP for giant cell arteritis if > 50 y
- 3 **Acute angle closure**: painful red eye, halos, mid-dilated fixed pupil, rock-hard globe, vomiting – IOP-lowering drops + acetazolamide; laser iridotomy (△ VERIFY)
- 4 **Open globe**: peaked pupil, Seidel positive, 360° subconjunctival hemorrhage – shield, no pressure, antibiotics, CT, surgery (△ VERIFY)
- 5 **Retrobulbar hemorrhage**: proptosis, tense orbit, vision loss after trauma – lateral canthotomy now (△ VERIFY)

Acetazolamide – acute angle closure (△ VERIFY)

IV / PO

500 mg once, then 250 mg q6h PO**Mannitol 20% – refractory IOP** (△ VERIFY)

IV over 30–60 min

1 – 2 g/kg

Red eye	Clue	Action
Corneal abrasion	FB sensation, fluorescein uptake	Topical antibiotic; no patching; contact lens → antipseudomonal drops, no lens
Keratitis / corneal ulcer	White infiltrate, contact lens	Same-day ophthalmology
Herpes zoster ophthalmicus	Rash on nose tip (Hutchinson)	Oral antiviral, ophthalmology
Orbital cellulitis	Pain on eye movement, proptosis, ↓ vision	IV antibiotics, CT, ENT/optho
Conjunctivitis	Discharge, normal vision	Hygiene; antibiotics if bacterial; hyperacute purulent → gonococcal

 Ear disorders

- **AOM**: bulging TM; amoxicillin high dose or observation in selected children (△ VERIFY)

- **Otitis externa**: topical fluoroquinolone drops (safe with perforation), wick if canal swollen; diabetic/elderly with severe pain + granulation tissue = **malignant otitis externa** → IV antipseudomonal, CT
- **Mastoiditis**: post-auricular swelling, protruding ear – IV antibiotics, CT, ENT
- **Sudden sensorineural hearing loss**: Weber lateralises to good ear – steroids within 2 weeks, urgent ENT; with vertigo/neuro signs consider posterior circulation stroke (△ VERIFY)
- **Foreign body**: live insect – drown with lidocaine/mineral oil first; button battery = emergency removal; do not irrigate vegetable matter
- TM perforation (trauma/barotrauma): keep dry, most heal in weeks

Amoxicillin – acute otitis media (△ VERIFY)

PO

500 mg – 1 g TID × 5–7 d (adult AOM/sinusitis)**Prednisone – sudden SNHL / Bell palsy** (△ VERIFY)

Sudden SNHL

60 mg PO daily × 7–14 d, then taper

Bell palsy (△ VERIFY)

Prednisolone 50 mg daily × 10 d (or 60 mg × 5 d then taper) **Epistaxis & nose**

- 1 Sit up, lean forward, firm pressure on the soft nose for **10–15 min** uninterrupted; ice to the bridge
- 2 Clear clots, topical vasoconstrictor + anesthetic (oxymetazoline/phenylephrine + lidocaine) for 10 min; identify the site
- 3 Visible anterior site: silver nitrate cautery (one side of the septum only); topical tranexamic acid option (△ VERIFY)
- 4 Persistent: anterior packing (nasal tampon/balloon) for 48–72 h; antibiotic prophylaxis not routinely needed
- 5 Posterior bleed (bilateral, down the throat despite anterior pack): double-balloon or Foley catheter, admit, ENT – risk of hypoxia/aspiration

Tranexamic acid – topical for epistaxis (△ VERIFY)

Topical

500 mg (5 mL of 100 mg/mL) soaked pledget, 10–15 min

- Anticoagulated: usually continue unless life-threatening; check INR; reverse only for severe bleeding
- Hypertension often reflects anxiety/pain – treat the bleed first
- **Septal hematoma** after nasal trauma (boggy purple swelling): drain the same day to prevent septal necrosis/abscess
- Nasal fracture: reduce within 5–10 days after swelling settles (ENT); CSF rhinorrhea → CT
- Acute sinusitis: mostly viral; antibiotics if > 10 days, severe, or worsening after improvement; orbital/intracranial complications are emergencies

 Oral & dental

- **Avulsed permanent tooth**: hold by the crown, rinse briefly, reimplant immediately (best < 60 min); otherwise store in milk/saliva/saline; splint and dentist; **never reimplant primary (baby) teeth**
- Dental fractures: exposed pulp (pink/bleeding) → urgent dentist; cover dentin
- Dental abscess: drainage + antibiotics; facial swelling crossing midline, trismus, floor-of-mouth swelling, voice change → **Ludwig angina**: airway (awake fiberoptic/surgical), IV antibiotics, CT, OR
- Post-extraction bleeding: bite on gauze (or gauze soaked in tranexamic acid) 20–30 min, local anesthetic with epinephrine, sutures, check anticoagulants

- Dry socket (2–4 days after extraction): irrigate, medicated dressing, analgesia (not an infection)

Odontogenic infection ⚠️ VERIFY

Outpatient

Amoxicillin 500 mg PO TID (or amoxicillin–clavulanate 875/125 BID) × 5–7 d

Ludwig / deep neck

Ampicillin–sulbactam 3 g IV q6h

Neck & upper airway

- 1 Upper-airway threat (stridor, drooling, tripod, muffled voice): keep patient upright and calm, call the most experienced airway operator, prepare for awake fiberoptic intubation and front-of-neck access; avoid lying flat or provoking with tongue depressors in children
- 2 **Angioedema**: allergic (urticaria, itching) → IM epinephrine (anaphylaxis); ACE-inhibitor/hereditary (no urticaria) → bradykinin-mediated: epinephrine works poorly; icatibant/C1-INH/FFP; airway early if tongue/floor of mouth/voice involved ⚠️ VERIFY
- 3 **Epiglottitis** (adults now more common than children – Hib vaccination): sore throat out of proportion, thumb sign on lateral neck X-ray; ceftriaxone 2 g IV (peds 50 mg/kg), airway in OR/resus room ⚠️ VERIFY
- 4 **Peritonsillar abscess**: trismus, uvular deviation, "hot potato" voice – aspiration/drainage, antibiotics, steroid ⚠️ VERIFY
- 5 **Retropharyngeal/deep neck abscess**: fever, neck stiffness/torticollis, dysphagia – CT neck, IV antibiotics, ENT drainage

Dexamethasone – pharyngitis/PTA pain, airway edema

⚠️ VERIFY

IV/PO

10 mg single dose

Icatibant / C1-esterase inhibitor (bradykinin angioedema)

⚠️ VERIFY

Icatibant SC

30 mg

C1-INH (plasma-derived) IV

20 IU/kg

Fresh frozen plasma

2 units (if nothing else)

Tracheostomy / laryngectomy emergency (NTSP approach)

- Apply high-flow O₂ to **face and stoma**; capnography; is the patient breathing?
- Laryngectomy = neck breather only – oxygenate via stoma, cannot intubate from above
- Blocked/displaced tube: remove inner cannula, pass suction catheter; if it will not pass → deflate cuff → remove tube; oxygenate via stoma (or orally if patent upper airway)
- Reinsert a new tube (same size or smaller) via stoma – bougie/fiberoptic; oral intubation if upper airway intact

Face & jaw

- **Bell palsy**: forehead involved (peripheral); steroids within 72 h; eye protection (lubricant, tape at night); forehead sparing = central lesion → stroke pathway ⚠️ VERIFY
- Look for Ramsay Hunt (vesicles in ear – add antiviral), Lyme, otitis media, parotid tumour

- **TMJ dislocation**: open locked jaw – reduce with thumbs on molars (wrapped), downward then backward pressure; procedural sedation if needed
- **Mandible fracture**: malocclusion, step, sublingual hematoma; open fractures (into mouth) need antibiotics and maxillofacial referral
- **Acute suppurative parotitis** (elderly, dehydrated): antistaphylococcal antibiotics, hydration, sialogogues

Prednisone – sudden SNHL / Bell palsy ⚠️ VERIFY

Sudden SNHL

60 mg PO daily × 7–14 d, then taper

Bell palsy ⚠️ VERIFY

Prednisolone 50 mg daily × 10 d (or 60 mg × 5 d then taper)

EENT sources

American Academy of Ophthalmology Preferred Practice Patterns (angle-closure, conjunctivitis, corneal edema/opacification) · Mac Grory et al., AHA scientific statement on CRAO 2021 · Lieberthal et al., AAP AOM guideline 2013 · Chandrasekhar et al., AAO-HNS sudden hearing loss 2019 · Baugh et al., AAO-HNS Bell palsy 2013 · Tunkel et al., AAO-HNS epistaxis 2020 · Reuben et al., NoPAC trial (Lancet Resp Med 2021) · Fouad et al., IADT dental trauma guidelines 2020 · McGrath et al., NTSP tracheostomy/laryngectomy algorithms 2012 · Maurer et al., WAO/EAACI HAE guideline 2021 · Shulman et al., IDSA pharyngitis 2012. Tintinalli 9e Section 19 used only as a topic map.

⚠ Items pending owner review in this section (12)

1. Epiglottitis: ceftriaxone 2 g IV daily (peds 50 mg/kg, max 2 g) ± vancomycin if MRSA risk

IDSA/expert; airway management takes priority.

2. Chemical eye injury: irrigate immediately with ≥ 1–2 L (≥ 30 min) normal saline/Ringer's lactate or tap water until pH 7.0–7.5 (recheck 20–30 min after stopping)

AAO/ED consensus; alkali needs longer irrigation.

3. Acute angle-closure: timolol 0.5% 1 drop, apraclonidine/brimonidine 1 drop, pilocarpine 2% 1 drop every 15 min × 2 once IOP begins to fall, acetazolamide 500 mg IV/PO, ± mannitol 1–2 g/kg IV (or oral glycerol)

AAO PPP 2020; avoid acetazolamide in sulfa allergy/sickle cell.

4. Retrobulbar hemorrhage with vision loss + IOP > 40 mmHg or proptosis/RAPD → lateral canthotomy and inferior cantholysis within ~ 90–120 min

Expert consensus; ophthalmology may not arrive in time.

5. Open globe: rigid shield, NPO, antiemetic, analgesia, tetanus, IV antibiotics (e.g. vancomycin 15 mg/kg + ceftazidime 50 mg/kg, or adult levofloxacin 750 mg) – no pressure, no topical drops

Regimens vary; ophthalmology preference.

6. AOM: amoxicillin 80–90 mg/kg/day + BID (max 4 g/day) × 10 d if < 2 y or severe, 5–7 d if older; penicillin allergy (non-anaphylactic) cefdinir/cefuroxime; observation option for non-severe AOM ≥ 2 y

AAP 2013.

7. Sudden sensorineural hearing loss (< 2 weeks): prednisone 1 mg/kg/day (max 60 mg) × 7–14 days then taper; ENT/audiology

AAO-HNS 2019 (option); exclude stroke (vertigo + hearing loss).

8. Epistaxis topical tranexamic acid: 500 mg (5 mL of 100 mg/mL) on pledget/packing for 10–15 min

NoPAC trial 2021 found no benefit vs placebo for packing need; earlier trials positive – owner to decide.

9. Bell palsy (within 72 h): prednisolone 50 mg daily × 10 days or 60 mg × 5 days then taper; antiviral (valacyclovir 1 g TID × 7 d) only as add-on for severe palsy; eye protection

AAO-HNS 2013 / Cochrane.

10. Odontogenic infection: amoxicillin 500 mg TID (or amoxicillin–clavulanate 875/125 BID) × 5–7 d with drainage; penicillin allergy clindamycin 300 mg QID; Ludwig angina: ampicillin–sulbactam 3 g IV q6h (or ceftriaxone 2 g + metronidazole 500 mg q8h)

ADA 2019 / IDSA; clindamycin C. difficile risk.

11. Peritonsillar abscess: needle aspiration or I&D; amoxicillin–clavulanate or clindamycin; single dexamethasone 10 mg IV (peds 0.6 mg/kg, max 10 mg) for pain/trismus

Steroid evidence modest.

12. ACE-inhibitor angioedema: stop ACEi; airway-first; icatibant 30 mg SC or FFP 2 units (evidence weak); hereditary angioedema: C1-INH 20 IU/kg IV or icatibant 30 mg SC

WAO/EAACI 2021 (HAE); icatibant/C1-INH rarely stocked in PH EDs.

 Dangerous rashes

Pattern	Think of	First steps
Petechiae/purpura + fever	Meningococcal, dengue (very common in PH), leptospirosis, sepsis/DIC, ITP/leukemia	Blood cultures, antibiotics within 1 h if meningococcus possible; CBC, platelets
Diffuse erythema + shock	Toxic shock syndrome (tampons, wounds), scarlet fever	Source removal, clindamycin + β -lactam/vancomycin
Mucosal erosions + skin pain/blisters	SJS/TEN	Stop drugs, SCORTEN, burn/ICU
Maculopapular + cough, coryza, conjunctivitis	Measles (outbreaks in PH) – Koplik spots	Airborne isolation, vitamin A, notify
Fever $\geq$ 5 days child + rash/eyes/lips/hands	Kawasaki disease	Echo, IVIG – see Pediatrics
Erythroderma (> 90% BSA)	Drug reaction, psoriasis, eczema, lymphoma	Fluids, temperature, admit

- Describe: distribution, morphology (macule, papule, vesicle, bulla, pustule, petechia), mucosa, Nikolsky sign, blanching
- Dengue: flushing/"islands of white in a sea of red" in recovery; tourniquet test; avoid NSAIDs/IM injections

 SJS/TEN & severe drug reactions

- ☐ Age $\geq$ 40
- ☐ Heart rate $\geq$ 120
- ☐ Cancer / hematologic malignancy
- ☐ Epidermal detachment > 10% BSA
- ☐ Urea > 10 mmol/L (BUN > 28 mg/dL)
- ☐ Bicarbonate < 20 mmol/L
- ☐ Glucose > 14 mmol/L (252 mg/dL)

0 SCORTEN (0–7)**Predicted mortality ~3%** 

- Stop every non-essential drug;** culprit usually started 4–28 days before (allopurinol, carbamazepine, phenytoin, lamotrigine, sulfonamides, nevirapine, NSAIDs, oxicams)



- Treat like a burn: warm room, fluids (less than Parkland), non-adherent dressings, analgesia, eye and genital care – ophthalmology early (lifelong scarring)

- Transfer to a burn unit/ICU if > 10% detachment; immunomodulators (cyclosporine, IVIG, steroids) per dermatology

- DRESS:** rash + fever + eosinophilia/atypical lymphocytes + organ involvement (hepatitis, nephritis) 2–8 weeks after drug (anticonvulsants, allopurinol, sulfa, vancomycin) – stop drug, steroids 
- Pemphigus vulgaris** (mucosal, flaccid bullae, Nikolsky +) vs **bullous pemphigoid** (elderly, tense bullae, itchy) – dermatology, steroids
- Staphylococcal scalded skin (young children): Nikolsky +, no mucosa – IV antistaphylococcal antibiotics

Prednisone – severe urticaria / DRESS 

Urticaria

40–50 mg daily $\times$ 3–5 daysDRESS with organ involvement **0.5–1 mg/kg/day, taper over weeks** **Skin & soft-tissue infections**

- Abscess:** incision and drainage (ultrasound helps); antibiotics if > 2 cm, cellulitis, systemic signs, immunosuppression, extremes of age
- Cellulitis:** mark margins; bilateral "cellulitis" is usually stasis dermatitis; elevate; treat tinea pedis portal
- Necrotising fasciitis:** pain out of proportion, rapid spread, bullae, crepitus, shock – **surgeon now**; imaging must not delay 

Cephalexin – non-purulent cellulitis 

PO

500 mg QID $\times$ 5 days (extend if not improving)**TMP-SMX – purulent / MRSA-risk SSTI** 

PO

1–2 DS tablets (160/800 mg) BID $\times$ 5–7 days**Clindamycin** 

PO / IV

300–450 mg PO TID · 600–900 mg IV q8h**Necrotising infection – empiric** 

IV

Piperacillin-tazobactam 4.5 g q6h + clindamycin 900 mg q8h + vancomycin 25 mg/kg load

- Scabies** (common in PH households, institutions): nocturnal itch, web-space burrows – treat everyone at once; wash bedding 
- Herpes zoster:** antiviral within 72 h; nose-tip lesions → ophthalmology 
- Impetigo:** topical mupirocin for limited; oral cephalexin for extensive
- Leprosy** (still endemic in PH): anesthetic hypopigmented patches, thickened nerves – refer to DOH leprosy programme (free MDT); reactions (painful nerves, ENL) need steroids
- Tinea corporis/capitis: topical azole/terbinafine; capitis needs oral therapy

Ivermectin – scabies 

PO day 1 and day 8

200 mcg/kg on days 1 and 8

Valacyclovir / acyclovir – herpes zoster ⚠️ VERIFY

PO

Valacyclovir 1 g TID × 7 d (or acyclovir 800 mg 5×/day)

IV (disseminated / immunocompromised)

10 mg/kg

q8h; hydrate; renal adjust.

 Urticaria & regional rashes

- Urticaria with any airway, breathing or circulation problem = anaphylaxis (IM epinephrine)
- Acute urticaria: non-sedating antihistamine regularly for 1–2 weeks; avoid triggers (NSAIDs, codeine) ⚠️ VERIFY

Cetirizine – urticaria ⚠️ VERIFY

PO

10 mg daily (may increase up to 4×)**Prednisone – severe urticaria / DRESS** ⚠️ VERIFY

Urticaria

40–50 mg daily × 3–5 daysDRESS with organ involvement ⚠️ VERIFY**0.5–1 mg/kg/day, taper over weeks**

Region	Common ED problems
Face & scalp	Erysipelas (raised sharp border), herpes zoster ophthalmicus, angioedema, impetigo, seborrheic dermatitis, tinea capitis/kerion

Region**Common ED problems**

Trunk

Pityriasis rosea (herald patch), drug exanthem, tinea versicolor, zoster (dermatomal), scabies

Groin & skinfolds

Intertrigo/candidiasis (satellite lesions), tinea cruris (spares scrotum), **Fournier gangrene** (pain, crepitus – surgical emergency), hidradenitis

Extremities

Cellulitis vs stasis dermatitis, erythema nodosum (TB, strep, sarcoid, leprosy), pressure injuries, contact dermatitis, diabetic foot ulcers

 Dermatology sources

Creamer et al., UK guidelines for SJS/TEN in adults (BAD 2016) · Bastuji-Garin et al., SCORTEN (J Invest Dermatol 2000) · Kardaun et al., RegiSCAR DRESS · Stevens et al., IDSA SSTI guideline 2014 · CDC scabies treatment guidance; WHO ivermectin mass drug administration guidance · Zuberbier et al., EAACI/GA²LEN/EuroGuiDerm/APAAACI urticaria guideline 2021 · Phillips et al., CPIC guideline HLA-B carbamazepine 2017 · DOH National Leprosy Control Program. Tintinalli 9e Section 20 used only as a topic map.

⚠️ Items pending owner review in this section (8)

1. SCORTEN (1 point each: age ≥ 40, HR ≥ 120, malignancy, detachment > 10% BSA day 1, urea > 10 mmol/L (28 mg/dL), bicarbonate < 20, glucose > 14 mmol/L (252 mg/dL)); predicted mortality 0–1: ~3%, 2: ~12%, 3: ~35%, 4: ~58%, ≥ 5: ~90%

Original 2000 derivation; recalibrated versions exist.

2. Cellulitis/abscess: drain abscesses; non-purulent cellulitis – cephalexin 500 mg QID (peds 12.5–25 mg/kg q6h, max 500 mg) × 5 d; MRSA risk/purulent – TMP-SMX 1–2 DS tab BID or clindamycin 300–450 mg TID; severe – cefazolin 2 g IV q8h ± vancomycin

IDSA 2014; PH community MRSA rates vary.

3. Necrotising soft-tissue infection: emergency surgical debridement; piperacillin–tazobactam 4.5 g IV q6h (or meropenem) + clindamycin 900 mg IV q8h (toxin suppression) + vancomycin

IDSA 2014; LRINEC is not sensitive enough to rule out.

4. Scabies: permethrin 5% cream neck down (include scalp/face in infants), wash off after 8–14 h, repeat in 7 days; or ivermectin 200 mcg/kg PO on days 1 and 8 (not < 15 kg or pregnancy); treat all household contacts

CDC/WHO; crusted scabies needs combined therapy.

5. Herpes zoster (within 72 h or new lesions/ophthalmic/immunocompromised): valacyclovir 1 g PO TID × 7 d or acyclovir 800 mg 5×/day × 7 d; disseminated/immunocompromised: acyclovir 10 mg/kg IV q8h

Renal dose adjustment required.

6. Urticaria: second-generation antihistamine (cetirizine 10 mg daily, up to 4× standard dose); short prednisone 0.5 mg/kg (max 40–50 mg) × 3–5 d for severe acute urticaria; peds cetirizine 2.5 mg (6 mo–2 y), 5 mg (2–5 y), 10 mg (≥ 6 y)

EAACI/GA²LEN 2021; up dosing off-label.

7. DRESS: stop culprit; systemic corticosteroids (prednisone 0.5–1 mg/kg/day, slow taper over weeks) when organ involvement

Expert consensus; no RCTs.

8. HLA-B*15:02 (carbamazepine/phenytoin SJS/TEN risk) is common in Southeast Asian populations incl. Filipinos – screen before carbamazepine where available

CPIC; local allele frequency estimates vary.

 Primary survey – <C>ABCDE

- 1 <C> Catastrophic hemorrhage:** direct pressure, tourniquet (note time), wound packing, pelvic binder
- 2 A – Airway + c-spine:** talking? obstruction, suction, jaw thrust, adjuncts; definitive airway if GCS ≤ 8 / failing; manual in-line stabilization
- 3 B – Breathing:** RR, SpO₂, chest rise, breath sounds, trachea. Tension PTX → needle decompression (4th/5th ICS anterior-to-mid axillary line in adults) → finger thoracostomy/chest tube; open PTX → 3-sided dressing
- 4 C – Circulation:** HR, BP, CRT, skin; 2 large-bore IV/IO, bloods + crossmatch, eFAST, pelvis, long bones; early blood products / MTP, limit crystalloid (≤ 1 L); TXA within 3 h
- 5 D – Disability:** GCS, pupils, lateralizing signs, glucose; treat hypoxia/hypotension (secondary brain injury)
- 6 E – Exposure/Environment:** fully undress, log-roll, prevent hypothermia (warm fluids, blankets, room)
- 7 Adjuncts:** ECG, ABG/lactate, urinary catheter (unless urethral injury), gastric tube, CXR/pelvic XR, eFAST → then secondary survey (AMPLE, head-to-toe)

 Burns – Parkland **VERIFY**

Enter weight and %TBSA (partial + full thickness only; exclude superficial/1st-degree).

Rule of nines – adult: head 9, each arm 9, anterior trunk 18, posterior trunk 18, each leg 18, perineum 1. Infant: head 18, each leg 14. Patient's palm + fingers $\approx 1\%$.

 Massive transfusion

- **Activate** if: ABC score ≥ 2 (penetrating, SBP ≤ 90 , HR ≥ 120 , +FAST), shock index ≥ 1 , ongoing hemorrhage, or > 4 units PRBC in 1 h expected **VERIFY**
- Ratio **1:1:1** PRBC : plasma : platelets (or whole blood); warmed, via rapid infuser
- Calcium: 1 g CaCl₂ 10% (or 3 g gluconate) with first units and ~every 4 units; target iCa > 1.1 mmol/L **VERIFY**
- **TXA** within 3 h (see TXA card)
- Permissive hypotension until hemorrhage control: SBP 80–90 mmHg (adult, no TBI); TBI: SBP ≥ 100 –110 mmHg **VERIFY**
- Lethal diamond: hypothermia, acidosis, coagulopathy, hypocalcemia. Targets: Hb 7–9 g/dL, INR < 1.5 , fibrinogen > 1.5 –2 g/L (cryo), platelets > 50 –100
- Stop MTP when bleeding controlled; notify blood bank

Tranexamic acid (TXA)**Tranexamic acid (within 3 h of injury)** **VERIFY**

Loading

1 g IV over 10 min

= 10 mL of 100 mg/mL in 100 mL NS. Best < 1 h; do not start > 3 h.

Maintenance

1 g IV over 8 h

≈ 125 mg/hr.

TBI / raised ICP**Hyperosmolar therapy** **VERIFY**

3% saline

250 mL IV over 10–20 min

Or 23.4% 30 mL via central line only.

Mannitol 20%**0.5 – 1 g/kg**

Over 15–20 min; avoid if hypotensive; Foley.

- Head up 30°, neck neutral, avoid hypoxia (SpO₂ $> 94\%$) and hypotension
- Normocapnia (PaCO₂ 35–45); brief hyperventilation only for herniation
- Seizure prophylaxis per local protocol

 Canadian C-spine rule

Alert (GCS 15), stable adult trauma patient with neck pain or mechanism.

High-risk factors (any → image)

- ☐ Age ≥ 65
- ☐ Dangerous mechanism (fall ≥ 1 m/5 stairs, axial load, high-speed/rollover/ejection MVC, motorized recreational vehicle, bicycle collision)
- ☐ Paresthesias in extremities

Low-risk factors (allow ROM assessment)

- ☐ Simple rear-end MVC
- ☐ Sitting position in ED
- ☐ Ambulatory at any time
- ☐ Delayed onset of neck pain
- ☐ Absence of midline c-spine tenderness
- ☐ Able to actively rotate neck 45° left and right

No low-risk factor → CT c-spine

- NEXUS alternative: no imaging if no midline tenderness, no focal deficit, normal alertness, no intoxication, no painful distracting injury
- CT is the first-line modality in adults; MRI for neurologic deficit or ligamentous concern
- Children: PECARN c-spine factors (altered mental status, abnormal airway/breathing/circulation, focal deficit, torticollis, substantial torso injury, high-risk MVC, diving) – prefer X-ray/observation to limit radiation
- Neurogenic shock: hypotension with bradycardia; avoid over-fluid; norepinephrine; MAP 85–90 often targeted for spinal cord injury (evidence weak)
- PH: motorcycle crashes (often unhelmeted, multiple riders) and falls from trees/roofs are common mechanisms

 Canadian CT head rule

GCS 13–15 with witnessed LOC, amnesia or disorientation; not for < 16 y, anticoagulated, seizure after injury.

High risk (neurosurgical intervention)

- ☐ GCS < 15 at 2 h after injury
- ☐ Suspected open or depressed skull fracture

- ☐ Sign of basal skull fracture (hemotympanum, raccoon eyes, Battle sign, CSF oto/rhinorrhea)
- ☐ Vomiting ≥ 2 episodes
- ☐ Age ≥ 65

Medium risk (brain injury on CT)

- ☐ Retrograde amnesia ≥ 30 min before impact
- ☐ Dangerous mechanism (pedestrian struck, ejection, fall > 1 m/5 stairs)

CT not required by rule

- Anticoagulated or on antiplatelets + head injury → low threshold for CT; reverse if bleed (reversal)
- Children: PECARN head rule
- Discharge with written concussion advice and a responsible adult; return for vomiting, worsening headache, confusion, seizures
- Moderate/severe TBI: avoid hypoxia and hypotension (SBP ≥ 100–110), raised ICP card

Chest & cardiac trauma

Injury	Key points
Tension pneumothorax	Clinical diagnosis – decompress (4th/5th ICS anterior axillary line or 2nd ICS MCL) then chest tube (PTX card)
Massive hemothorax	Chest tube + blood; surgery by output and physiology △ VERIFY
Flail chest / pulmonary contusion	Analgesia, NIV if appropriate, avoid fluid overload; intubate for failure
Cardiac tamponade	Penetrating precordial wound, hypotension, FAST effusion → OR/thoracotomy; pericardiocentesis only as bridge (pericardiocentesis)
Resuscitative thoracotomy	Penetrating trauma with recent loss of signs of life △ VERIFY
Blunt cardiac injury	ECG + troponin; normal both → excludes significant injury; arrhythmias → monitor, echo
Blunt aortic injury	Deceleration; widened mediastinum → CTA; HR < 100 and SBP < 120 (esmolol) to limit shear

Injury

Key points

Rib fractures (older adults)

High morbidity – analgesia and pulmonary toilet; incentive spirometry

[△ VERIFY](#)

Abdominal & GU trauma

- Blunt: eFAST in primary survey; unstable + positive FAST → OR/IR; stable → CT with IV contrast; seat-belt sign = hollow viscus/Chance fracture risk
- Penetrating: hypotension, peritonitis or evisceration → laparotomy; stable stab wounds → local wound exploration/CT/serial exams; gunshot to abdomen usually laparotomy
- Flank/buttock penetrating: triple-contrast CT (retroperitoneal colon, ureter, rectum); rectal/vaginal exam for blood
- **Urethral injury:** blood at meatus, perineal hematoma, high-riding prostate → retrograde urethrogram before catheter (or one gentle attempt per local practice)
- Gross hematuria/pelvic fracture → CT cystogram; renal injury mostly non-operative
- Solid organ injuries in stable patients: non-operative management with serial Hb and exams; angioembolization
- Hollow viscus injuries may be missed on early CT – re-examine
- Pregnancy: OB arrest / trauma

Face & neck trauma

- Penetrating neck: **hard signs** (expanding/pulsatile hematoma, active bleeding, bruit/thrill, air bubbling, stridor, shock, neurologic deficit) → OR; otherwise CTA neck ("no-zone" approach) regardless of zone
- Do not probe or explore beyond platysma in ED; direct pressure (Foley balloon tamponade for bleeding tracts)
- Airway: early intubation if expanding hematoma or voice change; prepare for surgical airway
- Blunt cerebrovascular injury (seat-belt neck, hanging, cervical fracture, Horner) → CTA
- Face: midface (Le Fort) fractures – airway and bleeding; mandible – malocclusion, open fractures need antibiotics; nasal septal hematoma – drain
- Orbit: retrobulbar hematoma (proptosis, ↑IOP, ↓vision) → lateral canthotomy within 90–120 min; blowout fracture with entrapment (esp. children – "white-eyed" blowout with vomiting/bradycardia) → urgent surgery (eye card)

Trauma in older adults

- Undertriage is common: SBP < 110 may be shock at age ≥ 65; HR may not rise (β-blockers)
- Low-energy falls cause serious injury (c-spine, subdural, hip, ribs); low threshold for CT including c-spine
- Check anticoagulants/antiplatelets early; lactate/base deficit for occult hypoperfusion
- Look for why they fell (syncope, arrhythmia, infection, stroke, hypoglycemia, medications) and for elder abuse (elder abuse)
- Delirium prevention, early mobility, early goals-of-care discussion; rib fracture care [△ VERIFY](#)

Trauma (extended) sources

Stiell et al., Canadian C-spine rule (JAMA 2001) and Canadian CT head rule (Lancet 2001) · Hoffman et al., NEXUS (NEJM 2000) · Leonard et al., PECARN cervical spine (Lancet Child Adolesc Health 2024) · ACS ATLS 11th edition · Seamon et al., EAST resuscitative thoracotomy 2015 · Galvagno et al., EAST rib fracture analgesia 2016 · Sperry et al., WTA penetrating neck (no-zone) 2013 · Morey et al., AUA urotrauma guideline 2020 · EAST geriatric trauma guideline. Tintinalli 9e Section 21 used only as a topic map.

⚠ Items pending owner review in this section (12)

1. Burn fluid formula: Parkland 4 mL/kg/%TBSA vs ATLS 10th (2 adult / 3 child / 4 electrical)

Starting rate only – titrate to urine output; local burn unit may prefer a different formula.

2. Calcium during massive transfusion (1 g CaCl₂ or 3 g gluconate per ~4 units)

Timing/ratio varies by MTP; guide by ionized calcium.

3. Pediatric MTP aliquots 10–20 mL/kg per component; activation > 40 mL/kg

Institution-specific pediatric MTP packs and ratios.

4. Pediatric TXA 15 mg/kg (max 1 g) then 2 mg/kg/hr × 8 h (max 1 g)

Pediatric regimens differ (e.g. RCPCH fixed doses, 10–15 mg/kg loads).

5. Hypertonic saline (3% 250 mL adult / 3–5 mL/kg peds; 23.4% 30 mL) and mannitol 0.5–1 g/kg

Concentration/stock and ICP protocols vary.

6. Permissive hypotension / TBI blood-pressure targets

ATLS/local trauma protocol dependent.

7. Tranexamic acid 1 g + 1 g (peds 15 mg/kg) (v1 dose, previously unflagged)

Carried over unchanged from the v1 site; needs owner sign-off like every other dose.

8. Primary survey numbers (fluid boluses, transfusion triggers)

Inline values from v1 / merged card, previously unflagged.

9. Massive transfusion activation criteria and ratios

Inline values from v1 / merged card, previously unflagged.

10. Rib fractures (older adults): multimodal analgesia – paracetamol 1 g q6h, NSAID if no contraindication, opioid PRN, early regional block (serratus anterior/erector spinae); ≥ 3 fractures or age ≥ 65 → admit/monitor

EAST 2016 / regional anesthesia practice; local block availability.

11. Resuscitative thoracotomy indications: penetrating thoracic trauma with signs of life lost < 15 min of prehospital CPR (strong), blunt trauma < 10 min (conditional); not for blunt trauma without signs of life

EAST 2015 / WTA.

12. Massive hemothorax: initial chest tube output > 1500 mL or > 200 mL/h for 2–4 h → thoracotomy consideration; tube size 28–32 Fr (or 14 Fr pigtail for stable patients)

ATLS 11th ed; small-bore increasingly acceptable.

🦴 Orthopedic injury basics

- 1 Check and document distal pulses, capillary refill, motor and sensory function **before and after** any reduction or splint
- 2 Pale/pulseless limb with deformity → immediate gentle realignment with traction, then reassess (knee/elbow dislocations: ankle-brachial index < 0.9 → CT angiography)
- 3 **Open fracture:** photograph, remove gross contamination, saline-soaked dressing, splint, IV antibiotics within 1 h, tetanus (table), orthopedics for debridement **△ VERIFY**
- 4 X-ray the joint above and below; repeat after reduction; children – compare views, consider Salter-Harris injuries
- 5 Analgesia early (see pain ladder and hematoma/fascia iliaca blocks)

Cefazolin – open fracture (within 1 h) △ VERIFY

IV q8h

2 g

≥ 120 kg: 3 g; Gustilo III: add gram-negative cover

Gentamicin – Gustilo III add-on △ VERIFY

IV daily

5 mg/kg

Use adjusted body weight if obese; renal function.

- Splint padding over bony prominences; check skin and circulation; elevate; written cast/splint care instructions (pain, numbness, colour change → return)
- Suspect non-accidental injury: fractures in non-ambulant infants, posterior rib, metaphyseal corner, inconsistent history (child abuse)

🦶 Ottawa rules

Ankle / foot

- ☐ Bone tenderness: posterior edge/tip of lateral malleolus (distal 6 cm)
- ☐ Bone tenderness: posterior edge/tip of medial malleolus
- ☐ Unable to bear weight 4 steps (immediately and in ED)
- ☐ Bone tenderness: base of 5th metatarsal
- ☐ Bone tenderness: navicular

Knee

- ☐ Age ≥ 55
- ☐ Tenderness at fibular head
- ☐ Isolated patellar tenderness
- ☐ Cannot flex to 90°
- ☐ Unable to bear weight 4 steps (immediately and in ED)

Region	Result
Ankle X-ray	Not needed by rule
Foot X-ray	Not needed by rule
Knee X-ray	Not needed by rule

Validated for adults (ankle rule also > 5 y); not for intoxicated, multiple injuries, decreased sensation, or > 10 days after injury. Clinical judgment overrides.

🦴 Upper limb injuries

Injury	Key points
Anterior shoulder dislocation	Pre-reduction X-ray (fracture?), check axillary nerve (regimental badge sensation); reduce with external rotation, FARES, Cunningham or scapular manipulation ± intra-articular lidocaine/sedation; sling, post-reduction X-ray
Posterior shoulder dislocation	Seizure/electrocution; arm held internally rotated – "lightbulb" sign; often missed
Clavicle	Sling; surgery if skin tenting, open, neurovascular injury
Supracondylar fracture (child)	Check radial pulse and AIN (OK sign); displaced/pulseless → urgent orthopedics
Radial head subluxation ("nursemaid")	Age 1–4, pulled arm; reduce by hyperpronation or supination-flexion; no X-ray if typical
Scaphoid	Snuffbox/tubercle tenderness with normal X-ray → thumb spica, repeat imaging/MRI in 10–14 d (avascular necrosis risk)
Distal radius	Hematoma block + reduction; neurovascular (median nerve) check
Hand	Boxer's (5th metacarpal neck) – check rotation; mallet finger – extension splint 6–8 weeks; gamekeeper's thumb (UCL) – thumb spica; fight bite (wound card)

Lower limb & pelvis

Injury

Key points

Pelvic fracture with shock	Binder at greater trochanters, MTP, eFAST, angio-embolisation/packing (trauma)
Hip fracture (elderly)	Shortened externally rotated leg; fascia iliaca block, delirium prevention, surgery within 24–48 h  ; occult → CT/MRI
Femoral shaft	Traction splint (not with pelvic/knee injury), blood loss 1–1.5 L
Knee dislocation	Often spontaneously reduced – check ABI/vascular imaging (popliteal artery injury), peroneal nerve
Patellar dislocation	Reduce by extending knee with medial pressure; X-ray for osteochondral fragment
Achilles rupture	Positive Thompson (calf squeeze) test; equinus cast/boot, orthopedics
Ankle fracture	Weber C/bimalleolar/talar shift → orthopedics; dislocated ankle with skin tenting → reduce immediately
Foot	Lisfranc (plantar ecchymosis, widened 1st–2nd MT gap – easily missed); 5th MT avulsion vs Jones fracture (non-weight-bearing, higher non-union)

3 Measure pressures in obtunded/uncertain patients; ΔP (diastolic – compartment) < 30 mmHg supports fasciotomy 

4 Emergency fasciotomy by surgeon; check CK, potassium, renal function (rhabdomyolysis)

- High risk: tibial and forearm fractures, crush (earthquake/collapse crush card), snakebite (rarely true compartment syndrome – measure first), burns, IV infiltration, reperfusion after vascular repair

Orthopedics sources

Stiell et al., Ottawa ankle rules (JAMA 1993) and Ottawa knee rule (JAMA 1996) · BOAST open fractures 2017 · Hoff et al., EAST open fracture antibiotic guideline 2011 · AAOS hip fracture in older adults guideline 2021 · NICE CG124 hip fracture · BOAST compartment syndrome 2016; McQueen & Court-Brown pressure threshold · NICE NG38 non-complex fractures · ATLS 10th ed. Tintinalli 9e Section 22 used only as a topic map.

Compartment syndrome

- 1 **Pain out of proportion**, pain on passive stretch, tense compartment – earliest signs; pulselessness and paralysis are late
- 2 Remove/split casts and dressings to skin, limb at heart level (not elevated), treat hypotension, avoid excessive opioids masking pain

⚠ Items pending owner review in this section (3)

1. Open fracture antibiotics within 1 h: cefazolin 2 g IV (3 g if ≥ 120 kg; peds 30 mg/kg, max 2 g) q8h; add gram-negative cover (e.g. gentamicin 5 mg/kg daily or ceftriaxone) for Gustilo III; farm/soil contamination – add penicillin/metronidazole for clostridia

EAST 2011 / BOAST 2017; regimens vary by institution.

2. Compartment syndrome: fasciotomy if compartment pressure within 30 mmHg of diastolic BP ($\Delta P < 30$ mmHg) with clinical signs

McQueen threshold; clinical diagnosis in alert patients.

3. Hip fracture: analgesia within 30 min incl. fascia iliaca block; surgery within 24–48 h

AAOS 2021 / NICE CG124.

 Neck & back pain

- 1 **Cauda equina:** bilateral sciatica, saddle anesthesia, urinary retention/incontinence, reduced anal tone → emergency MRI and spinal surgery △ VERIFY
 - 2 **Spinal epidural abscess:** back pain + fever ± neuro deficit; risks diabetes, IV drug use, recent procedure, bacteremia; ESR/CRP, MRI, blood cultures, antibiotics after cultures (vancomycin + ceftriaxone/cefepime), surgery
 - 3 **Malignancy:** history of cancer, weight loss, night pain, age > 50 – MRI; cord compression (oncology card)
 - 4 **Fracture:** elderly/osteoporosis/steroids, trauma; **AAA** in older patients with back pain + hypotension (aorta)
 - 5 **Spinal TB (Pott disease)** – relevant in PH: insidious back pain, gibbus, constitutional symptoms, paraparesis – MRI, TB work-up
- No red flags: reassure, stay active, NSAID/acetaminophen, heat; avoid routine imaging and opioids; most improve in 4–6 weeks
 - Radiculopathy: same plus follow-up; progressive motor deficit → MRI
 - Neck pain: look for cervical artery dissection (headache, Horner, neuro deficits), meningitis, retropharyngeal abscess

 Hot swollen joint

Child with limp / irritable hip – Kocher criteria

- ☐ Non-weight-bearing
- ☐ Fever > 38.5 °C
- ☐ ESR > 40 mm/h
- ☐ WBC > 12 × 10⁹/L

0 Kocher (0–4)

Septic hip probability ≈ < 1% △ VERIFY – ≥ 2 → urgent ultrasound/aspiration, orthopedics

- 1 A single hot swollen joint is **septic until proven otherwise** – aspirate before antibiotics (unless septic shock); send cell count, Gram stain, culture, crystals
- 2 Synovial WBC > 50,000/mm³ (often > 90% neutrophils) strongly suggests infection; crystals do not exclude co-existing infection
- 3 Young sexually active: disseminated gonococcal infection (tenosynovitis, pustules, migratory arthralgia) – ceftriaxone
- 4 Prosthetic joint: do not aspirate in the ED – orthopedics

Septic arthritis – empiric (after aspiration) △ VERIFY

IV

Cefazolin 2 g q8h (vancomycin 25 mg/kg load if MRSA risk); gonococcal: ceftriaxone 1 g daily

Gout / pseudogout

- Gout: monosodium urate (needle, negative birefringence); first MTP, ankle, knee; triggers – alcohol, seafood/organ meat, diuretics, dehydration △ VERIFY
- Pseudogout (CPPD): knee/wrist, elderly, rhomboid positively birefringent crystals; same anti-inflammatory approach

Colchicine – acute gout (low-dose) △ VERIFY

PO

1.2 mg, then 0.6 mg after 1 h (max 1.8 mg day 1); then 0.6 mg daily–BID

Halve/avoid in CKD, with clarithromycin, azoles, verapamil, cyclosporine.

Naproxen / prednisone – acute gout △ VERIFY

Naproxen

500 mg PO BID × 5–7 d (avoid CKD, GI bleed, HF)

or Prednisone

30–40 mg PO daily × 5 days

 Rheumatic emergencies

- **Giant cell arteritis** (> 50 y, new headache, jaw claudication, scalp tenderness, visual loss, high ESR/CRP): steroids immediately – do not wait for biopsy △ VERIFY
- **SLE flare:** fever in lupus = infection until proven (immunosuppressed); lupus nephritis, serositis, cytopenias, CNS lupus, antiphospholipid thrombosis
- **Vasculitis:** pulmonary-renal syndrome (hemoptysis + AKI) → ICU, plasma exchange/steroids
- **Rheumatoid arthritis:** atlantoaxial instability – careful neck handling during intubation; drug toxicity (methotrexate: cytopenias, pneumonitis)
- **Chikungunya** (PH outbreaks): fever + severe symmetric polyarthralgia; supportive (acetaminophen until dengue excluded); arthralgia may persist months
- Steroid-treated patients: adrenal crisis risk with acute illness (adrenal crisis)

Steroids – giant cell arteritis △ VERIFY

No visual loss

Prednisone 40–60 mg PO daily

Visual symptoms

Methylprednisolone 500–1,000 mg IV daily × 3 d

 Regional musculoskeletal pain

Region	ED-relevant problems
Shoulder	Rotator cuff tendinopathy/tear, adhesive capsulitis, calcific tendinitis; referred pain (MI, diaphragmatic irritation, Pancoast)
Hip	Older: occult fracture, OA, trochanteric bursitis; child: transient synovitis vs septic hip vs SCFE (obese adolescent, knee pain) vs Perthes
Knee	Effusion: aspirate if septic concern; prepatellar bursitis (septic in kneelers – aspirate); Baker cyst rupture mimics DVT

Region	ED-relevant problems
Hand	Flexor tenosynovitis (Kanavel: fusiform swelling, flexed posture, pain on passive extension, sheath tenderness) → IV antibiotics + hand surgeon; felon (drain), paronychia (drain), herpetic whitlow (no incision)
Foot	Diabetic foot infection – grade severity, probe to bone, X-ray, offload  ; plantar fasciitis; ingrown toenail (partial avulsion); gout

Diabetic foot infection

Mild

Cephalexin 500 mg QID or amoxicillin-clavulanate 875/125 BID

Moderate-severe

Piperacillin-tazobactam 4.5 g IV q6h ± vancomycin

MSK sources

Qaseem et al., ACP low back pain guideline 2017 · NICE NG59 low back pain and sciatica · FitzGerald et al., 2020 ACR gout guideline · Richette et al., EULAR gout 2016 · Coakley et al., BSR/BHPR hot swollen joint guideline 2006 · Kocher et al., septic hip criteria (JBJS 1999) · Mackie et al., BSR GCA guideline 2020 · Senneville et al., IWGDF/IDSA diabetic foot infection 2023 · Berbari et al., IDSA native vertebral osteomyelitis 2015 · WHO TB guidelines. Tintinalli 9e Section 23 used only as a topic map.

Items pending owner review in this section (6)

1. Acute gout: colchicine 1.2 mg then 0.6 mg 1 h later (then 0.6 mg daily-BID; reduce with CKD/interacting drugs); or naproxen 500 mg BID; or prednisone 30–40 mg daily × 5 d; intra-articular steroid for single joint; do not stop allopurinol

ACR 2020 low-dose colchicine; EULAR uses 1 mg then 0.5 mg.

2. Septic arthritis empiric: cefazolin 2 g IV q8h (or vancomycin if MRSA risk); gonococcal: ceftriaxone 1 g IV daily; peds: cefazolin 50 mg/kg q8h (MRSA risk: clindamycin/vancomycin)

BSR/IDSA practice; Kingella in < 4 y responds to cefazolin/cefuroxime.

3. Kocher criteria (child with irritable hip): non-weight-bearing, fever > 38.5 °C, ESR > 40 mm/h, WBC > 12 × 10⁹/L – septic arthritis probability ~3% (1), 40% (2), 93% (3), 99% (4); CRP > 2 mg/dL often added

Kocher 1999 derivation; lower in validation cohorts.

4. Giant cell arteritis: prednisone 40–60 mg daily immediately (before biopsy); visual symptoms → IV methylprednisolone 500–1,000 mg daily × 3 days

BSR 2020 / ACR 2021.

5. Cauda equina: post-void residual > 200 mL (with symptoms) increases likelihood – emergency MRI regardless

Observational; normal PVR does not exclude.

6. Diabetic foot infection: mild – cephalexin/amoxicillin-clavulanate; moderate-severe – piperacillin-tazobactam 4.5 g IV q6h ± vancomycin; probe-to-bone/X-ray/MRI for osteomyelitis

IWGDF/IDSA 2023; local MDR prevalence.

 Mental health evaluation

- 1 **Safety first:** search/remove dangerous items, safe room, observation level; staff never alone with an agitated patient
- 2 **Medical evaluation:** vital signs, glucose, focused history/exam; look for delirium (fluctuating attention), intoxication/withdrawal, head injury, hypoxia, infection, thyroid disease. Routine labs are not required when history/exam are reassuring
- 3 **Red flags for organic cause:** age > 40 with first psychiatric episode, abnormal vitals, visual hallucinations, disorientation, focal neurology, recent medication change
- 4 **Capacity** is decision-specific: understands, retains, weighs, communicates – document
- 5 PH RA 11036 Mental Health Act: rights, informed consent, least-restrictive care; emergency treatment without consent only to prevent imminent harm, time-limited and documented **△ VERIFY**
- 6 Crisis support: NCMH Crisis Hotline 1553 **△ VERIFY**

 Acute agitation

- 1 **Verbal de-escalation:** one calm speaker, respect personal space, listen, offer choices (food, drink, oral medication), set clear limits
- 2 Offer **oral** medication if cooperating (lorazepam 1–2 mg, olanzapine ODT 5–10 mg, risperidone 2 mg) **△ VERIFY**
- 3 Danger to self/others → IM medication; physical restraint only as a last resort, minimum time, continuous observation, vital signs every 15 min, document
- 4 Methamphetamine ("shabu") intoxication: benzodiazepines first, cool, check temperature/CK/ECG (Tox: sympathomimetic)

Midazolam IM – agitation **△ VERIFY**

IM

5 mg (2.5 mg if elderly/frail); repeat after 10–15 min if needed**Haloperidol (± benzodiazepine)** **△ VERIFY**

IM

5 mg (2.5 mg elderly), may combine with midazolam 5 mg; ECG when feasible**Olanzapine** **△ VERIFY**

IM / ODT

10 mg (5 mg elderly); no IM benzodiazepine within 1 h**Ketamine – severe dangerous agitation** **△ VERIFY**

IM

4 – 5 mg/kg

Or 1–2 mg/kg IV. Airway-capable clinician, monitoring; laryngospasm, emergence.

Diphenhydramine – mild agitation (child) / acute dystonia**△ VERIFY**

IV/IM

25 – 50 mg

- After sedation: airway position, SpO₂, capnography if deep, glucose, temperature, search for cause
- Excited delirium features (hyperthermia, extreme agitation, superhuman strength) → rapid sedation, cooling, avoid prone restraint (asphyxia)

 Suicide risk & self-harm

- 1 Treat the medical injury/overdose first (Tox approach); paracetamol level in every intentional overdose
- 2 Screen: ask directly about suicidal thoughts, plan, intent, means, previous attempts (C-SSRS screener; ASQ for 10–24 y) – asking does not increase risk
- 3 High risk: plan/intent, recent attempt, access to lethal means (pesticides, firearms, medication stockpiles), psychosis, intoxication, hopelessness, social isolation, recent loss
- 4 Constant observation, remove belongings with potential for harm, psychiatric assessment before discharge
- 5 Discharge only if low risk with a safety plan, means restriction counselling (family to secure medicines/pesticides), follow-up within days, crisis numbers **△ VERIFY**

 Psychosis, mood, anxiety & older adults

- New psychosis: exclude delirium, substances (methamphetamine, cannabis, steroids), encephalitis (autoimmune/HSV), HIV, thyroid, seizures; urine drug screen helps little
- Antipsychotic side effects: acute dystonia/oculogyric crisis, akathisia → diphenhydramine/benzotropine **△ VERIFY**; NMS (Tox); QT prolongation
- Mania: sleep loss, grandiosity – lithium toxicity check if on lithium (Tox: lithium)
- Panic attack is a diagnosis of exclusion: ECG, SpO₂, glucose; consider PE, ACS, thyrotoxicosis, arrhythmia, asthma
- **Older adults:** delirium (acute, fluctuating, inattention – CAM) is a medical emergency; avoid benzodiazepines/anticholinergics; low-dose haloperidol only if distressing/dangerous; look for urinary retention, constipation, infection, drugs, pain
- **Eating disorders:** bradycardia, hypotension, hypokalemia, hypophosphatemia, hypoglycemia – admit if unstable; refeeding syndrome risk **△ VERIFY**

Diphenhydramine – mild agitation (child) / acute dystonia**△ VERIFY**

IV/IM

25 – 50 mg **Alcohol withdrawal & substance use****0** CIWA-Ar (0–67)**< 10: minimal – supportive care, reassess q4–8h****△ VERIFY**

- Timeline: tremor/anxiety 6–24 h; seizures 12–48 h; hallucinosis 12–48 h; delirium tremens 48–96 h (mortality without treatment)
- CIWA requires a communicating patient; if sedated/delirious use RASS-guided treatment
- **Thiamine before or with glucose** in every malnourished/alcohol-dependent patient; Wernicke = confusion, ataxia, ophthalmoplegia (often incomplete) **△ VERIFY**
- Also replace Mg, K, phosphate; look for head injury, GI bleed, pancreatitis, infection, hypoglycemia, methanol from adulterated lambrano (Tox: toxic alcohols)

Diazepam – alcohol withdrawal (symptom-triggered)**△ VERIFY**

PO / IV

10 – 20 mg q1–2h while CIWA-Ar ≥ 10 (IV 5–10 mg q5–10 min for severe)

Lorazepam (liver failure, elderly)

1 – 4 mg PO/IV q1–2h

Phenobarbital – refractory alcohol withdrawal

IV over 30 min

10 mg/kg

Monitored setting; additive respiratory depression with benzodiazepines.

Thiamine

Prophylaxis

100 – 250 mg IV/IM daily

Suspected Wernicke

500 mg IV TID × 3 days, then 250 mg daily

- Other substance use: methamphetamine (most common illicit drug in PH), cannabis, inhalants ("rugby" solvent sniffing in street children), opioids (naloxone) – brief intervention and referral to DOH-accredited treatment and rehabilitation centres

Psychosocial sources

Holloman & Zeller, Project BETA consensus statements (West J Emerg Med 2012) · NICE NG10 violence and aggression 2015 · ACEP clinical policy: severe agitation 2023 · Gerson et al., BETA pediatric agitation consensus 2019 · ASAM clinical practice guideline on alcohol withdrawal management 2020 · Sullivan et al., CIWA-Ar (Br J Addict 1989) · Thomson et al., RCP Wernicke guidance · Posner et al., C-SSRS; NIMH ASQ toolkit · Philippine RA 11036 Mental Health Act and IRR (2018) · National Center for Mental Health crisis hotline · APA eating disorders guideline 2023; MARSIPAN. Tintinalli 9e Section 24 used only as a topic map.

Items pending owner review in this section (8)

1. Agitation (adult): midazolam 5 mg IM (2.5 mg elderly); haloperidol 5 mg IM ± midazolam 5 mg or lorazepam 2 mg; olanzapine 10 mg IM (no parenteral benzodiazepine within 1 h); droperidol 5–10 mg IM; severe/dangerous: ketamine 4–5 mg/kg IM (1–2 mg/kg IV)

BETA 2012 / NICE NG10 / ACEP 2023; elderly and intoxicated need lower doses; ECG for QT with antipsychotics.

2. Pediatric agitation: diphenhydramine 1 mg/kg (max 50 mg) PO/IM; midazolam 0.1–0.15 mg/kg IM (max 5 mg); olanzapine 2.5–5 mg (< 12 y) / 5–10 mg (≥ 12 y); haloperidol 0.05–0.1 mg/kg IM (max 5 mg) in ≥ 6 y

AACAP/BETA pediatric consensus 2019; specialist input.

3. Alcohol withdrawal: CIWA-Ar ≥ 10 → diazepam 10–20 mg PO/IV q1–2h (or lorazepam 2–4 mg if liver failure/elderly); refractory: phenobarbital 10 mg/kg IV over 30 min (or escalating doses) in monitored setting

ASAM 2020; phenobarbital loading regimens vary.

4. Thiamine: prophylaxis 100–250 mg IV/IM daily; suspected Wernicke encephalopathy 500 mg IV TID × 3 days then 250 mg daily

RCP/EFNS; give before or with glucose (do not delay glucose).

5. Acute dystonia: diphenhydramine 25–50 mg IV/IM (peds 1 mg/kg, max 50 mg) or benztropine 1–2 mg IV/IM; akathisia: propranolol 10–20 mg or diphenhydramine

Standard practice.

6. PH Mental Health Act (RA 11036, 2018): voluntary, informed consent is the default; emergency treatment without consent is permitted only to prevent imminent harm, for up to 72 h, documented and reviewed

Summary of IRR provisions – confirm details with hospital legal/psychiatry.

7. PH NCMH Crisis Hotline: 1553 (Luzon-wide toll-free landline), 0917-899-8727 / 0966-351-4518 / 0908-639-2672

Numbers from public NCMH listings – confirm before posting.

8. Eating disorder admission thresholds (adult): HR < 40–50, SBP < 90, K < 3.0, phosphate < 0.8 mmol/L, BMI < 13–15, syncope, temperature < 35.5 °C; start refeeding cautiously (thiamine, phosphate monitoring)

MARSIPAN/APA 2023 thresholds vary; adolescents have stricter criteria.

Sexual assault

- 1 Treat injuries first; private room, chaperone, trauma-informed approach (explain each step, allow the survivor to stop)
- 2 Consent for each part: medical care, forensic evidence, photography, police notification (adult survivors decide; mandatory reporting for minors)
- 3 **PH:** refer to the hospital **Women and Children Protection Unit (WCPU)** / medico-legal officer (PNP Crime Lab / NBI); RA 8505 rape crisis centres △ VERIFY
- 4 Forensic evidence best collected ≤ 72 h (up to 5–7 days for some samples); avoid washing/changing clothes before exam if possible; document injuries with body map, measurements, photos
- 5 Baseline tests: pregnancy test, HIV, HBsAg, syphilis, hepatitis C; NAAT for gonorrhea/chlamydia/trichomonas if available
- 6 Emergency contraception, STI prophylaxis, HIV PEP (≤ 72 h), hepatitis B vaccine △ VERIFY, tetanus if wounds (tetanus)
- 7 Mental health first aid, safety assessment, follow-up at 2 weeks, 6 weeks, 3 and 6 months (HIV/syphilis serology)

Levonorgestrel – emergency contraception △ VERIFY

PO

1.5 mg single dose as soon as possible (≤ 72 h; reduced efficacy to 120 h)

Ulipristal (if available)

30 mg PO single dose ≤ 120 h

STI prophylaxis after assault △ VERIFY

Ceftriaxone

500 mg IM once

Doxycycline

100 mg PO BID × 7 days (azithromycin 1 g once if pregnant)

Metronidazole (women)

500 mg PO BID × 7 days

HIV post-exposure prophylaxis (28 days) △ VERIFY

PO daily

TDF 300 mg + 3TC 300 mg (or FTC 200 mg) + DTG 50 mg × 28 days; start ≤ 72 h, ideally ≤ 2 h

Intimate partner violence

- Ask privately (without partner/family): "Has anyone at home hurt you or made you feel afraid?" – screen all women of reproductive age and pregnant women
- Clues: delayed presentation, injuries inconsistent with story, central injuries (face, breasts, abdomen), injuries in pregnancy, repeated visits, controlling partner
- **Strangulation:** may have few external signs – voice change, petechiae, dysphagia, neck pain → consider CTA neck (dissection), observe for airway swelling
- High lethality risk: strangulation, weapon threats, escalating frequency, pregnancy, recent separation, threats to kill → urgent safety planning
- Document verbatim statements, body map, photos (with consent); records may be used in court
- **PH:** RA 9262 (VAWC) – Barangay Protection Order (BPO) from barangay VAW desk; Temporary/Permanent Protection Orders from court; PNP Women and Children Protection Desk; WCPU referral; DSWD shelters △ VERIFY
- Children in the household: assess for abuse (child maltreatment)

Elder abuse

- Types: physical, sexual, psychological, financial, neglect (including self-neglect)
- Clues: unexplained bruises (especially arms, trunk, ears), pressure injuries, dehydration/malnutrition, delays in seeking care, caregiver who answers for the patient, medication non-adherence or over-sedation
- Interview patient alone; assess cognition and capacity; check medication levels where possible
- Admit if unsafe discharge; involve social services
- **PH:** refer to hospital social worker, DSWD and the local Office of Senior Citizens Affairs (OSCA); RA 9994 Expanded Senior Citizens Act △ VERIFY

Abuse & assault sources

WHO Clinical management of rape and intimate partner violence survivors (2020) · WHO Responding to intimate partner violence and sexual violence against women (2013) · CDC Sexually Transmitted Infections Treatment Guidelines 2021 · WHO Guidelines for HIV post-exposure prophylaxis 2024; CDC nPEP guideline 2025 · FIGO/WHO emergency contraception statements · Philippine RA 8353, RA 8505, RA 9262, RA 7610, RA 11648, RA 9994 · DOH Administrative Order 2013-0011 (Women and Children Protection Units) · Child abuse content: see Pediatrics. Tintinalli 9e Section 25 used only as a topic map.

△ Items pending owner review in this section (5)

1. Emergency contraception: levonorgestrel 1.5 mg PO single dose ≤ 72 h (some efficacy to 120 h; consider 3 mg if BMI > 30); ulipristal 30 mg ≤ 120 h if available; copper IUD ≤ 5 days most effective
WHO/FIGO; ulipristal availability in PH is limited – confirm local formulary.
2. STI prophylaxis after sexual assault (adolescents/adults): ceftriaxone 500 mg IM once (1 g if ≥ 150 kg) + doxycycline 100 mg PO BID × 7 d + metronidazole 500 mg PO BID × 7 d (women); azithromycin 1 g once if doxycycline contraindicated/pregnant
CDC STI 2021; WHO alternative regimens differ.
3. HIV PEP ≤ 72 h: tenofovir DF 300 mg + lamivudine 300 mg (or emtricitabine 200 mg) + dolutegravir 50 mg daily × 28 days; children: weight-band dosing per DOH/WHO
WHO 2024 / CDC nPEP; PH supply via DOH treatment hubs – confirm local protocol and starter packs.
4. Hepatitis B after assault: start/complete HBV vaccine if not immune; HBIG if source known HBsAg-positive
CDC 2021.
5. PH laws: RA 8353 (rape as crime against persons), RA 11648 (age of sexual consent raised to 16), RA 9262 (VAWC – barangay protection order), RA 7610 (child abuse), RA 9994 (senior citizens); WCPU referral (DOH AO 2013-0011)
Legal summary – confirm with hospital WCPU/legal officer.

 People who inject drugs

- Fever in PWID = endocarditis until proven otherwise: ≥ 2 –3 sets of blood cultures before antibiotics, echocardiography (TTE then TEE); right-sided (tricuspid) disease → septic pulmonary emboli
- Back pain + fever → spinal epidural abscess / vertebral osteomyelitis: MRI whole spine
- Skin abscess: incision and drainage; necrotizing infection signs (pain out of proportion, crepitus) → surgery (skin infections)
- Wound botulism (descending paralysis, ptosis, diplopia, clear sensorium) → antitoxin, airway watch; tetanus (tetanus)
- Also: retained needle fragments, pseudoaneurysm (do not incise a pulsatile groin mass), HIV/HBV/HCV testing, naloxone for take-home
- PH: injected nalbuphine ("Nubain") and shared needles drove an HIV outbreak among PWID in Cebu – offer HIV testing and link to treatment hubs

Vancomycin – suspected endocarditis/sepsis in PWID**△ VERIFY****IV load****25 – 30 mg/kg, max 3,000 mg**

Then dose by levels/AUC; add ceftriaxone 2 g IV daily if Gram-negative cover needed.

 Transplant recipient

- Call the transplant team early (PH: National Kidney and Transplant Institute for many kidney recipients)
- Fever: blunted signs; timeline – < 1 month nosocomial/surgical, 1–6 months opportunistic (CMV, PCP, fungal, TB reactivation), > 6 months community-acquired; low threshold for cultures, imaging, broad antibiotics
- Rejection: graft tenderness, rising creatinine, fever – ultrasound with Doppler; do not stop immunosuppression without specialist advice
- Drug interactions: macrolides, azoles, diltiazem/verapamil raise calcineurin inhibitor levels; rifampicin, phenytoin, carbamazepine lower them; avoid NSAIDs and IV contrast where possible
- Chronic steroids → adrenal suppression: stress-dose hydrocortisone in shock (adrenal crisis) **△ VERIFY**

 Obesity & dosing weights

Uses weight, height and sex from the patient bar.

Enter weight, height and sex in the patient bar.

- Airway: ramp position (ear to sternal notch), preoxygenate with head-up 25–30° + PEEP/HFNO; shorter safe apnea time
- Imaging may exceed scanner weight limits – check table limits early

 Transgender & gender-diverse patients

- Use the patient's name and pronouns; ask respectfully only what is clinically relevant
- **Organ inventory:** anatomy present determines differential and screening – prostate in transfeminine people, pregnancy and ovarian/uterine pathology possible in transmasculine people (pregnancy test when relevant)
- Estrogen: VTE risk (higher with ethinyl estradiol, smoking), hyperkalemia with spironolactone
- Testosterone: polycythemia, hypertension, acne; does not reliably prevent pregnancy
- Lab reference ranges (creatinine, hemoglobin) depend on hormonal status – interpret with care
- Post-operative complications of gender-affirming surgery: neovaginal stenosis/fistula, urethral stricture, flap problems → contact surgical team
- Do not stop hormone therapy routinely; screen for self-harm risk (higher prevalence)

 Palliative care

- Goals-of-care conversation: what does the patient know, what matters most, what are they afraid of, what trade-offs are acceptable; involve family as the patient wishes (family-centred decisions are common in PH)
- Record code status (DNR/DNI) and treatment limits clearly; time-limited trials are an option
- Dyspnea: opioid, fan to face, positioning; oxygen only if hypoxemic and helps
- Pain: regular plus breakthrough opioid; laxative with every opioid
- Last days: stop non-essential drugs/monitoring, mouth care, treat secretions (reposition, antisecretory), delirium, nausea **△ VERIFY**

Morphine – palliative dyspnea/pain (opioid-naïve) △ VERIFY**PO****2.5 – 5 mg q4h + PRN****IV / SC****1 – 2 mg q1h PRN (elderly/renal: lower dose, consider hydromorphone or fentanyl)****Antisecretory – terminal secretions △ VERIFY****Glycopyrrolate****0.2 mg SC/IV q4h PRN****Hyoscine butylbromide****20 mg SC q4h PRN****Terminal delirium / agitation △ VERIFY****Haloperidol****0.5 – 1 mg SC q2–4h PRN****Midazolam****2.5 – 5 mg SC q1h PRN** **Death notification**

- 1 Prepare: confirm identity and facts, private room, sit down, bring a nurse/social worker, silence phones
- 2 Introduce yourself; find out what the family knows; give a warning shot ("I have bad news")
- 3 Say "died" or "dead" clearly; avoid euphemisms; then pause and allow silence and emotion
- 4 Explain what happened briefly; answer questions; offer to let them see the body; respect religious/cultural practices (PH: prayers, priest/pastor/imam, wakes)
- 5 Offer family presence during resuscitation where feasible – it does not worsen outcomes
- 6 Organ/tissue donation: follow local protocol (PH: Philippine Network for Organ Sharing; RA 7170) **△ VERIFY**
- 7 Violent, suspicious or unexplained deaths, deaths on arrival, and deaths within 24 h of injury → medico-legal case; notify PNP; do not remove lines/tubes before medico-legal review
- 8 Look after the team: brief debrief after difficult deaths

 Legal & ethical issues (PH)

- Consent: informed, voluntary, by a person with capacity; emergency doctrine permits life-saving treatment when consent cannot be obtained
- Minors (< 18 y in PH): parent/guardian consent; treat emergencies without delay; child protection concerns override confidentiality
- Refusal / leaving against advice ("HAMA" – home against medical advice): assess capacity, explain risks in plain language, offer alternatives, document, invite return

- **RA 10932** (Anti-Hospital Deposit Law): hospitals may not demand deposits or refuse/detain emergency patients for inability to pay
- **RA 10173** Data Privacy Act: disclose only to those entitled; avoid patient details in messaging groups
- **RA 11332**: report notifiable diseases to the local epidemiology surveillance unit (e.g., measles, dengue clusters, leptospirosis)
- Police cases (assault, gunshot, stab, vehicular): medico-legal certificate, chain of custody for evidence (△ **VERIFY**)

Military & tactical medicine

- TCCC phases: care under fire (return fire, tourniquet only), tactical field care, tactical evacuation
- **MARCH**: Massive hemorrhage (tourniquet, wound packing, junctional devices), Airway, Respiration (needle decompression, vented chest seal), Circulation (TXA, blood/fluids), Hypothermia/Head injury

- Tourniquet high and tight; note time; reassess for conversion within 2 h (△ **VERIFY**)
- Blast and penetrating injury patterns (blast injury); PH relevance: armed-conflict areas in Mindanao, mass-casualty events

Special situations sources

Baddour et al., AHA infective endocarditis scientific statements (2015) and PWID statement (2022) · AST Infectious Diseases Community of Practice guidelines 2019 · Erstad, dosing in obesity reviews; Janmahasatian et al., lean body weight (Clin Pharmacokinet 2005) · WPATH Standards of Care v8 (2022) · NICE NG31 Care of dying adults in the last days of life · Committee on TCCC guidelines 2024 · Philippine Family Code; RA 10932, RA 10173, RA 11332, RA 7170 · DOH HIV surveillance reports (Cebu PWID). Tintinalli 9e Section 26 used only as a topic map.

△ **Items pending owner review in this section (6)**

- 1. Suspected endocarditis in PWID (after ≥ 2–3 blood cultures): vancomycin 25–30 mg/kg IV load (max 3 g) then per levels ± ceftriaxone 2 g or cefazolin per local flora; do not delay in sepsis**
AHA 2015/2022; local MRSA rates.
- 2. Obesity dosing weights: IBW (Devine), adjusted BW = $IBW + 0.4 \times (TBW - IBW)$, lean BW (Janmahasatian); drug-weight pairings (sux TBW, rocuronium IBW/LBW, propofol induction LBW, ketamine LBW/IBW, enoxaparin TBW with anti-Xa if > 150 kg, vancomycin load TBW max 3 g, aminoglycosides AdjBW)**
ASHP/Erstad reviews; pairings vary by source.
- 3. Palliative (opioid-naïve): morphine 2.5–5 mg PO q4h or 1–2 mg IV/SC q1h PRN for dyspnea/pain; on opioids: 10–15 % of daily dose as breakthrough; secretions: glycopyrrolate 0.2 mg SC q4h or hyoscine butylbromide 20 mg SC q4h; terminal delirium: haloperidol 0.5–1 mg SC, midazolam 2.5–5 mg SC**
NICE NG31 / palliative formularies; titrate to effect.
- 4. Transplant: stress-dose steroids if on chronic prednisone and shocked (hydrocortisone 100 mg IV); avoid nephrotoxins; check tacrolimus/cyclosporine level and interacting drugs (macrolides, azoles, rifampicin)**
AST 2019; contact transplant team before changing immunosuppression.
- 5. PH legal points: age of majority 18 (Family Code); RA 10932 bans hospital deposit/refusal in emergencies; RA 10173 Data Privacy Act; RA 11332 mandatory notifiable disease reporting; RA 7170 organ donation; medico-legal cases reported to PNP**
Legal summary – confirm with hospital legal office.
- 6. Tactical care: tourniquet high and tight for massive limb bleeding, convert/reassess within 2 h; TXA 2 g IV/IO ≤ 3 h; ketamine 20–30 mg IV / 50–100 mg IM for analgesia**
TCCC 2024.

Reference aid only – verify clinically. Organised by the 26 major sections of *Tintinalli's Emergency Medicine* 9e as a **topic map only**; all text is original and summarises public guidelines (AHA/ACC 2025 ACS, ESC 2023 ACS / 2021 HF, ACC/AHA 2017 HTN & 2022 aortic, ERC 2021, WMS, WHO SEARO 2016 snakebite, DOH/RITM, UP-PGH NPMCC, UHMS/ACEP CO, ACOG, WHO PPH 2023, FIGO, ADA/EASD/JBDS 2024, ATA 2014/2016, Endocrine Society, AACT/EAPCCT, EXTRIP). Doses follow common guideline values (AHA 2020/2025 ACLS/PALS, SSC 2021, ATLS, AES 2016, ADA 2024, ISPAD 2022, ASRA, KDIGO 2012 AKI, European 2014 / US 2013 hyponatremia, UK Kidney Assoc. 2020/2023 hyperkalaemia, ASHP/IDSA 2020 vancomycin, EXTRIP, BTS/SIGN 2019 & GINA 2024 asthma, GOLD 2025, ATS/IDSA 2019 & PCAP 2016 CAP, ESC 2019 PE, BTS 2023 pleural, ARDS Berlin 2012 / global 2023) but must be checked against the actual product concentration, your institution's protocols and the patient. Pediatric doses should not exceed adult doses. Clinical judgment always overrides. Not a medical device.